

# Desarrollo de Soluciones de IA

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# Agenda

1. Introducción asignatura
2. Introducción a los datos
3. Arquitecturas de procesamiento de datos
4. Datos y modelos matemáticos
5. Introducción al aprendizaje automático
6. Introducción al aprendizaje profundo
7. Casos de uso en la realidad
8. Productos basados en datos

# Programa de la asignatura

1. Tema 1: Programación avanzada en Python
2. Tema 2: Pruebas y despliegue
3. Tema 3: Automatización de tareas e IA
4. Tema 4: Entornos de desarrollo con IA generativa
5. Tema 5: Desarrollo End-to-End

<https://github.com/dmartincc/dsia-26-27>

# **Evaluación asignatura**

**Proyecto I (10%)**

**Proyecto II (20%)**

**Proyecto III (40%)**

**Examen final (30%)**



# Planificación semanal

| Fecha                 | Clase                                                                                                                                              | Entregas             |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 8 de septiembre 2026  | Presentación + Entornos virtuales python y Control de versiones (Git/GitHub)                                                                       |                      |
| 15 de septiembre 2026 | Programación avanzada: procesamiento de datos (pandas) + Ejercicios + Presentación Proyectos                                                       |                      |
| 22 de septiembre 2026 | Diseño de proyectos de software: Arquitectura modular, patrones de diseño, Clean Code y SOLID + Ejercicios + Trabajar en proyecto                  |                      |
| 29 de septiembre 2026 | Pruebas unitarias (pytest), test suites y aserciones + Ejercicios + Trabajar en proyectos                                                          | Entrega Proyecto I   |
| 6 de octubre 2026     | Automatización de tareas: Construcción de flujos de datos (data flows) + Trabajar en proyecto                                                      |                      |
| 13 de octubre 2026    | Integración de servicios de IA mediante APIs (OpenAI, Anthropic, Hugging Face) + Trabajar en proyecto                                              |                      |
| 20 de octubre 2026    | Desarrollo End-to-End I: Integración del flujo completo (Ingesta -> Procesamiento -> API de IA) + Trabajar en proyecto                             |                      |
| 27 de octubre 2026    | Desarrollo End-to-End II: Robustez del sistema, gestión de errores y logging + Trabajar en proyecto                                                | Entrega Proyecto II  |
| 3 de noviembre 2026   | Despliegue y puesta en producción: Estrategias de empaquetado y preparación para producción + Trabajar en proyecto                                 |                      |
| 10 de noviembre 2026  | Introducción a herramientas de desarrollo con IA generativa (Claude Code, Gemini CLI, Cursor) aplicadas al código existente + Trabajar en proyecto |                      |
| 17 de noviembre 2026  | Revisión de código, refactorización, generación asistida de pruebas y documentación con IA + Taller de integración final (Release Candidate)       |                      |
| 24 de noviembre 2026  | Trabajar en el proyecto final                                                                                                                      |                      |
| 1 de diciembre 2026   | Presentación Trabajo Final (1/2)                                                                                                                   |                      |
| 8 de diciembre 2026   | Presentación Trabajo Final (2/2)                                                                                                                   | Entrega Proyecto III |
| TBD                   | Al ser master no lo considero a no ser que me digais lo contrario                                                                                  |                      |

**Lunes de 19:15 a 20:55**

85% Asistencia obligatoria

# Motivación profesional de la asignatura

**Motivación.** Hay datos de sobra; lo que falta es convertir un problema de negocio en algo que se pueda ver, explorar y compartir (no un Excel ni un gráfico muerto).

**Perfil egresado.** Alguien que prototipa y despliega cuadros de mando / apps de datos con criterio de producto: qué mostrar, a quién y cómo dejarlo usable en una URL.

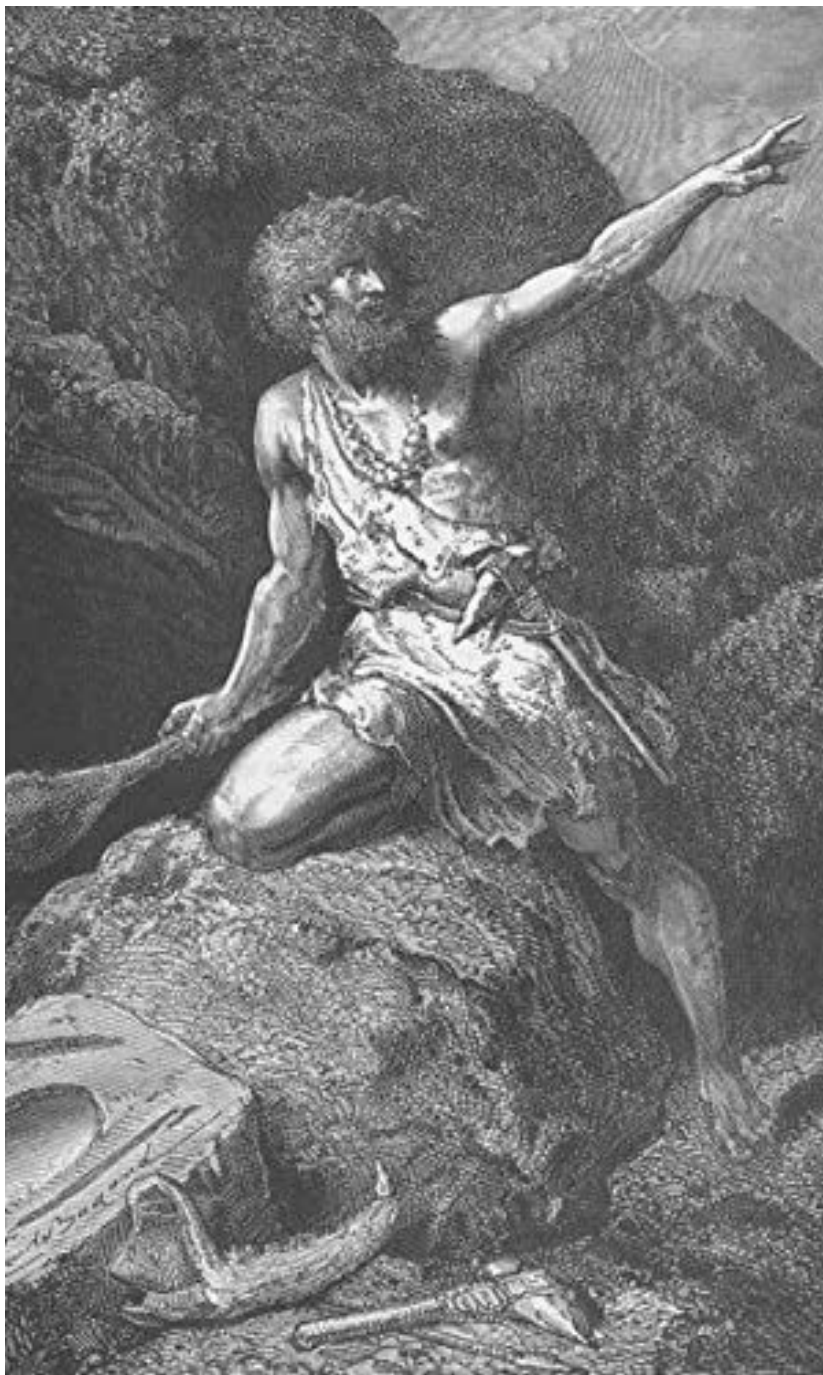
**Data Scientist, Product Managers, AI Engineers, Business analysts, ...**

**Casos de uso.** Panel de ventas o churn; KPIs operativos; indicadores públicos/ESG; explorador interno de un dataset o de un modelo.

# Sobre David Martín-Corral

- ▶ Cofundador de **D3LF**, High Frequency Trading Fund
- ▶ Cofundador de **Zensei**, **Politibot** y **Souly**
- ▶ He trabajado en **IBM**, **CaixaBank**, **Bankinter**, **BBVA**, **Smartvel** y **Sanitas** en las primeras plataformas de Big Data de España y como data scientist
- ▶ **Ingeniero Eléctrico** por la UCLM
- ▶ **Ingeniero Industrial** por la UPM
- ▶ **Máster en Investigación Operativa** por la Universidad de Strathclyde (UK)
- ▶ **Doctor en Ingeniería Matemática** por la UC3M
- ▶ Imparte o ha impartido clases en **Instituto de Empresa**, **Universidad Complutense de Madrid**, **The Valley**, **ThePowerMBA**, **Universidad Carlos III de Madrid** y **Universidad de Alcalá**. David ha publicado en revistas científicas de alto nivel como **Nature** y **PNAS** y ha sido investigador invitado en el laboratorio de dinámica humana del **MIT Media Lab** (USA).

# Introducción a los datos y su analítica

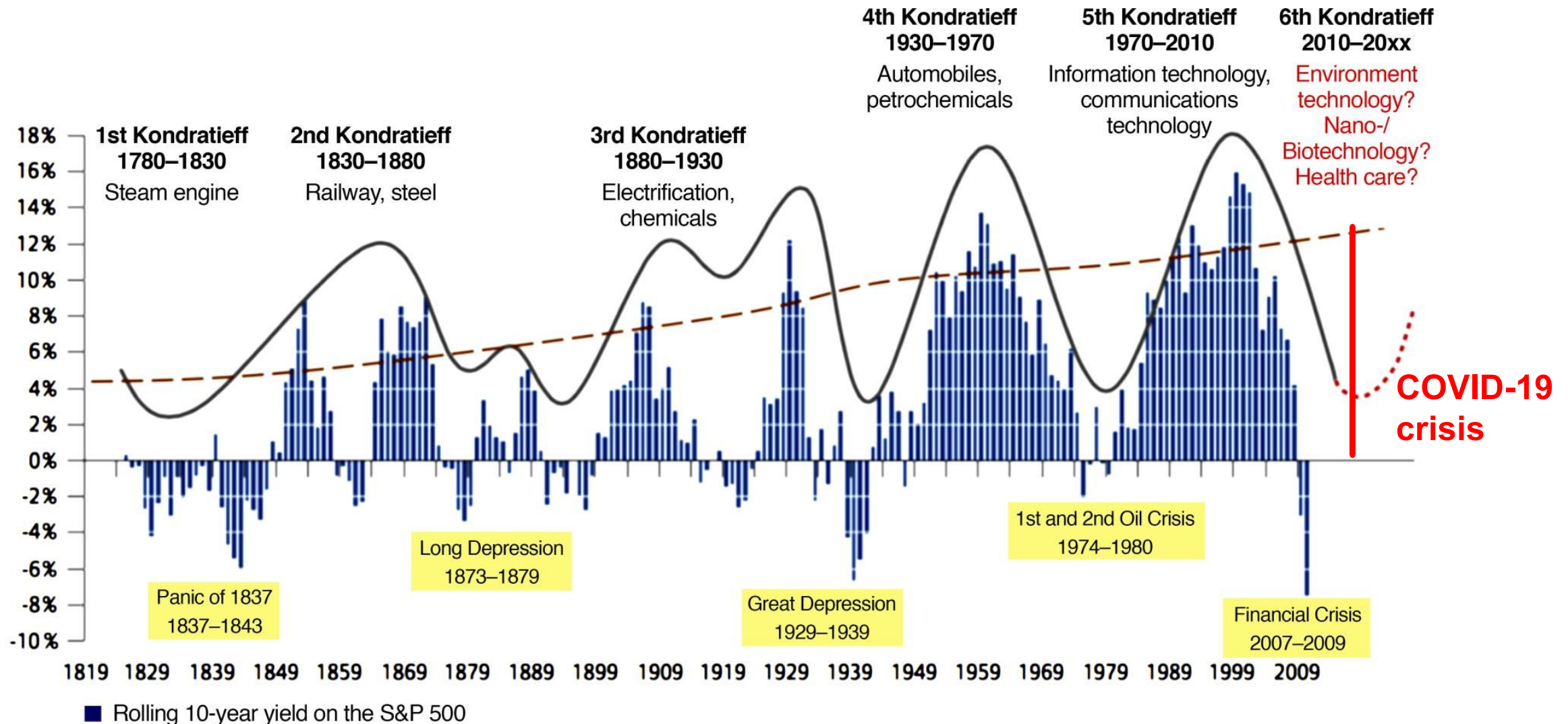


# Situación de los datos y social

|                                                                                                            | Datos       | Conexión social |
|------------------------------------------------------------------------------------------------------------|-------------|-----------------|
|  <b>Neolítico</b>         | No existían | Muy baja        |
|  <b>Principios s. XX</b>  | Escasos     | Media alta      |
|  <b>Principios s. XXI</b> | Exponencial | Hiperconectados |



Estamos en el inicio de la sexta Kondratieff (ola de innovación), donde existen oportunidades de grandes retornos en salud y clima. Pero tenemos que cambiar.



Fuente: [The ownership paradox: nurturing continuity and change for the future ASCI. 2019. Marr](#)

# ¿De donde podemos extraer los datos?

## ► Fuentes de datos propias

Bases de datos

Datos de comportamiento

Sensores

## ► Proveedores de datos

## ► Datos abiertos

## ► Scrapeado de internet

The collage consists of three overlapping images. The top-right image is a screenshot of the Google search homepage, showing the Google logo, a search bar, and navigation links like 'Gmail', 'Images', and 'Log In'. The bottom-left image is an advertisement for the Foursquare Swarm app, featuring an orange background, the 'Foursquare Swarm' logo, and a smartphone displaying the app's interface with a map of New York and a list of check-ins. The bottom-right image is a screenshot of the Foursquare website, showing the 'FOURSQUARE' header, navigation links, and a large headline that reads 'If it tells you where, it's probably built on Foursquare'. Below the headline is a paragraph about Foursquare's commitment to accuracy and privacy, and a 'Contact Sales' button.



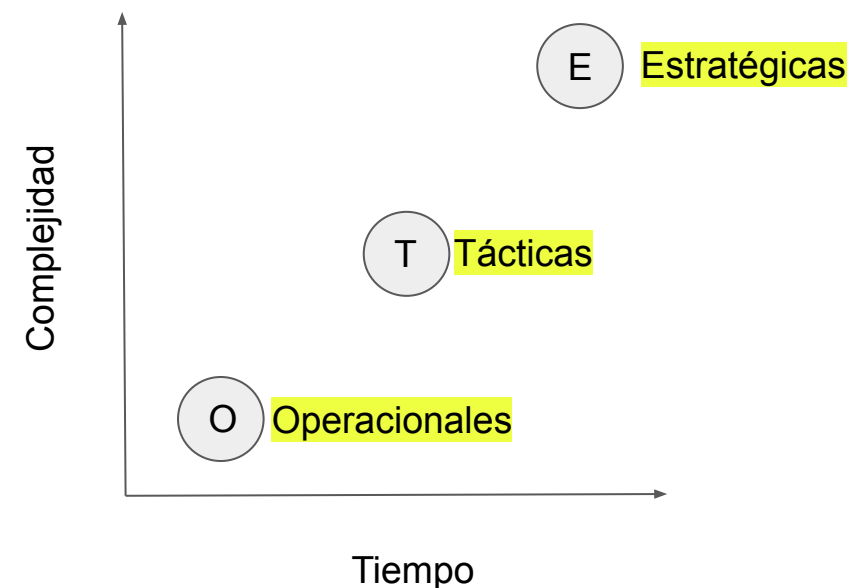
# ¿Para qué podemos utilizar los datos?

## Toma de decisiones

- ▶ Operacionales
- ▶ Tácticas
- ▶ Estratégicas

|                | Operativa        | Táctica   | Estratégica      |
|----------------|------------------|-----------|------------------|
| Opciones       | Binarias o Bajas | Múltiples | Miles o millones |
| Incertidumbre  | Baja             | Media     | Alta             |
| Automatización | Alta             | Media     | Baja             |

En términos de escala temporal y complejidad podemos dividir las en:

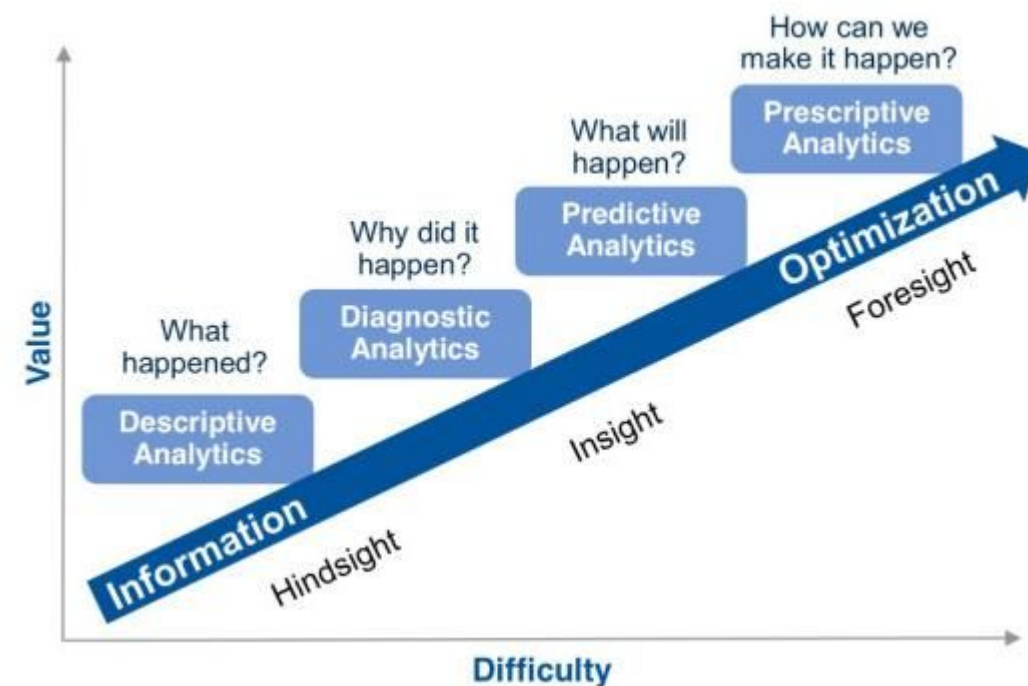


# Análisis que podemos realizar sobre los datos

## Analítica

- ▶ Descriptiva
  - ¿Qué sucedió?
- ▶ Diagnóstica
  - ¿Por qué sucedió?
- ▶ Predictiva
  - ¿Qué sucederá?
- ▶ Prescriptiva
  - ¿Cómo hacemos que suceda?

En términos de valor y dificultad podemos dividir las en:



# Tipos de datos

## ► Estructurados

Tabulares (csv, excel,...)

Diccionarios (json, xml,...)

## ► No estructurados

Texto

Imágenes

Sonido

Vídeo

dreams.csv

```
1 subreddit,date,num_comments,score,title,description,replies
2 DreamInterpretation,2018-03-09 11:42:21,0,1,Pulling Worms and bugs out of my brea
3 DreamInterpretation,2018-03-09 15:58:52,0,1,Doing barrel rolls in a older airplan
4 DreamInterpretation,2018-03-09 17:53:18,1,3,"Accidentally taking car that wasn't m
5 DreamInterpretation,2018-03-10 15:36:41,0,2,Friend Dropped His Phone In The Harbo
6 DreamInterpretation,2018-03-10 17:46:20,1,3,Glen Campbell imparts cryptic life les
7 DreamInterpretation,2018-03-10 22:00:16,0,2,My mom dies in my dreams.,[deleted],
8 DreamInterpretation,2018-03-10 22:09:42,0,3,"When I realize I am dreaming, I know
9 DreamInterpretation,2018-03-10 22:24:41,0,2,The guest house and black hooded figur
10 DreamInterpretation,2018-03-11 02:45:59,1,3,Need help filling in the blanks.,[dele
11 DreamInterpretation,2018-03-11 05:25:23,0,2,Aztec Ritual,"Hi, I had a weird dream
12 DreamInterpretation,2018-03-11 05:43:46,2,3,An indoor beach,"I went walking in a p
13 DreamInterpretation,2018-03-12 01:12:13,0,2,Family Members torturing people,"I ha
14 DreamInterpretation,2018-03-12 04:36:26,0,3,NSFW. I (female) often dream of havin
15 DreamInterpretation,2018-03-12 07:45:56,0,1,Being treated like a child,[deleted],
16 DreamInterpretation,2018-03-12 09:59:46,0,2,Dreamt of a long time crush and being
17 DreamInterpretation,2018-03-12 10:16:57,0,1,"Cemetery, crypt, butchered bodies,,[
18 DreamInterpretation,2018-03-12 12:24:30,0,1,Random Ex-Coworker messaged me this -
19 DreamInterpretation,2018-03-12 18:34:51,0,1,transforming assassin,"I took some Va
20 DreamInterpretation,2018-03-12 21:39:03,0,1,Lucid dreaming Discord x),,[
21 DreamInterpretation,2018-03-12 22:21:18,1,1,Floating.. a recurring dream,"So, I ke
22 DreamInterpretation,2018-03-13 00:20:03,2,1,"In my dream, I saw myself watching my
23 DreamInterpretation,2018-03-13 06:04:33,3,1,BLACK CAT DREAM,[deleted],[{
24 DreamInterpretation,2018-03-13 06:21:38,0,1,Another to interpret,[deleted],[]
25 DreamInterpretation,2018-03-13 18:18:10,4,2,"I had the most awful dream last night
26 DreamInterpretation,2018-03-14 12:49:34,0,1,I can never get through to 911/Nobody
27 DreamInterpretation,2018-03-14 01:35:25,0,1,Small black scorpion stinging very lar
28 DreamInterpretation,2018-03-14 05:08:35,0,1,Recurring dream?,"Just curious if anyo
29 DreamInterpretation,2018-03-14 13:39:10,2,1,Reoccurring Dream about Classes,[deleted],[{
31 DreamInterpretation,2018-03-14 13:47:53,2,2,"Saw my ""ancestors""?",[deleted],[{
32 DreamInterpretation,2018-03-14 23:12:09,1,2,idek?,[deleted],[{
}
```

Period

Saxo

001-102 I:XX To

001-008 I:Enfer

009-041 II:Turno

042-043 III:Enf.

044-045 IV:Enfer

046-049 V:Trast

050-052 VI-VIII:I

053-061 IX:Enle

062-067 X:En

068-071 XI:Enle

072-075 XII:Enle

076-079 XIII:Enle

080-083 XIV:Enle

084-087 XV:Enle

088-091 XVI:Enle

092-095 XVII:Enle

096-099 XVIII:Enle

100-103 XIX:Enle

104-107 XX:Enle

108-111 XXI:Enle

112-115 XXII:Enle

116-119 XXIII:Enle

120-123 XXIV:Enle

124-127 XXV:Enle

128-131 XXVI:Enle

132-135 XXVII:Enle

136-139 XXVIII:Enle

140-143 XXIX:Enle

144-147 XXX:Enle

148-151 XXXI:Enle

152-155 XXXII:Enle

156-159 XXXIII:Enle

160-163 XXXIV:Enle

164-167 XXXV:Enle

168-171 XXXVI:Enle

172-175 XXXVII:Enle

176-179 XXXVIII:Enle

180-183 XXXIX:Enle

184-187 XL:Enle

188-191 XLI:Enle

192-195 XLII:Enle

196-199 XLIII:Enle

200-203 XLIV:Enle

204-207 XLV:Enle

208-211 XLVI:Enle

212-215 XLVII:Enle

216-219 XLVIII:Enle

220-223 XLIX:Enle

224-227 L:Enle

228-231 LI:Enle

232-235 LII:Enle

236-239 LIII:Enle

240-243 LIV:Enle

244-247 LV:Enle

248-251 LVI:Enle

252-255 LVII:Enle

256-259 LVIII:Enle

260-263 LIX:Enle

264-267 LX:Enle

268-271 LXI:Enle

272-275 LXII:Enle

276-279 LXIII:Enle

280-283 LXIV:Enle

284-287 LXV:Enle

288-291 LXVI:Enle

292-295 LXVII:Enle

296-299 LXVIII:Enle

300-303 LXIX:Enle

304-307 LXX:Enle

308-311 LXXI:Enle

312-315 LXXII:Enle

316-319 LXXIII:Enle

320-323 LXXIV:Enle

324-327 LXXV:Enle

328-331 LXXVI:Enle

332-335 LXXVII:Enle

336-339 LXXVIII:Enle

340-343 LXXIX:Enle

344-347 LXXX:Enle

348-351 LXXXI:Enle

352-355 LXXXII:Enle

356-359 LXXXIII:Enle

360-363 LXXXIV:Enle

364-367 LXXXV:Enle

368-371 LXXXVI:Enle

372-375 LXXXVII:Enle

376-379 LXXXVIII:Enle

380-383 LXXXIX:Enle

384-387 XC:Enle

388-391 XCI:Enle

392-395 XCII:Enle

396-399 XCIII:Enle

400-403 XCIV:Enle

404-407 XCV:Enle

408-411 XCVI:Enle

412-415 XCVII:Enle

416-419 XCVIII:Enle

420-423 XCIX:Enle

424-427 C:Enle

428-431 CI:Enle

432-435 CII:Enle

436-439 CIII:Enle

440-443 CIV:Enle

444-447 CV:Enle

448-451 CVI:Enle

452-455 CVII:Enle

456-459 CVIII:Enle

460-463 CIX:Enle

464-467 CL:Enle

468-471 CLI:Enle

472-475 CLII:Enle

476-479 CLIII:Enle

480-483 CLIV:Enle

484-487 CLV:Enle

488-491 CLVI:Enle

492-495 CLVII:Enle

496-499 CLVIII:Enle

500-503 CLIX:Enle

504-507 CLX:Enle

508-511 CLXI:Enle

512-515 CLXII:Enle

516-519 CLXIII:Enle

520-523 CLXIV:Enle

524-527 CLXV:Enle

528-531 CLXVI:Enle

532-535 CLXVII:Enle

536-539 CLXVIII:Enle

540-543 CLXIX:Enle

544-547 CLXX:Enle

548-551 CLXXI:Enle

552-555 CLXXII:Enle

556-559 CLXXIII:Enle

560-563 CLXXIV:Enle

564-567 CLXXV:Enle

568-571 CLXXVI:Enle

572-575 CLXXVII:Enle

576-579 CLXXVIII:Enle

580-583 CLXXIX:Enle

584-587 CLXXX:Enle

588-591 CLXXXI:Enle

592-595 CLXXXII:Enle

596-599 CLXXXIII:Enle

600-603 CLXXXIV:Enle

604-607 CLXXXV:Enle

608-611 CLXXXVI:Enle

612-615 CLXXXVII:Enle

616-619 CLXXXVIII:Enle

620-623 CLXXXIX:Enle

624-627 CCCC:Enle

628-631 CCCC1:Enle

632-635 CCCC2:Enle

636-639 CCCC3:Enle

640-643 CCCC4:Enle

644-647 CCCC5:Enle

648-651 CCCC6:Enle

652-655 CCCC7:Enle

656-659 CCCC8:Enle

660-663 CCCC9:Enle

664-667 CCCC10:Enle

668-671 CCCC11:Enle

672-675 CCCC12:Enle

676-679 CCCC13:Enle

680-683 CCCC14:Enle

684-687 CCCC15:Enle

688-691 CCCC16:Enle

692-695 CCCC17:Enle

696-699 CCCC18:Enle

700-703 CCCC19:Enle

704-707 CCCC20:Enle

708-711 CCCC21:Enle

712-715 CCCC22:Enle

716-719 CCCC23:Enle

720-723 CCCC24:Enle

724-727 CCCC25:Enle

728-731 CCCC26:Enle

732-735 CCCC27:Enle

736-739 CCCC28:Enle

740-743 CCCC29:Enle

744-747 CCCC30:Enle

748-751 CCCC31:Enle

752-755 CCCC32:Enle

756-759 CCCC33:Enle

760-763 CCCC34:Enle

764-767 CCCC35:Enle

768-771 CCCC36:Enle

772-775 CCCC37:Enle

776-779 CCCC38:Enle

780-783 CCCC39:Enle

784-787 CCCC40:Enle

788-791 CCCC41:Enle

792-795 CCCC42:Enle

796-799 CCCC43:Enle

800-803 CCCC44:Enle

804-807 CCCC45:Enle

808-811 CCCC46:Enle

812-815 CCCC47:Enle

816-819 CCCC48:Enle

820-823 CCCC49:Enle

824-827 CCCC50:Enle

828-831 CCCC51:Enle

832-835 CCCC52:Enle

836-839 CCCC53:Enle

840-843 CCCC54:Enle

844-847 CCCC55:Enle

848-851 CCCC56:Enle

852-855 CCCC57:Enle

856-859 CCCC58:Enle

860-863 CCCC59:Enle

864-867 CCCC60:Enle

868-871 CCCC61:Enle

872-875 CCCC62:Enle

876-879 CCCC63:Enle

880-883 CCCC64:Enle

884-887 CCCC65:Enle

888-891 CCCC66:Enle

892-895 CCCC67:Enle

896-899 CCCC68:Enle

900-903 CCCC69:Enle

904-907 CCCC70:Enle

908-911 CCCC71:Enle

912-915 CCCC72:Enle

916-919 CCCC73:Enle

920-923 CCCC74:Enle

924-927 CCCC75:Enle

928-931 CCCC76:Enle

932-935 CCCC77:Enle

936-939 CCCC78:Enle

940-943 CCCC79:Enle

944-947 CCCC80:Enle

948-951 CCCC81:Enle

952-955 CCCC82:Enle

956-959 CCCC83:Enle

960-963 CCCC84:Enle

964-967 CCCC85:Enle

968-971 CCCC86:Enle

972-975 CCCC87:Enle

976-979 CCCC88:Enle

980-983 CCCC89:Enle

984-987 CCCC90:Enle

988-991 CCCC91:Enle

992-995 CCCC92:Enle

996-999 CCCC93:Enle

untitled

```
1 <?xml version="1.0" encoding="utf-8"?>
2
3 <root_element>
4
5   <child_element_1>
6     <child_element_2>Content</child_element_2>
7   </child_element_1>
8
9   <child_element_1>
10     <child_element_2>Content</child_element_2>
11   </child_element_1>
12
13   <child_element_1>
14     <child_element_2>Content</child_element_2>
15     <child_element_2>Content</child_element_2>
16   </child_element_1>
17
18 </root_element>
```

s3-event.json

```
1 {
2   "Records": [
3     {
4       "eventVersion": "2.0",
5       "eventSource": "aws:s3",
6       "awsRegion": "eu-west-1",
7       "eventTime": "2018-04-09T13:39:25.298Z",
8       "eventName": "ObjectCreated:Put",
9       "userIdentity": {
10         "principalId": "AWS:AROAI4UHMFLMCF60KUC:thunder-dev-refinery"
11       },
12       "requestParameters": {
13         "sourceIPAddress": "34.244.58.59"
14       },
15       "responseElements": {
16         "x-amz-request-id": "5F87B1DF8246EAA4",
17         "x-amz-id-2": "0uLqe9041By99GCTGr+ArtKsGJ561kv/NYMBb1Xr0Xm#XodJre5jKdkZsus8VG++csi"
18       },
19       "s3": {
20         "s3SchemaVersion": "1.0",
21         "configurationId": "28baf0a1-170b-4530-8d0b-8bdbab865bf7",
22         "bucket": {
23           "name": "dev.refinery.eu-west-1.zenseiapp.com",
24           "ownerIdentity": {
25             "principalId": "A61RV6AN93VA"
26           },
27           "arn": "arn:aws:s3:::dev.refinery.eu-west-1.zenseiapp.com"
28         },
29         "object": {
30           "key": "refinery/6359304/2020-09-21/airquality.json",
31           "size": 208,
32           "eTag": "075902cdc66242cc9628286156701303",
33           "sequencer": "005ACB608036801D97"
34         }
35       }
36     }
37   ]
38 }
```

# Tipos de datos

## ► Estructurados

Tabulares (csv, excel,...)

Diccionarios (json, xml,...)

## ► No estructurados

Texto

Imágenes

Sonido

Vídeo

Cualitativas (No numéricos)

Cuantitativos (Numéricos)

| Nombre      | Color de pelo | Estatura     |
|-------------|---------------|--------------|
| Pedro       | Rubio         | 1,86m        |
| Ana         | Morena        | 1,76m        |
|             |               |              |
| Cualitativo | Cualitativo   | Cuantitativo |



# Tipos de datos

## ► Estructurados

Tabulares (csv, excel,...)

Diccionarios (json, xml,...)

## ► No estructurados

Texto

Imágenes

Sonido

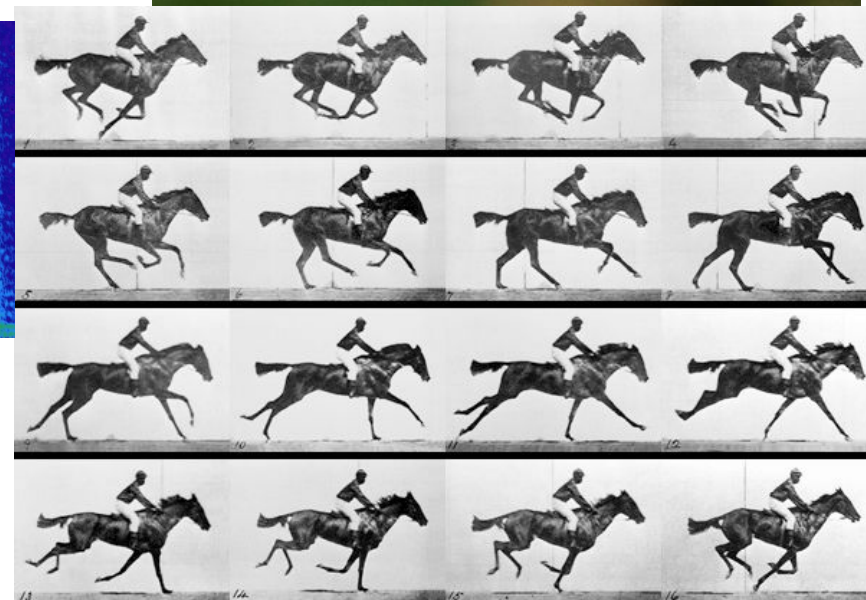
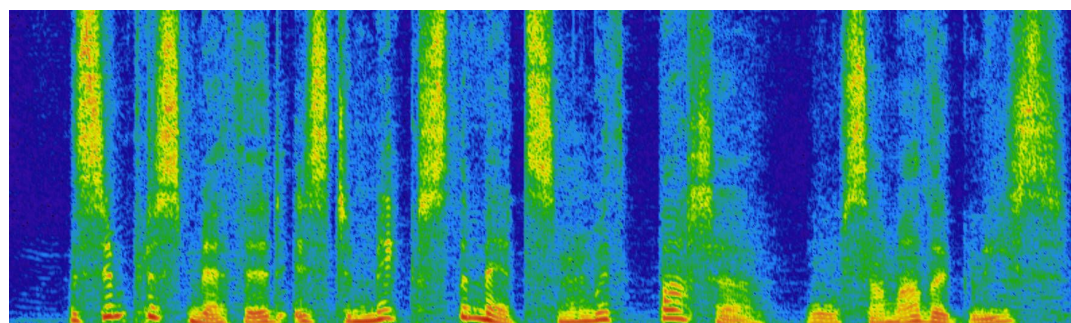
Vídeo



**David Perell** @david\_perell · 1m

Talented kids are more capable than we think:

- Leonardo da Vinci was an apprentice to Verrocchio at 14
- Andrew Carnegie finished schooling at 12. At 13, he became a office boy. By 16, he earned most of the family income.



# Tipos de variables

## ► Contenido

1. Texto
  - a. Cadenas de caracteres
  - b. Mensajes
2. Categóricas
  - a. Color, marca, producto, ranking
3. Numérica

## ► Origen

1. Dummy
2. Latente o sintética

## ► Función

1. Independientes
  - a. Precio
  - b. Marketing
  - c. Leads
2. Dependientes
  - a. Ventas

# ¿Cómo podemos trabajar con datos?

## ► No estructurados

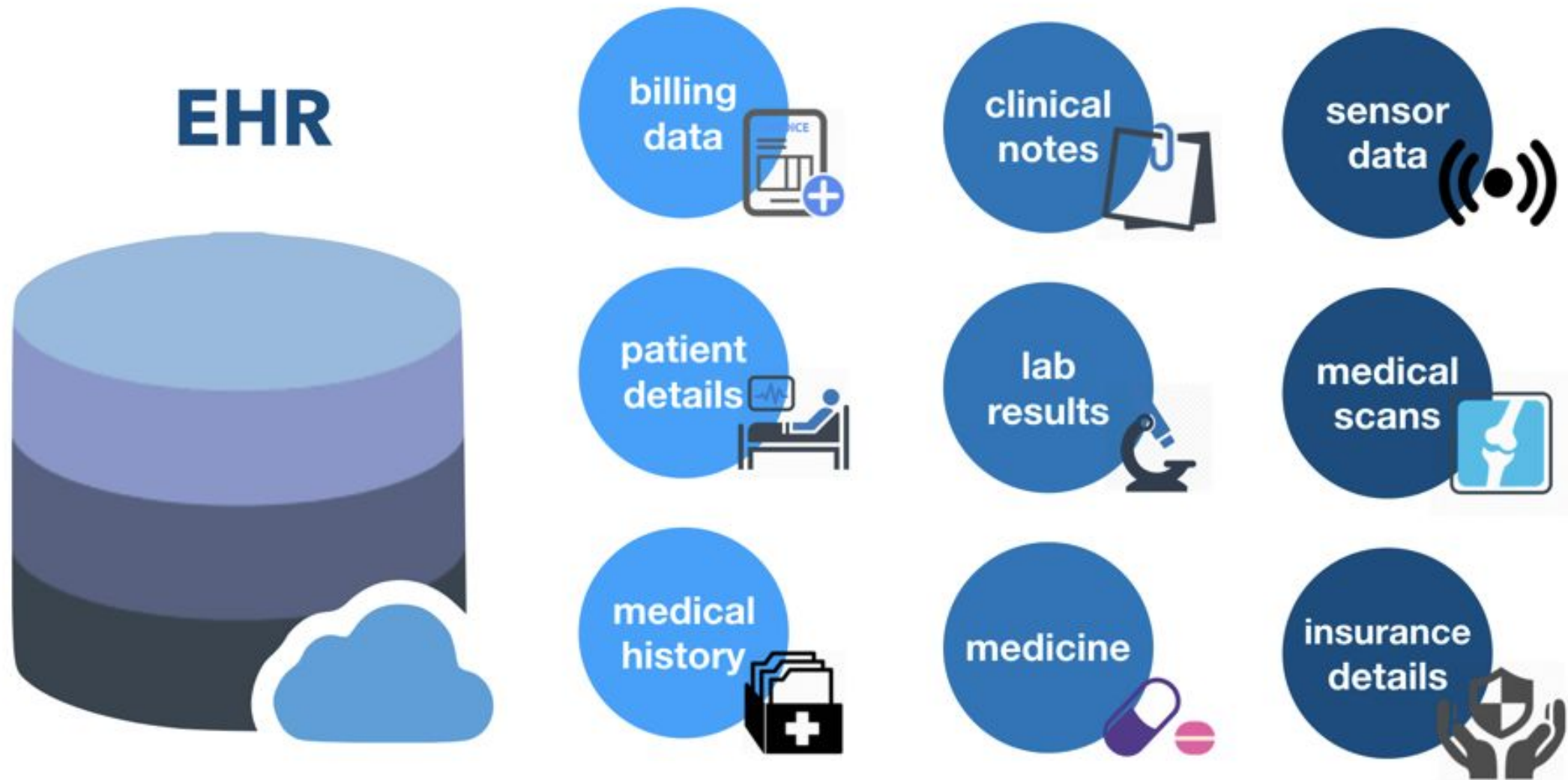
1. Técnicas de procesamiento del...
  - a. Lenguaje
  - b. Visión
  - c. Sonidos

## ► Estructurados

1. Técnicas estadísticas y matemáticas

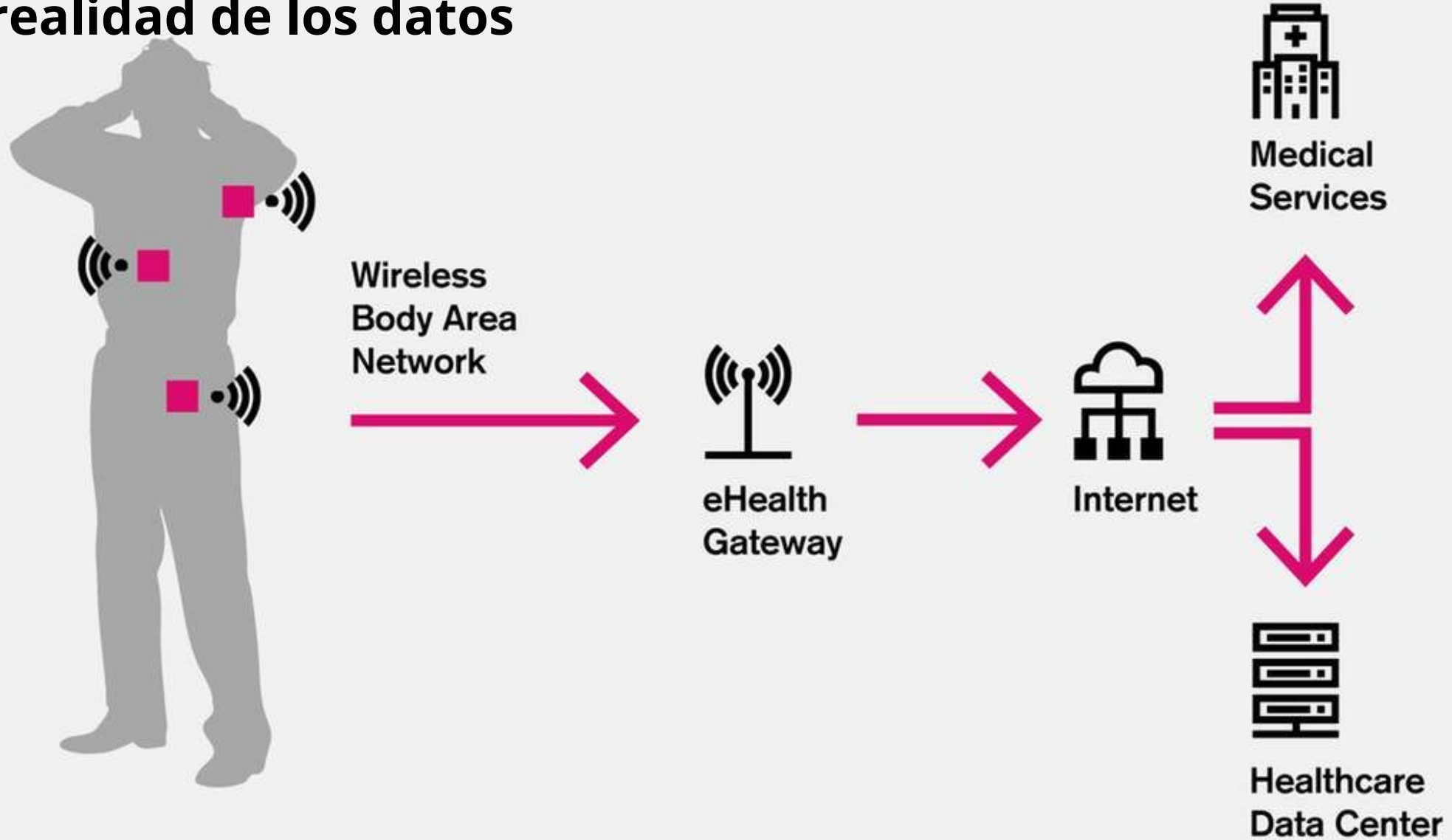
## ► Plataformas de procesamiento Big Data

# La realidad de los datos

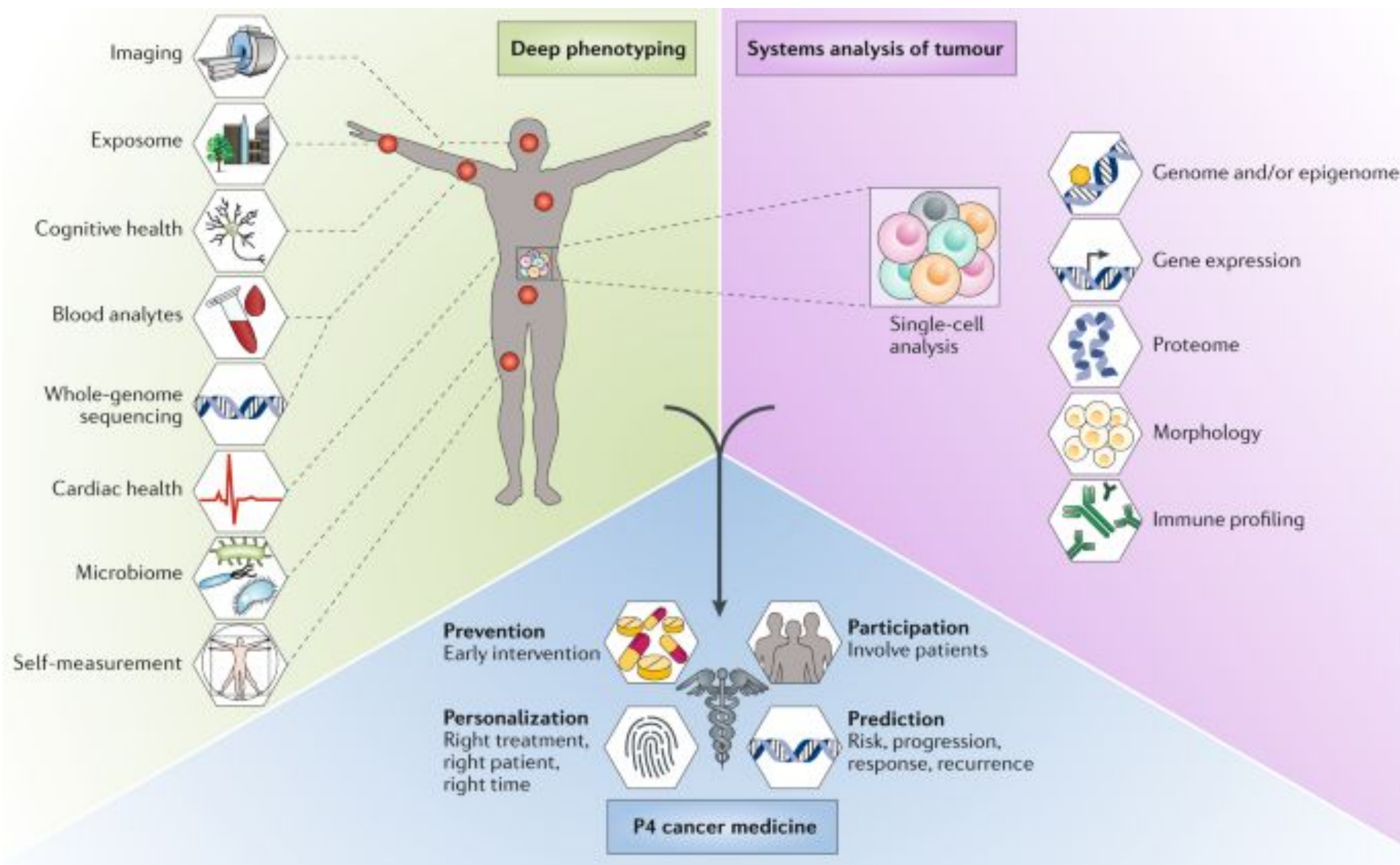




# La realidad de los datos



# La realidad de los datos



# Arquitecturas de procesamiento de datos

## **Tipos de tareas de una plataforma de Big data**

- ▶ Procesamiento batch
- ▶ Procesamiento en tiempo real
- ▶ Exploración interactiva de datos
- ▶ Analítica

## Qué es el famoso data lake

► *Es una base de datos distribuida donde todas las fuentes de datos de una organización están integradas y en su formato original para posteriores procesamientos . Permitiendo eliminar silos de información.*

## **Tipos de arquitecturas**

- ▶ Tradicional
- ▶ Lambda
- ▶ Kappa

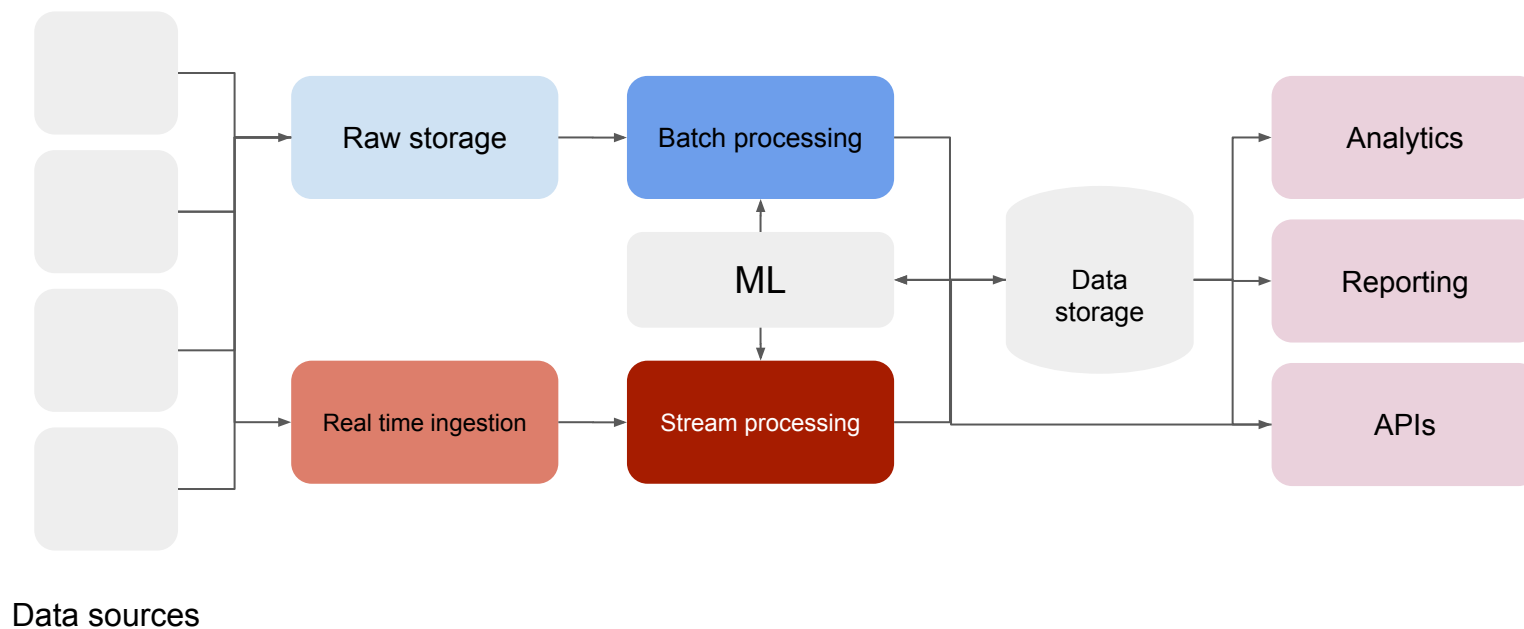


# Tipos de arquitecturas

## ► Tradicional

## ► Lambda

## ► Kappa

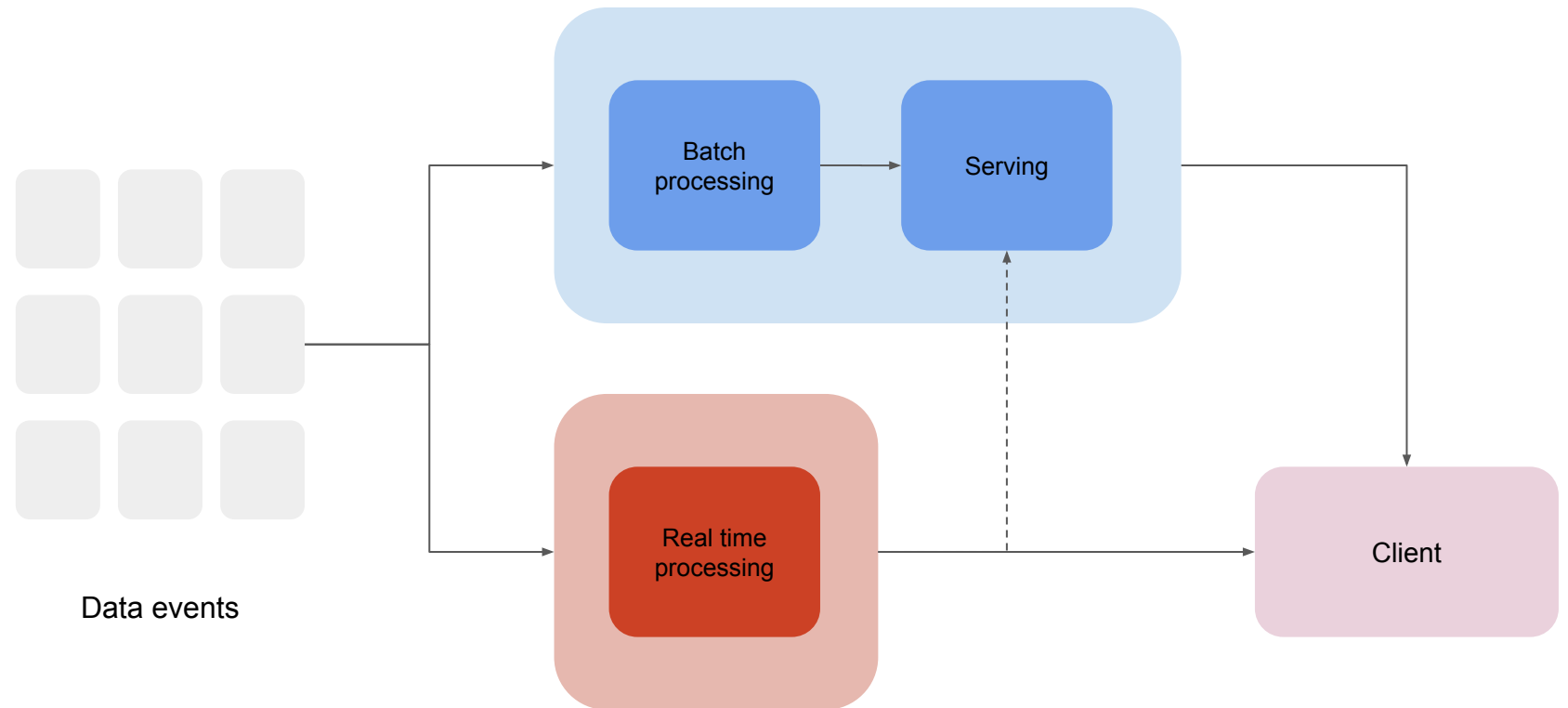


# Tipos de arquitecturas

► Tradicional

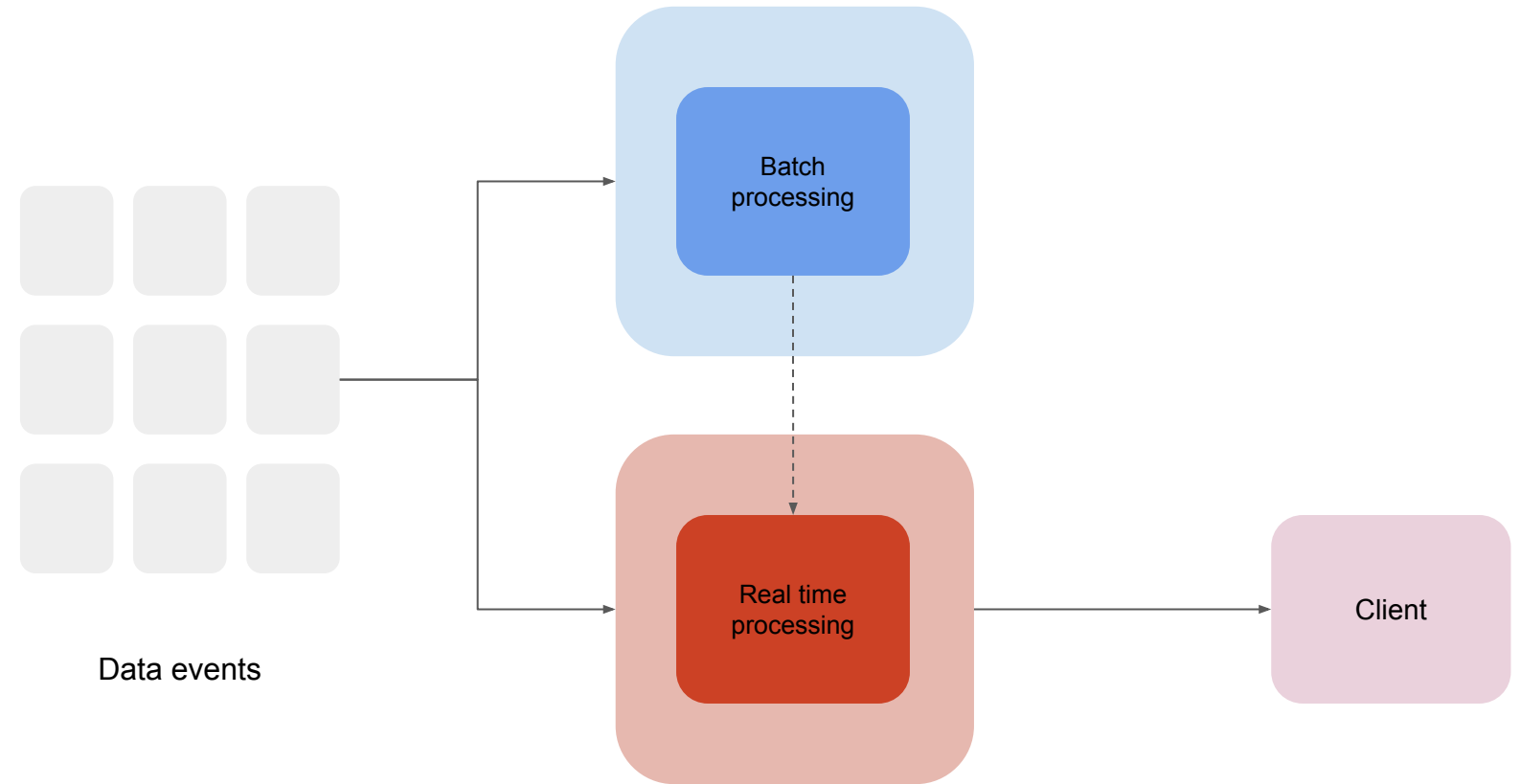
► Lambda

► Kappa



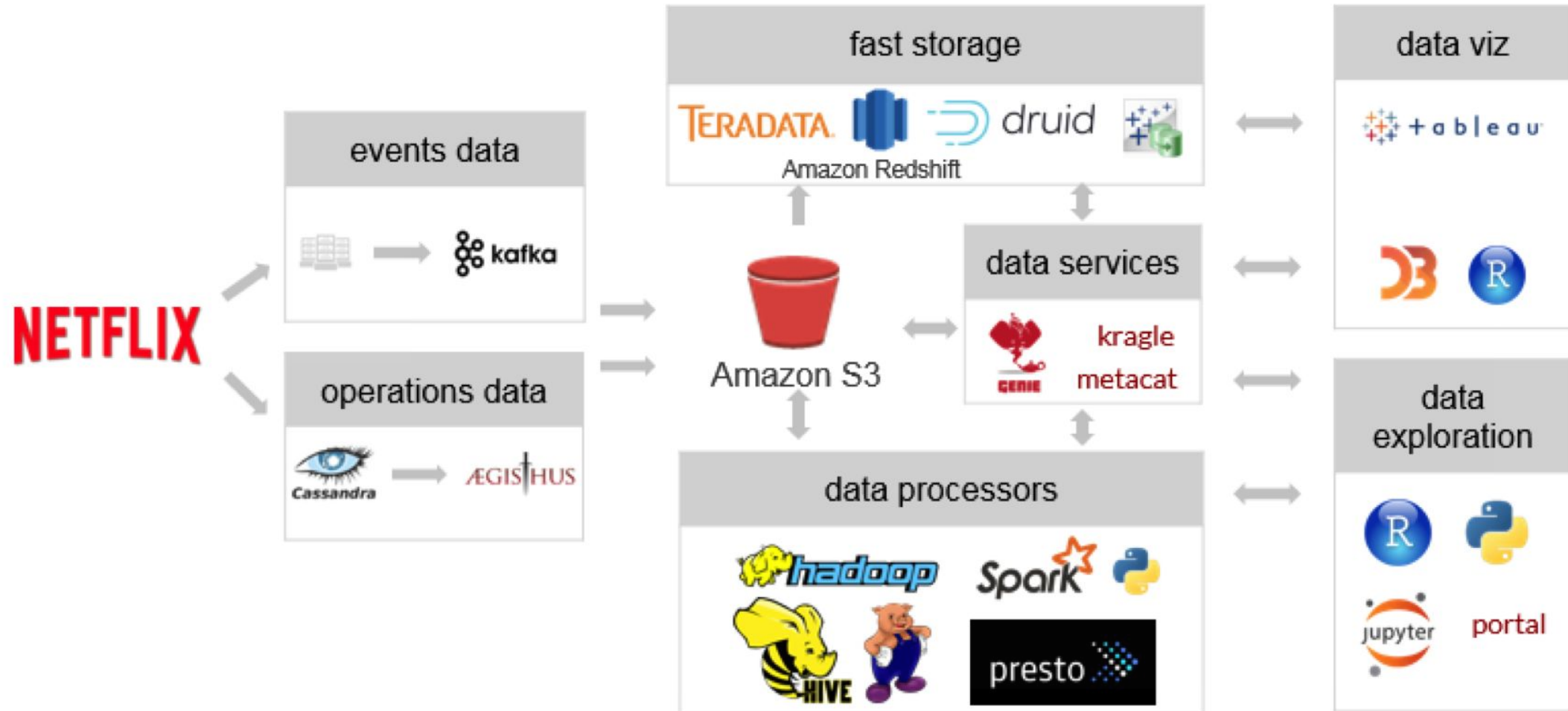
## Tipos de arquitecturas

- ▶ Tradicional
- ▶ Lambda
- ▶ **Kappa**



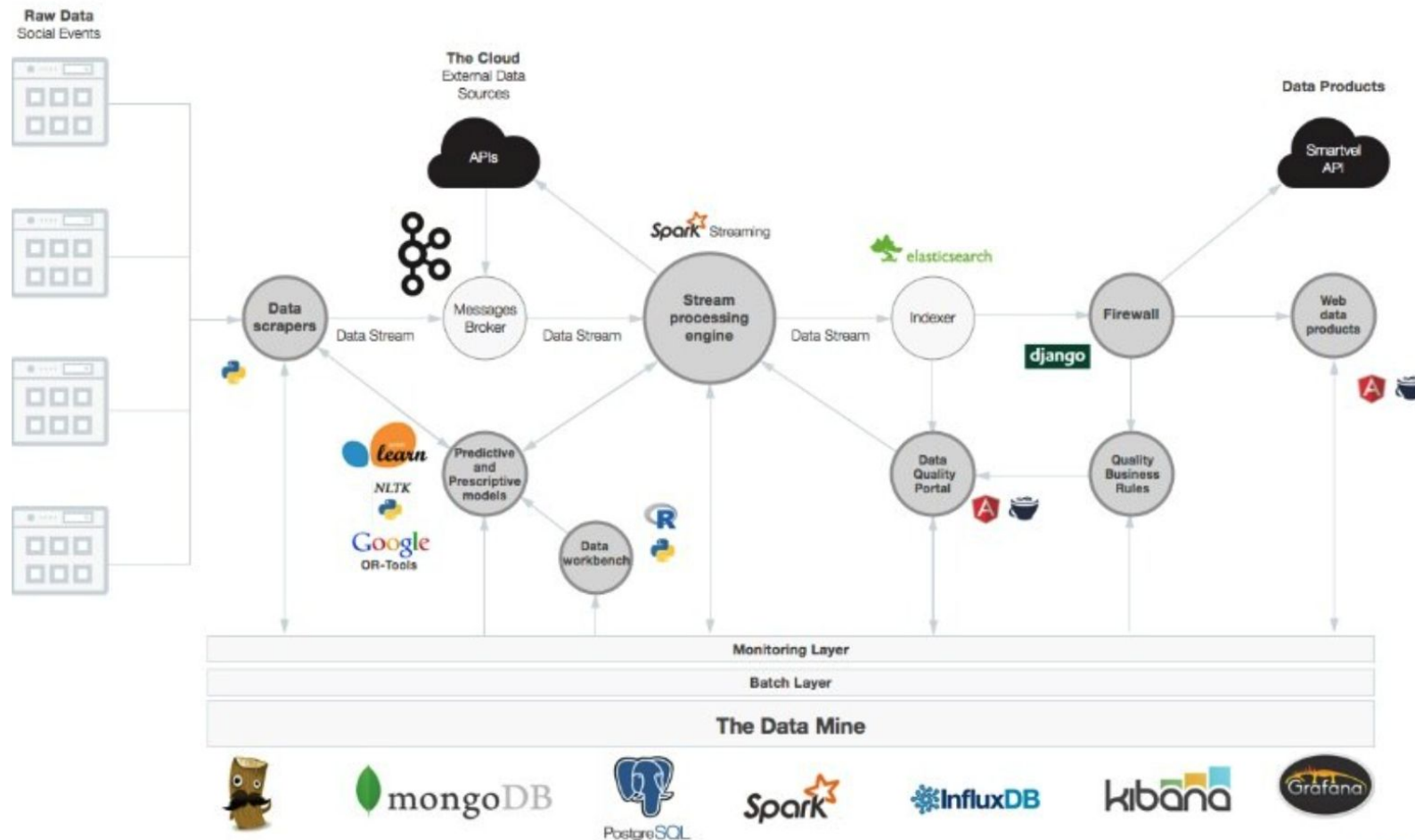
## Tipos de arquitecturas

### ► Arquitectura de microservicios de Netflix



# Tipos de arquitecturas

## ► Arquitectura de microservicios (Otro ejemplo)



## **Trabajo previo a la implantación**

- ▶ Casos de uso y de negocio
- ▶ Análisis de fuentes de datos

Problema - Qué problemas tiene el negocio.

Solución - Qué solución tiene impacto.

Tecnología - Cómo podemos implementarla.

Cuidado con los arcos de iglesia y las culturas de técnicos tiranos.

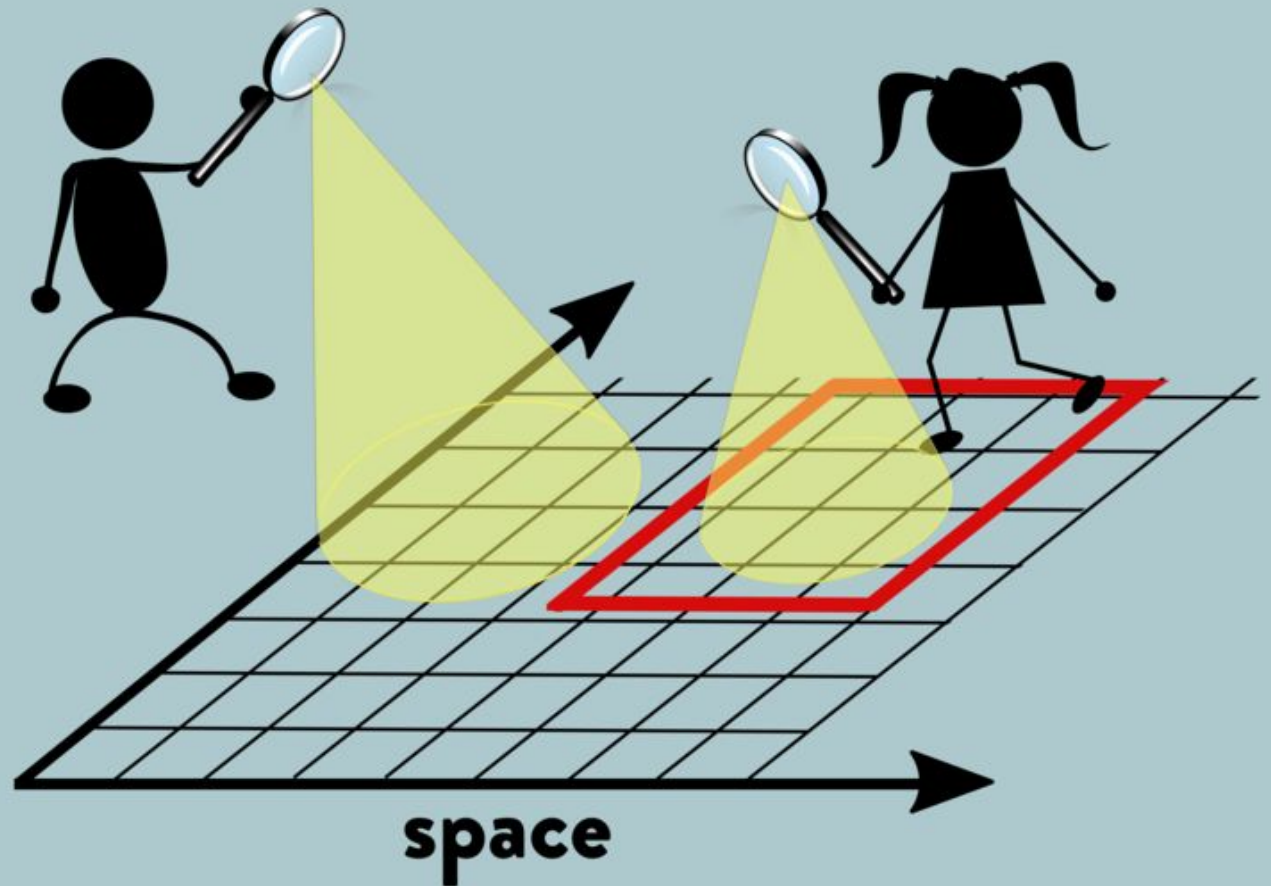
# Datos y modelos matemáticos



Datos

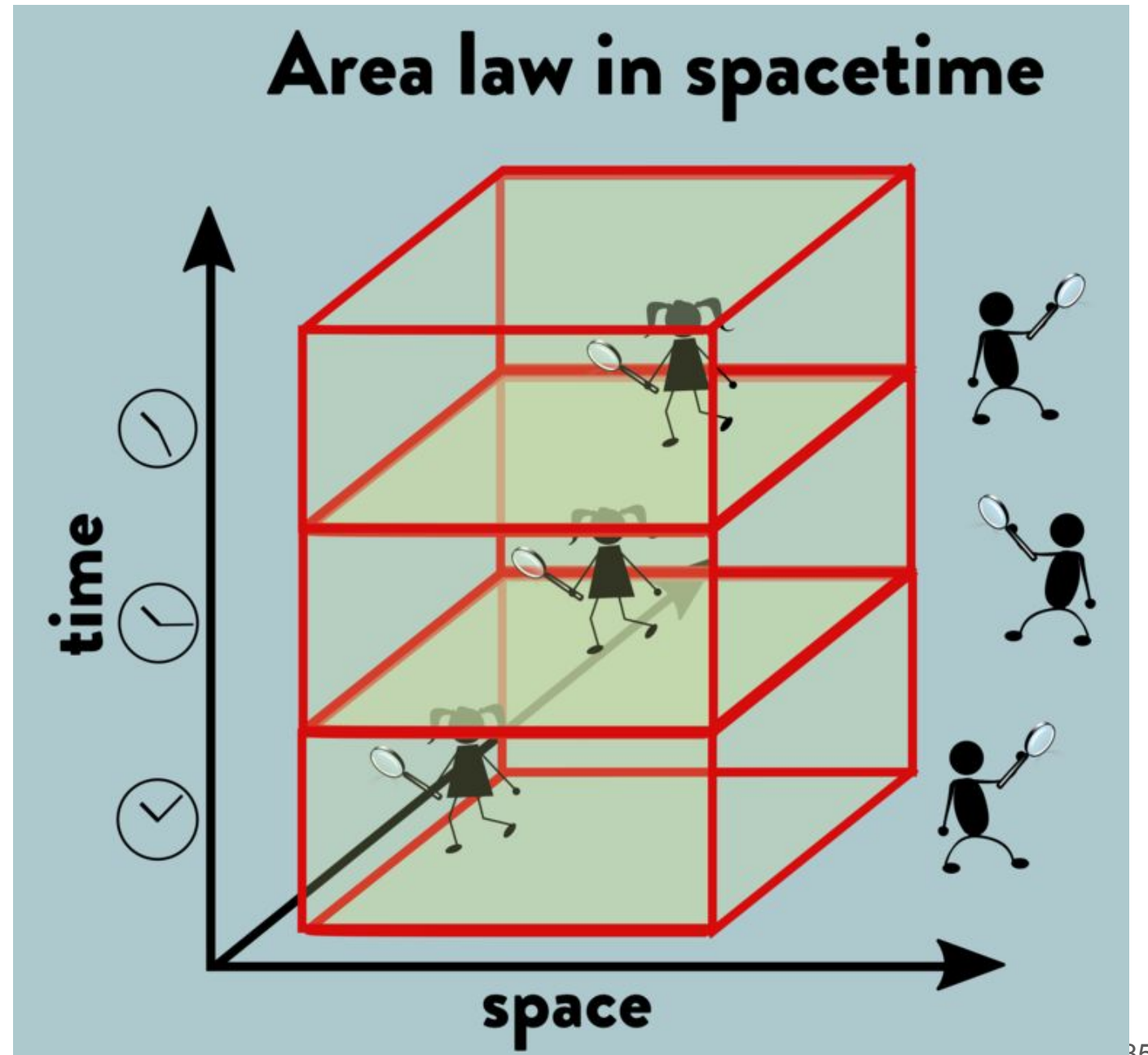
Espacio

## Area law in space



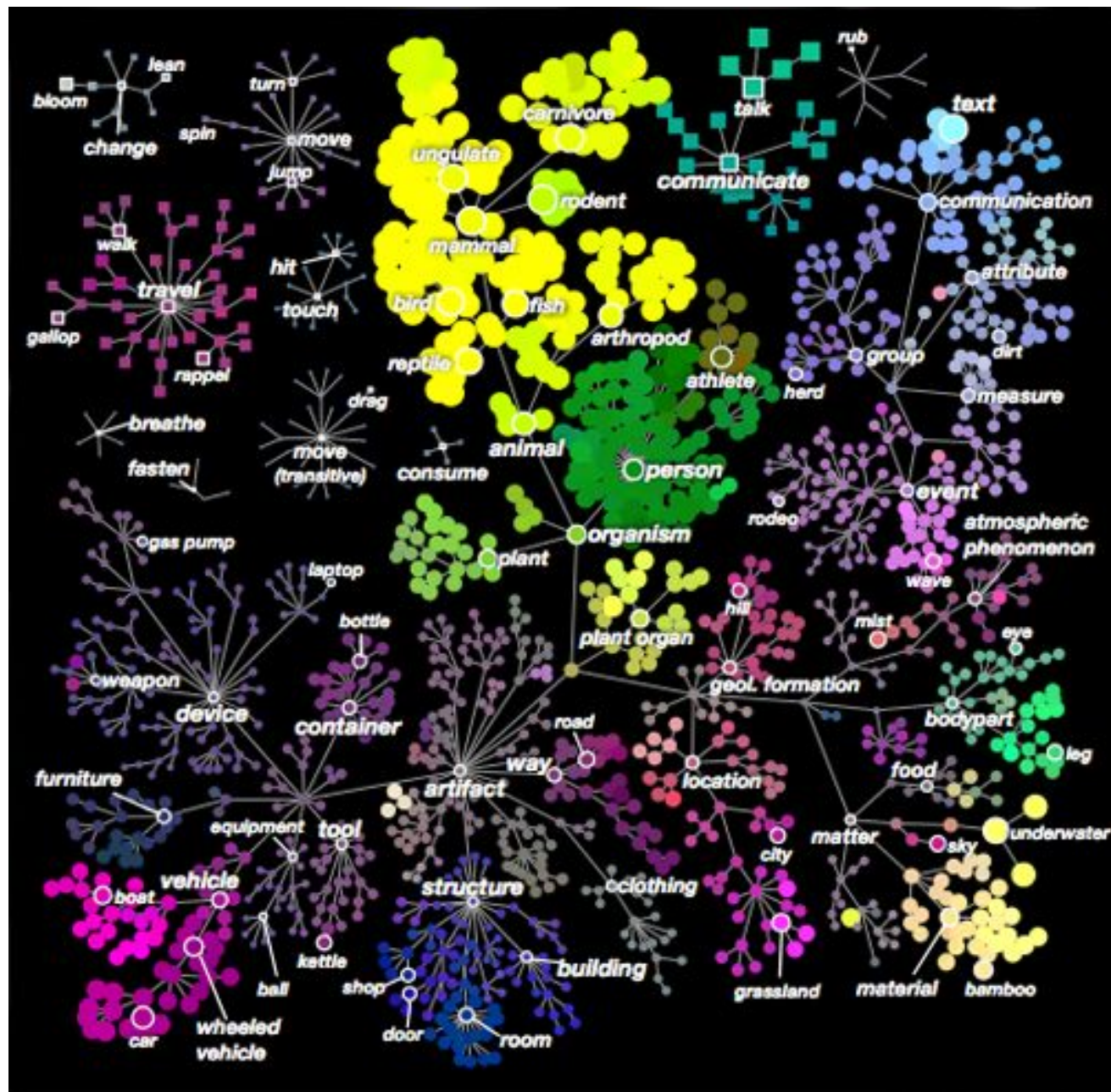
Datos

Espacio-tiempo



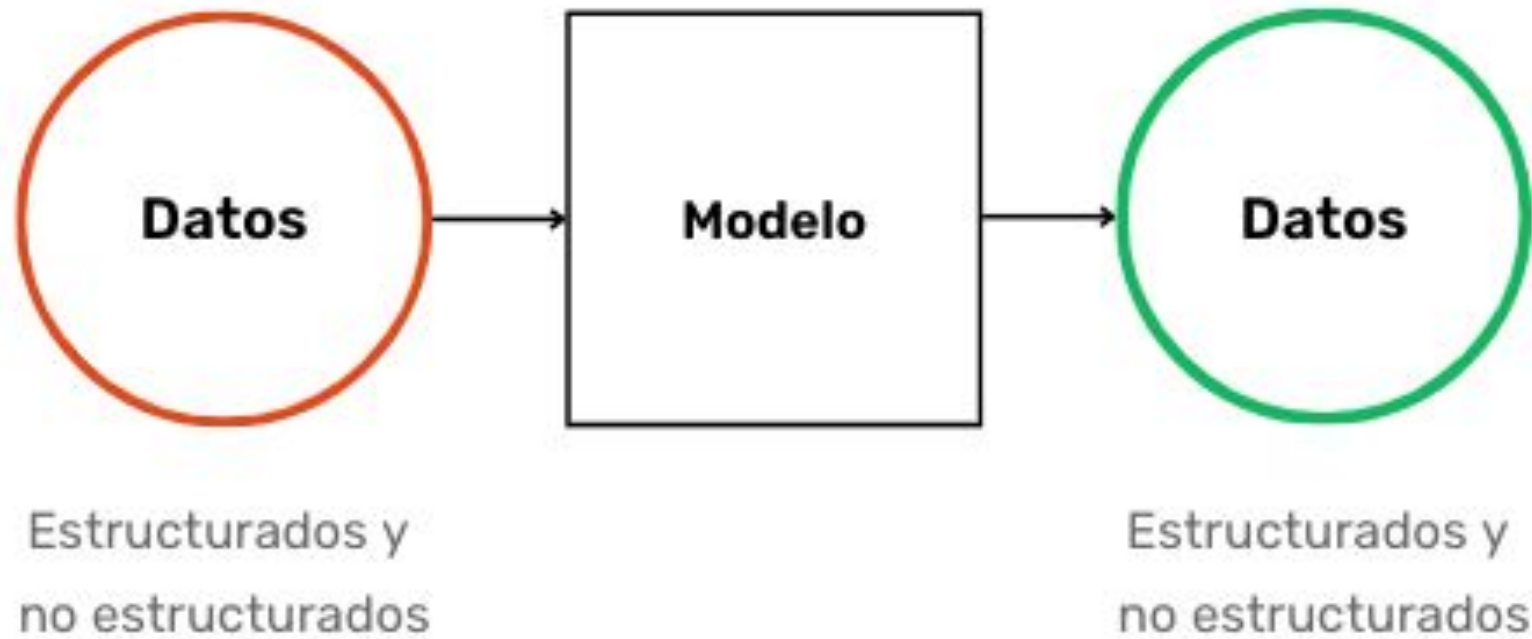
# Datos

Representación simbólica,  
fonológica, semántica y  
gramatical = Lenguaje



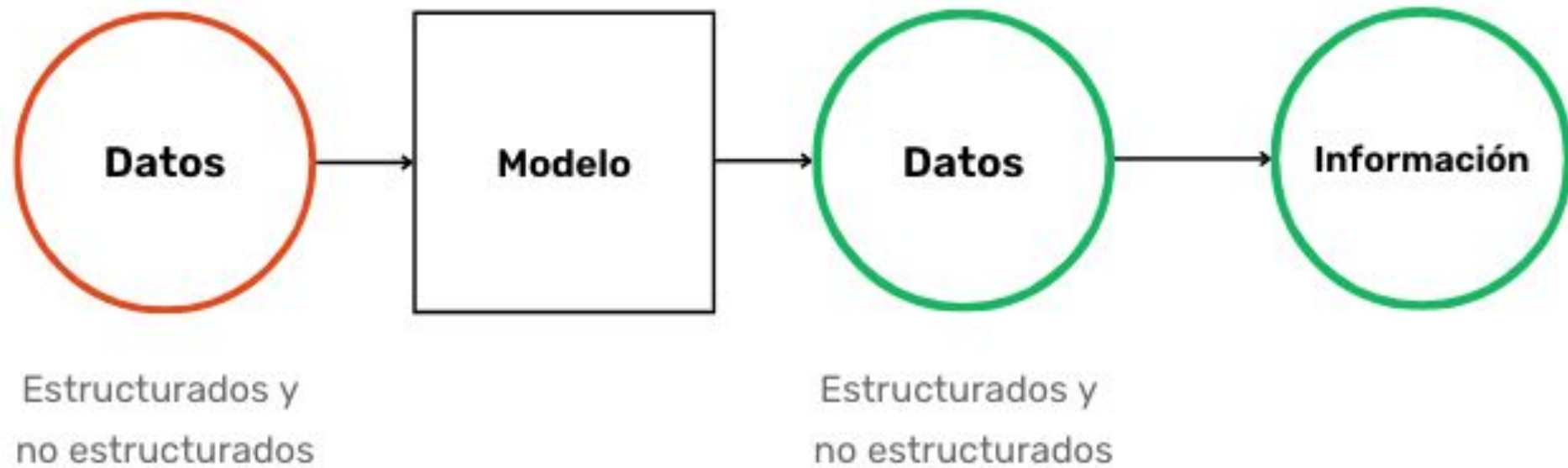
# Modelos

Datos + Modelo = Datos



# Modelos

Datos + Modelo = Datos

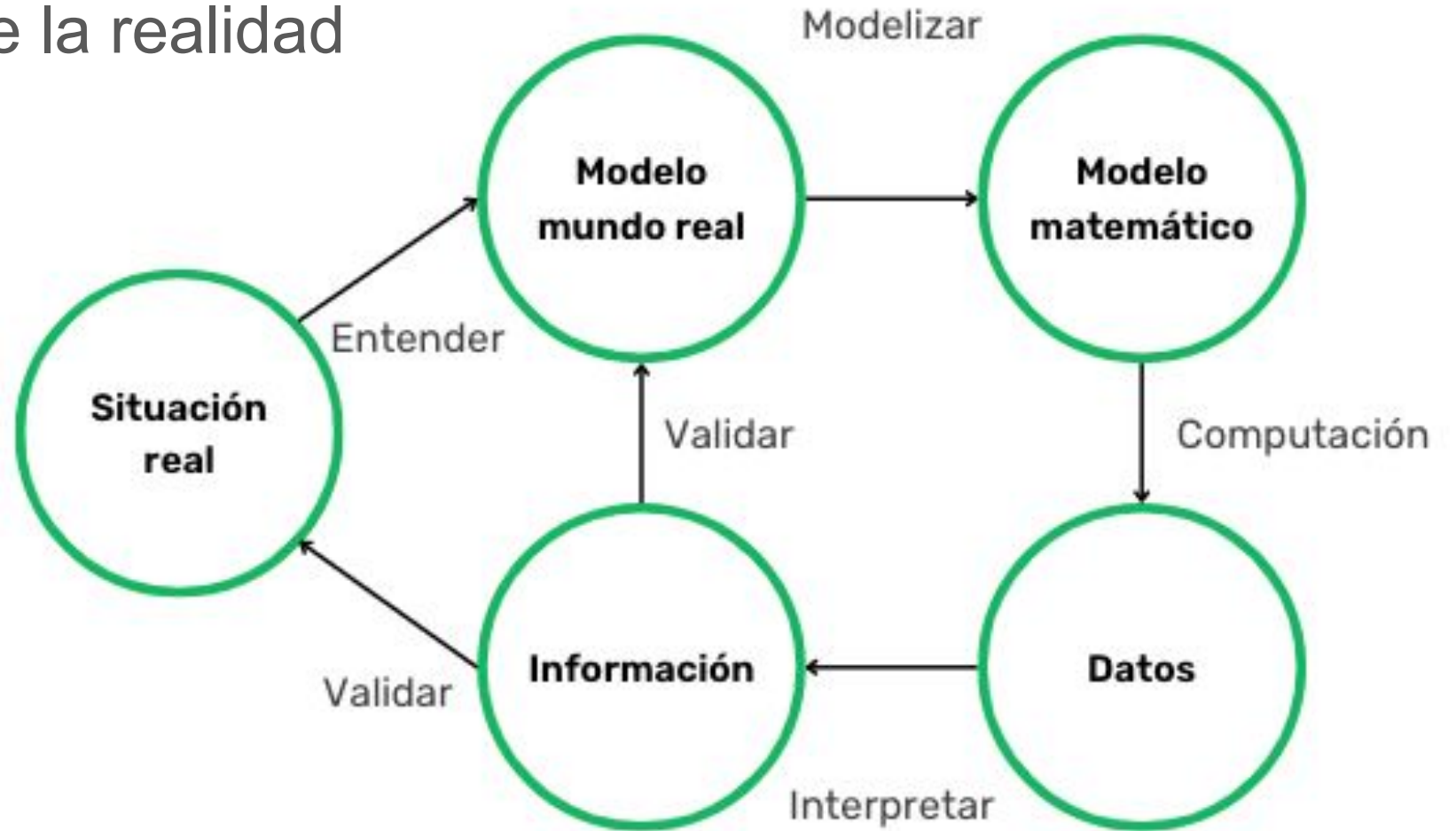




# Modelos

¿Qué es un modelo?

Una representación de la realidad



# Introducción al aprendizaje automático

## Cognición

*"Proceso mental que incluye el pensamiento, el aprendizaje, la memorización, la conciencia del entorno y el sentido de realidad o juicio."*



# Cognición

Nos creemos dioses y queremos crear inteligencias artificiales que sean capaces de ...

- ▶ Percibir = Sensores = Datos
- ▶ Aprender = Representación de la información de los datos
- ▶ Memorizar = Recordar en el corto y largo plazo
- ▶ Razonar = Lógica, asociación, inducción, deducción
- ▶ Sentir emociones
- ▶ Ser conscientes

# Aprendizaje

El aprendizaje es el proceso de adquirir nuevos conocimientos, comportamientos, habilidades, valores, actitudes y preferencias.

- ▶ Supervisado
- ▶ Semi supervisado
- ▶ No supervisado
- ▶ Refuerzo

# Aprendizaje

## Supervised Learning

**Data:**  $(x, y)$

$x$  is data,  $y$  is label

**Goal:** Learn function to map  
 $x \rightarrow y$

**Examples:** Classification, regression, object detection, semantic segmentation, etc.

## Unsupervised Learning

**Data:**  $x$

$x$  is data, no labels!

**Goal:** Learn some *hidden* or *underlying structure* of the data

**Examples:** Clustering, feature or dimensionality reduction, etc.

## **Modelos matemáticos para resolver problemas**

- ▶ Regresión = Variable numérica
- ▶ Clasificación = Variable categórica
- ▶ Clusterización = Similitud
- ▶ Series temporales = Variables temporales
- ▶ Análisis de redes = Interdependencia
- ▶ Reducción de la dimensionalidad = Condensación de la información
- ▶ Optimización = Resolución óptima o factible de problemas
- ▶ Simulaciones = Solo tenemos datos de distribuciones estadísticas
- ▶ Agentes = Toma de acciones en mundos virtuales
- ▶ Recomendadores = Busca elementos similares

## TIPOS DE APRENDIZAJE

### Tareas



# Introducción al aprendizaje profundo



# El despertar de la inteligencia artificial

## 'Deep Voice' Software Can Clone Anyone's Voice With Just 3.7 Seconds of Audio

Using snippets of voices, Baidu's 'Deep Voice' can generate new speech, accents, and tones.



Let There Be Sight: How Deep Learning Is Helping the Blind 'See'



## Technology outpacing security measures

Facial Recognition | Features and Interviews

## AI beats docs in cancer spotting

A new study provides a fresh example of machine learning as an important diagnostic tool. Paul Biegler reports.



## AI Can Help In Predicting Cryptocurrency Value



## 'Creative' AlphaZero leads way for chess computers and, maybe, science

Former chess world champion Garry Kasparov likes what he sees of computer that could be used to find cures for diseases



## How an A.I. 'Cat-and-Mouse Game' Generates Believable Fake Photos

By GIDE MEYER and KEITH COLLINS JAN. 3, 2018



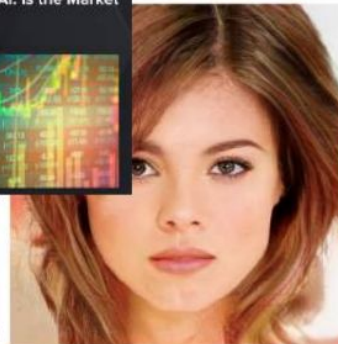
## Stock Predictions Based On AI: Is the Market Truly Predictable?



Complex of bacteria-infecting viral proteins modeled at CASP-13. The complex consists of 10 subunits that were modeled individually. PROTEIN DATA BANK

## Google's DeepMind acers protein folding

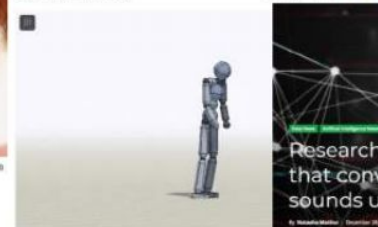
By Robert F. Service | Dec. 6, 2018, 12:05 PM



To create the final image in this set, the system generated 10 million revisions over 18 days.

## After Millions of Trials, These Simulated Humans Learned to Do Perfect Backflips and Cartwheels

George Dvorsky 4/10/18 11:34am • Filed to AI



## Neural networks everywhere

New chip reduces neural networks' power consumption by up to 95 percent, making them practical for battery-powered devices.

Wed, 05/16/2018 - 8:00am 1 Comment by Kenny Walter - Digital Reporter - @RandIMagazine



## Researchers introduce a deep learning method that converts mono audio recordings into 3D sounds using video scenes

By Marissa Matthei | December 26, 2018 10:22 am

## AI faces show how far AI image generation has come in just four years

Images on the right aren't real; they're the product of machine learning



## Automation And Algorithms: De-Risking Manufacturing With Artificial Intelligence

Sarah Goehrke Contributor Manufacturing Focus on the industrialization of additive manufacturing.

TWEET THIS

The two key applications of AI in manufacturing are pricing and manufacturability feedback

## ARTIFICIAL INTELLIGENCE

Any technique that enables computers to mimic human behavior



## MACHINE LEARNING

Ability to learn without explicitly being programmed



## DEEP LEARNING

Extract patterns from data using neural networks



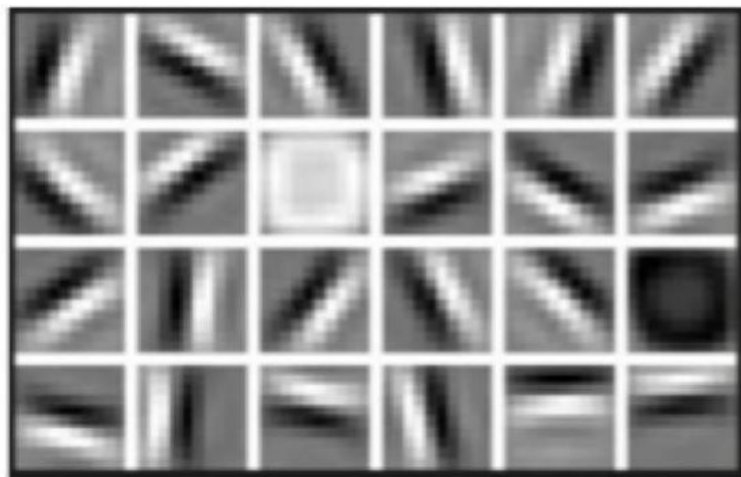


# Por qué el aprendizaje profundo

Hand engineered features are time consuming, brittle and not scalable in practice

Can we learn the **underlying features** directly from data?

Low Level Features



Lines & Edges

Mid Level Features



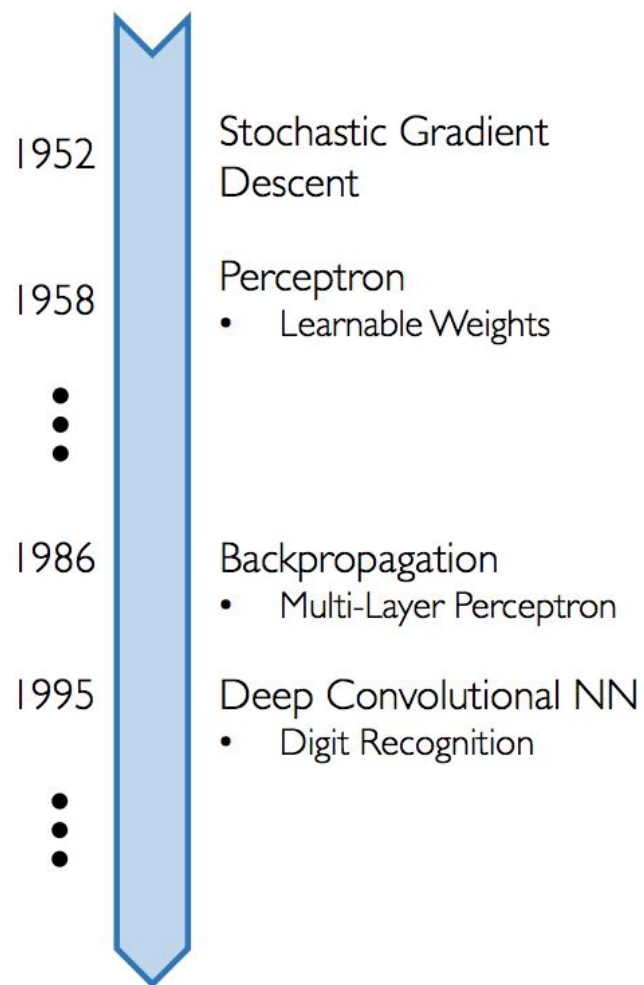
Eyes & Nose & Ears

High Level Features



Facial Structure

# Por qué ahora



Neural Networks date back decades, so why the resurgence?

## 1. Big Data

- Larger Datasets
- Easier Collection & Storage

IMAGENET



## 2. Hardware

- Graphics Processing Units (GPUs)
- Massively Parallelizable



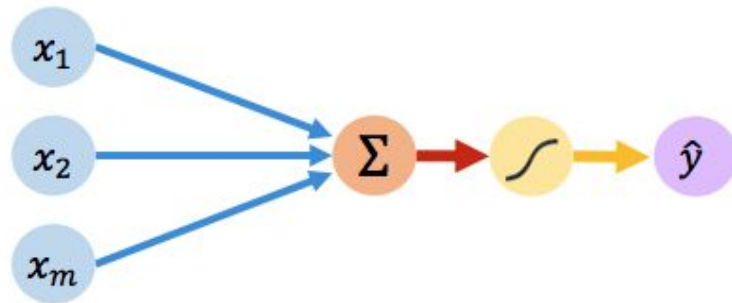
## 3. Software

- Improved Techniques
- New Models
- Toolboxes



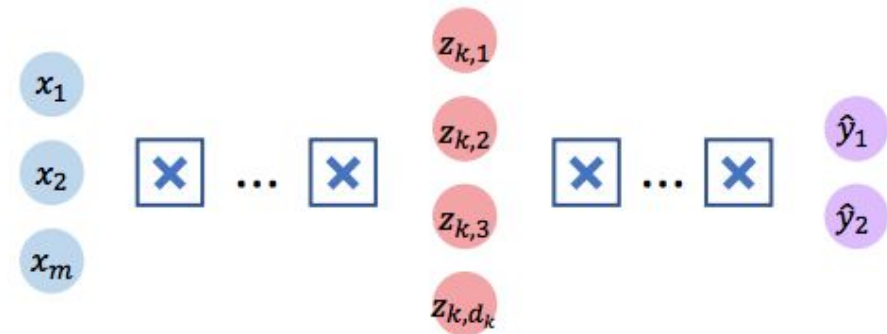
## The Perceptron

- Structural building blocks
- Nonlinear activation functions



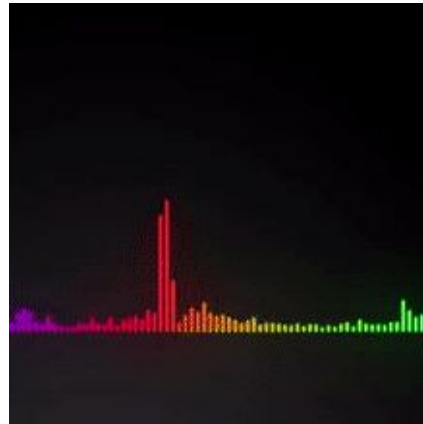
## Neural Networks

- Stacking Perceptrons to form neural networks
- Optimization through backpropagation





**Vision**



**Secuencias**



**Generativa**



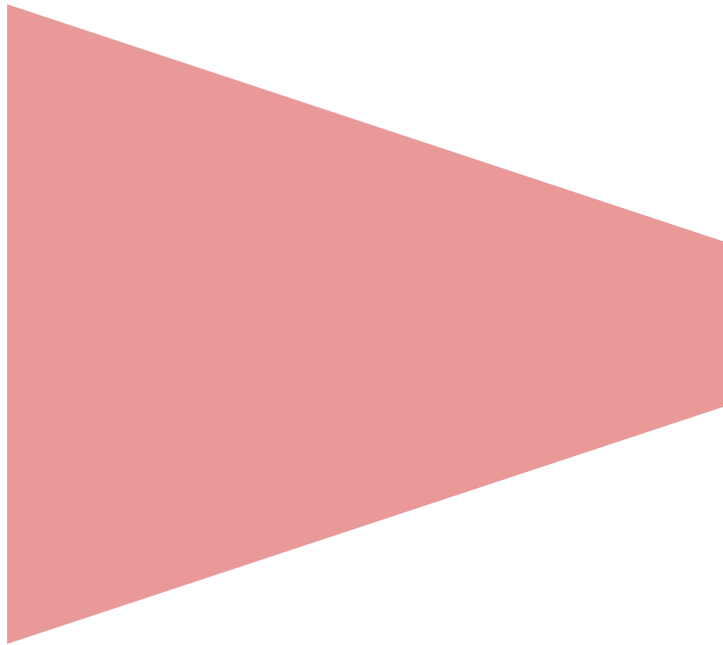
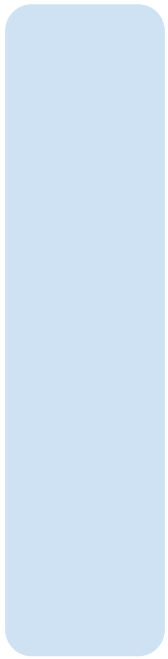
**Refuerzo**

# Convolución

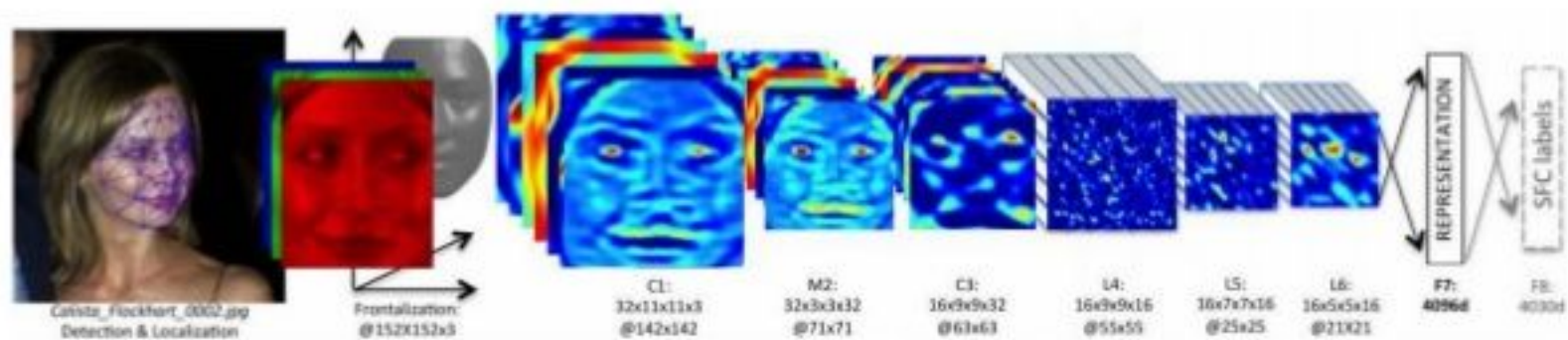
Entrada  
X

Aprende  
patrones

Predicción  
y

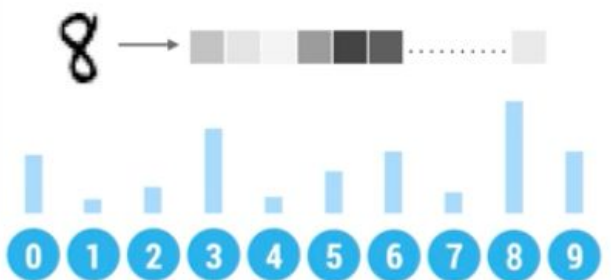
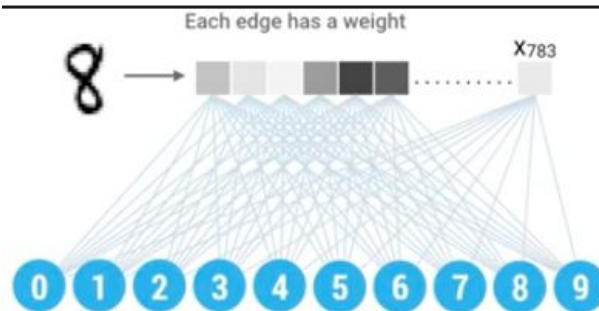
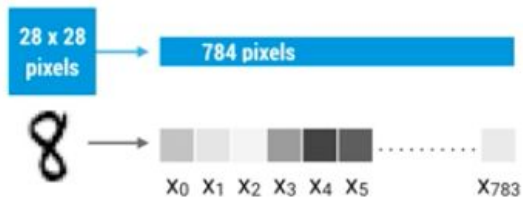
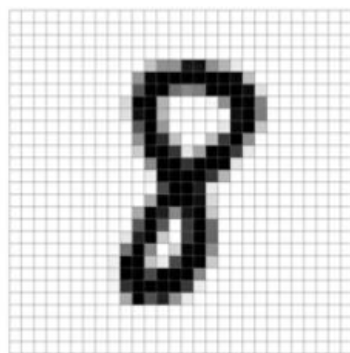


## DeepFace Architecture



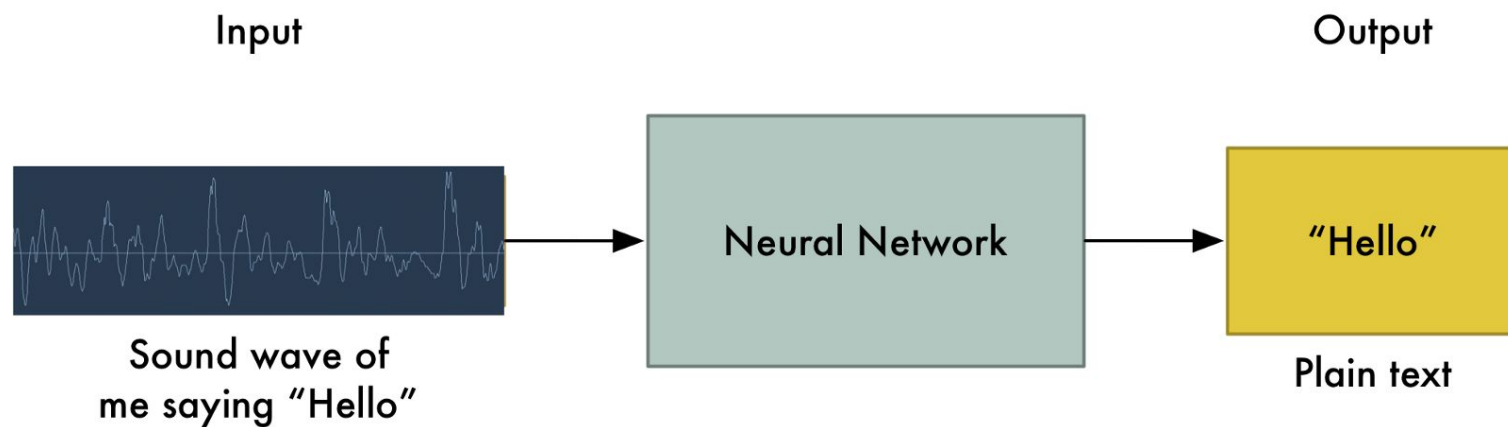
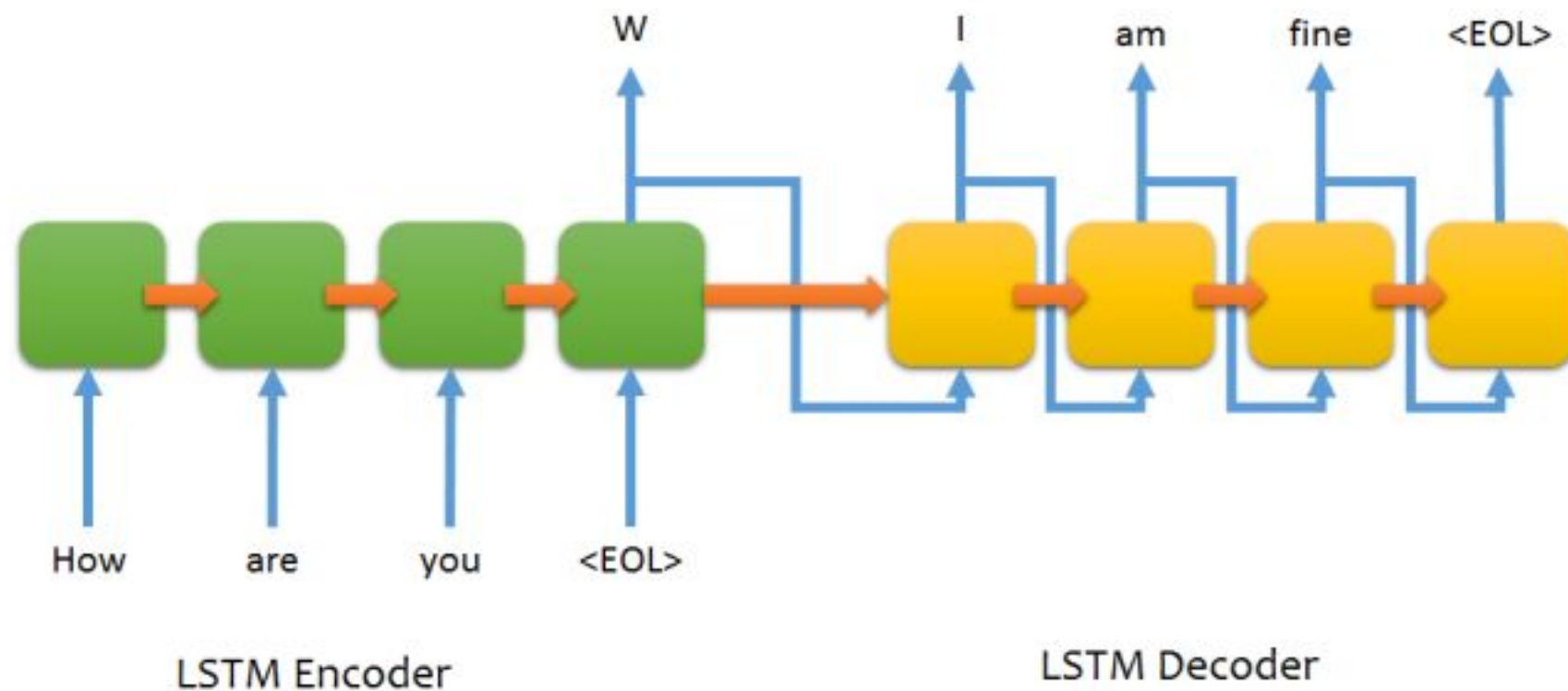
Yaniv Taigman, etc (Facebook) . [DeepFace: Closing the Gap to Human-Level Performance in Face Verification](#), CVPR 2014





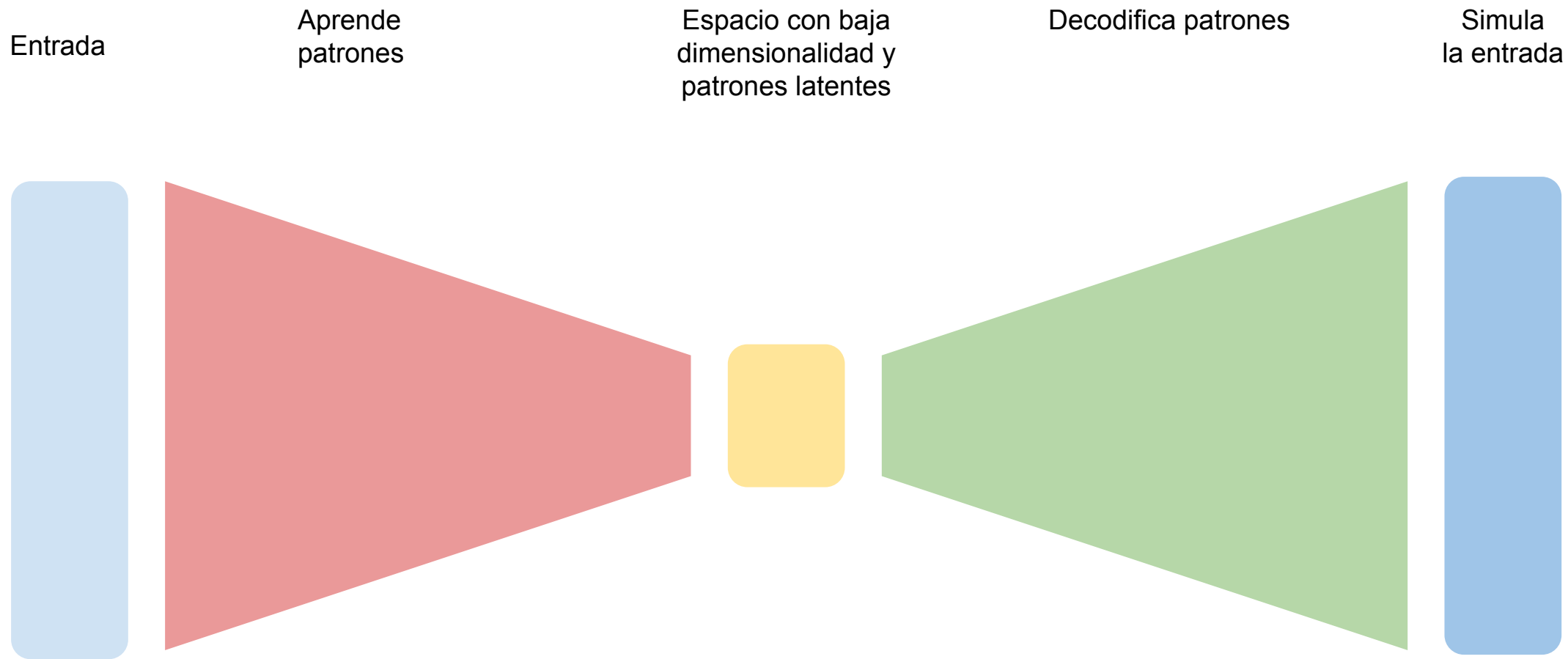
$$\text{evidence } y_i = \sum_j W_{i,j} x_j$$



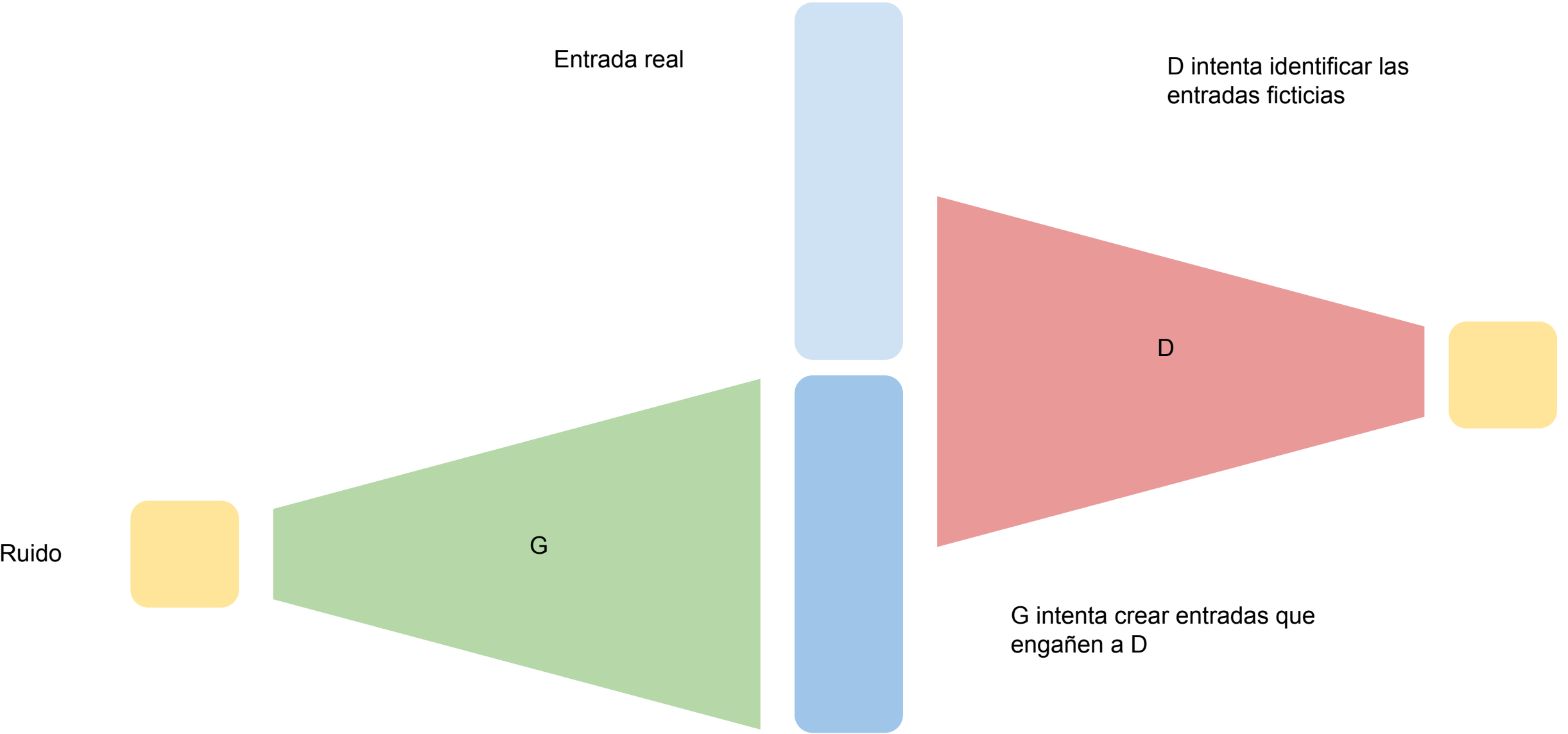


# Autoencoders

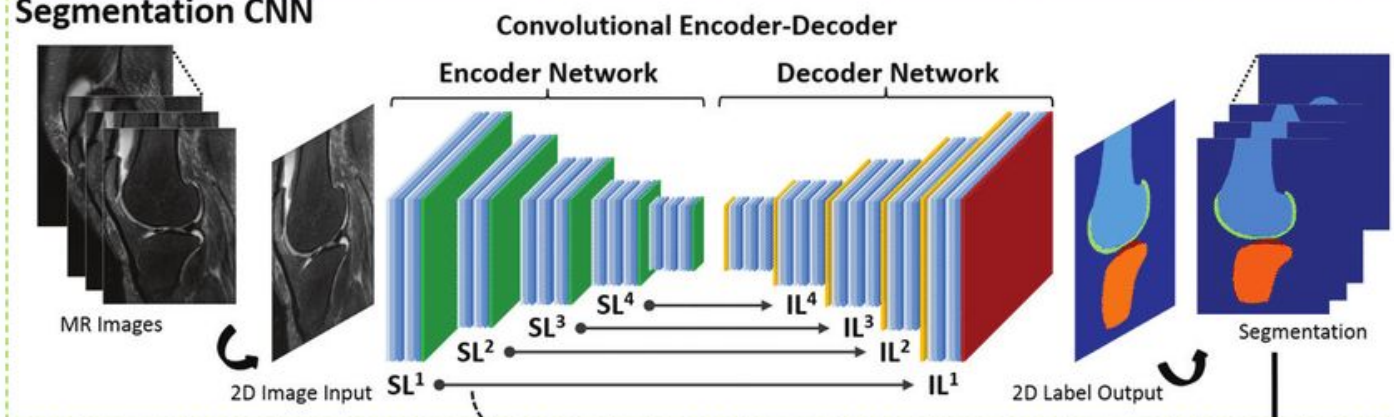
Permite reconstruir datos originales, es como una comprensión de los datos.



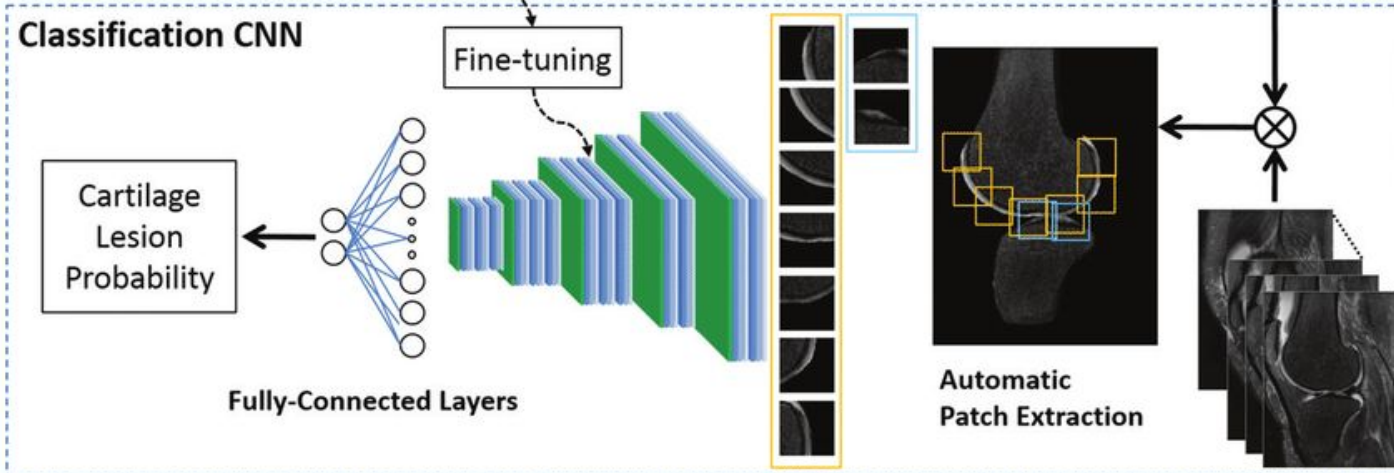
# Generative Adversarial Networks



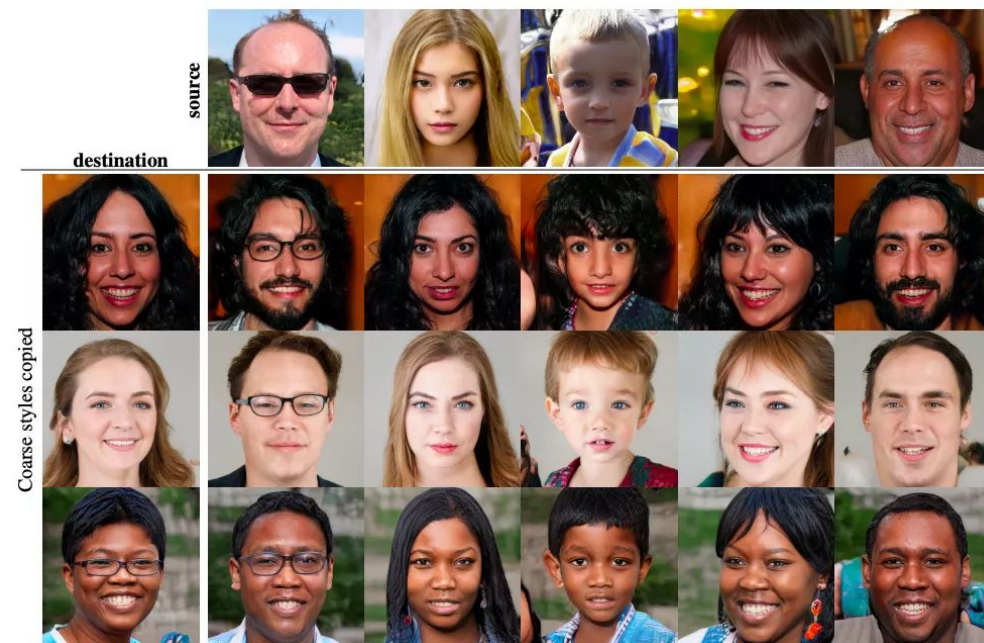
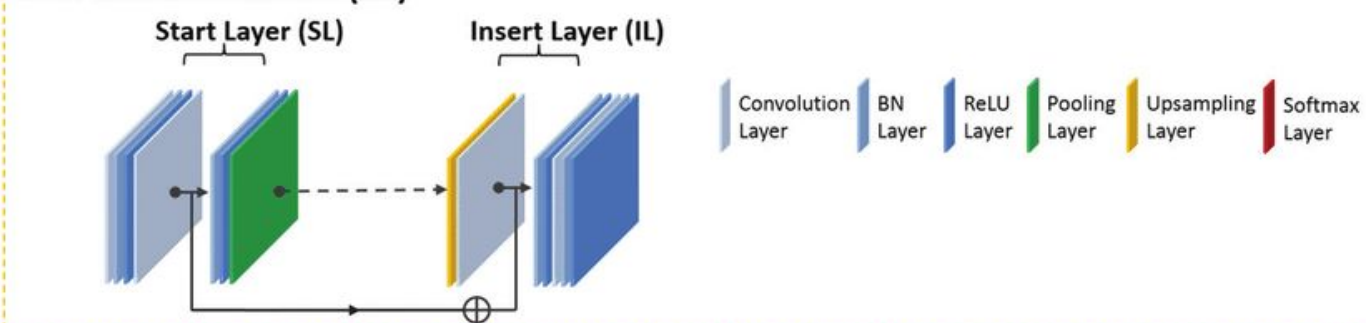
## Segmentation CNN



## Classification CNN



## Shortcut Connection (SC)





Monet ↔ Photos



Monet → photo



photo → Monet

Zebras ↔ Horses



zebra → horse

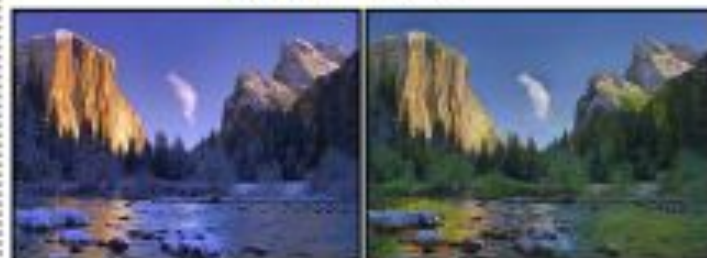


horse → zebra

Summer ↔ Winter



summer → winter



winter → summer



Photograph



Monet



Van Gogh



Cezanne

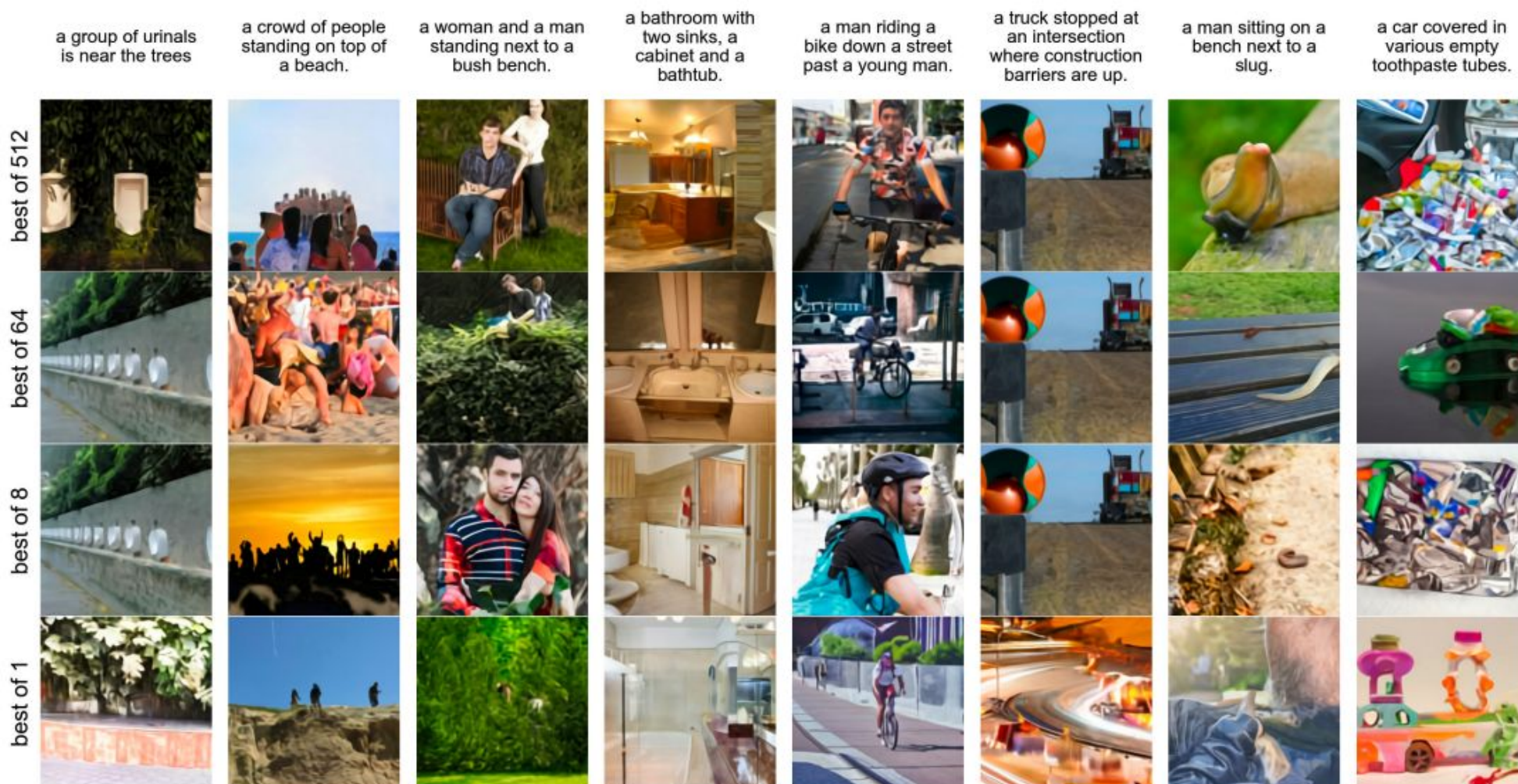


Ukiyo-e



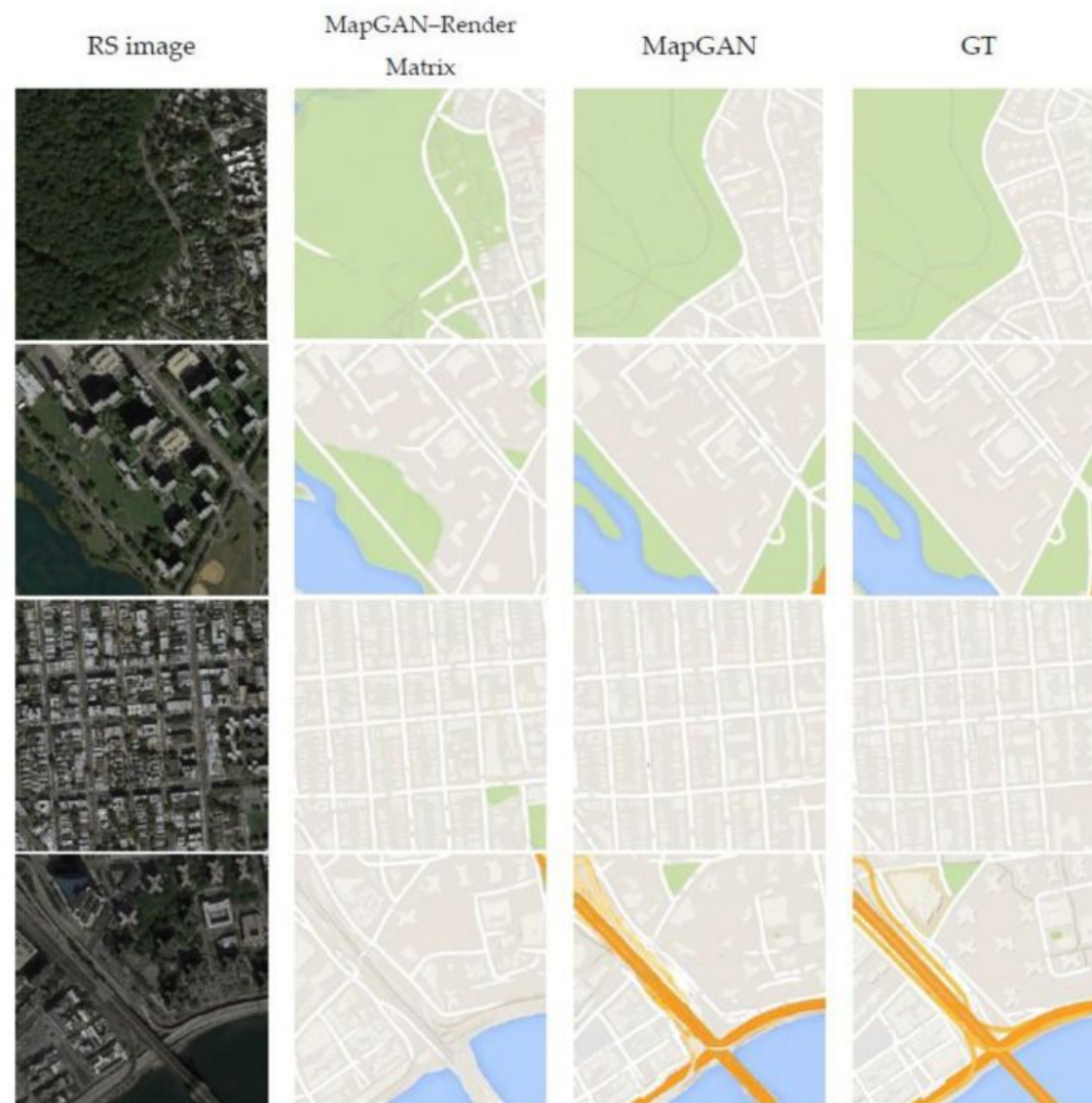
# Text to image

Unpaired Image-to-Image Translation  
using Cycle-Consistent Adversarial Network



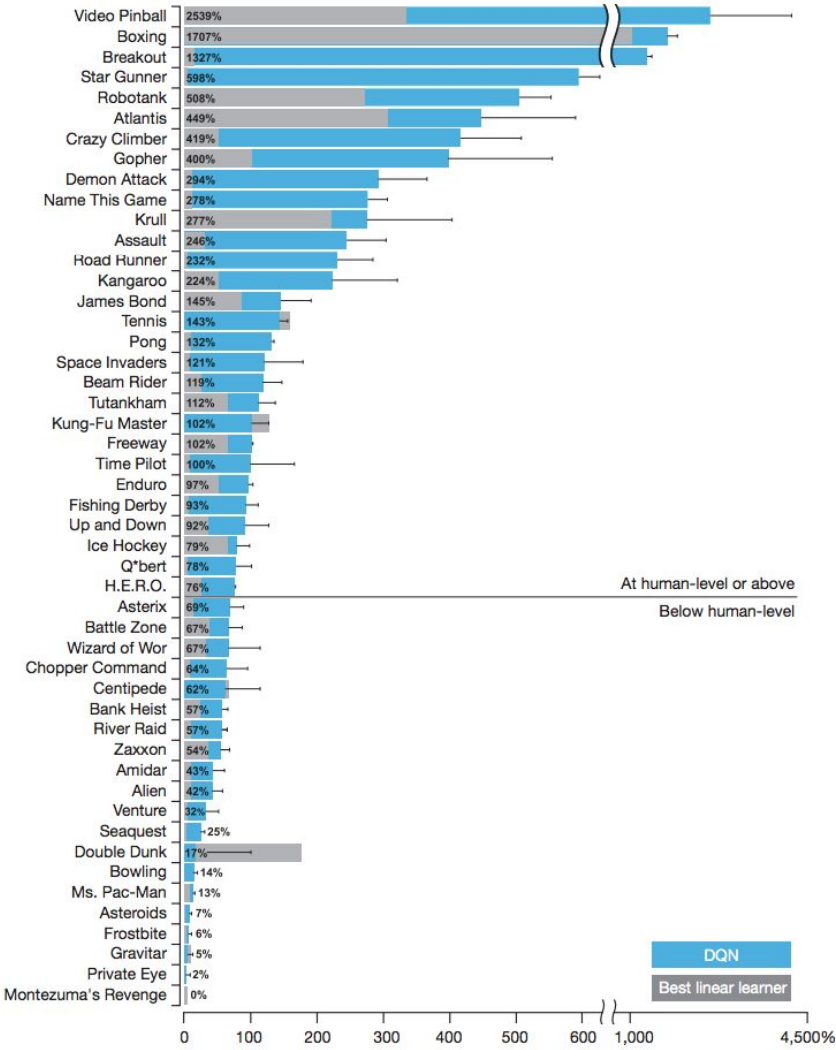
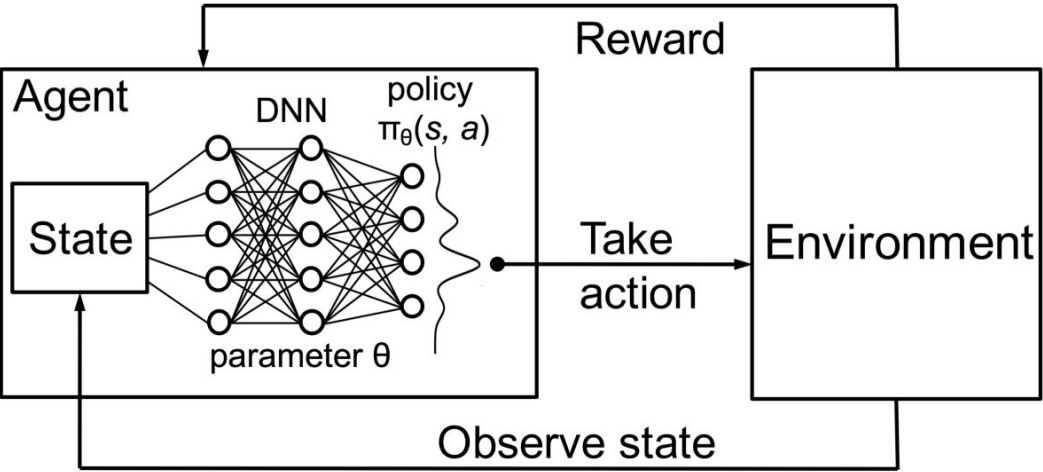
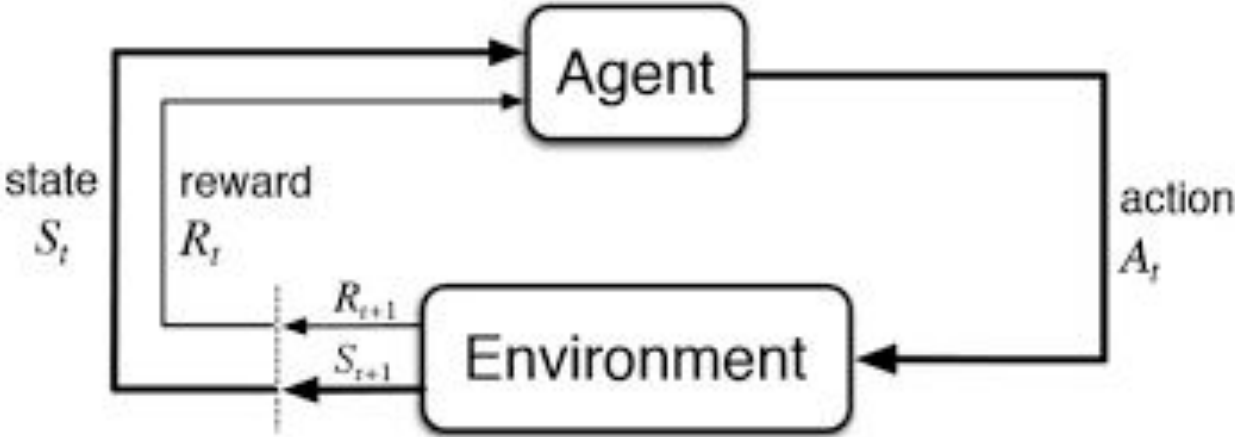
# Creación de mapas

MapGAN: An Intelligent Generation Model for Network Tile Maps





# Refuerzo



# Atención

Efecto de realzar algunos patrones de los datos de entrada y permite transferir información entre redes neuronales

## Attention Is All You Need

Ashish Vaswani\*  
Google Brain  
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Noam Shazeer\*  
Google Brain  
noam@google.com

Niki Parmar\*  
Google Research  
nikip@google.com

Jakob Uszkoreit\*  
Google Research  
uszko@google.com

Llion Jones\*  
Google Research  
llion@google.com

Aidan N. Gomez\*<sup>†</sup>  
University of Toronto  
aidan@cs.toronto.edu

Illia Polosukhin\*<sup>‡</sup>  
illia.polosukhin@gmail.com

### Abstract

The dominant sequence transduction models are based on convolutional neural networks that include an encoder performing models also connect the encoder and decoder mechanism. We propose a new simple network architecture based solely on attention mechanisms, dispensing with recurrently. Experiments on two machine translation tasks are superior in quality while being more parallelizable and less time to train. Our model achieves 28.4 BLEU on the to-German translation task, improving over the existing ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art training for 3.5 days on eight GPUs, a small fraction of the best models from the literature. We show that the Transformer can be applied to other tasks by applying it successfully to English constituency parsing and limited training data.

### 1 Introduction

Recurrent neural networks, long short-term memory [13] and generative models, have been firmly established as state of the art approaches for sequence transduction.

\*Equal contribution. Listing order is random. Jakob proposed replacing the effort to evaluate this idea. Ashish, with Illia, designed and implemented the model. Illia has been crucially involved in every aspect of this work. Noam proposed the attention mechanism and the parameter-free position representation and became the main contributor to the implementation.

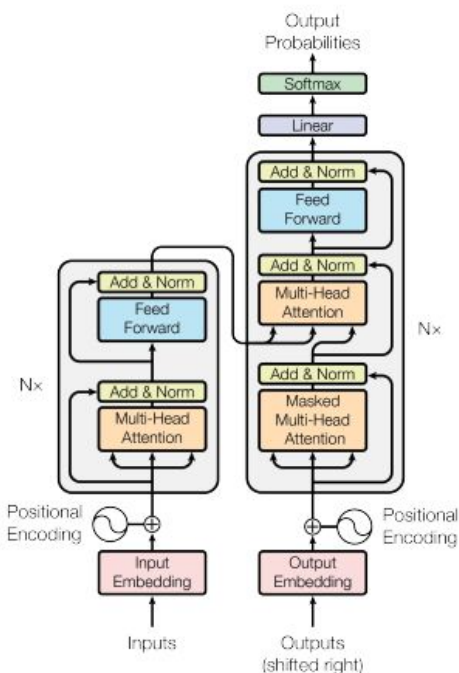
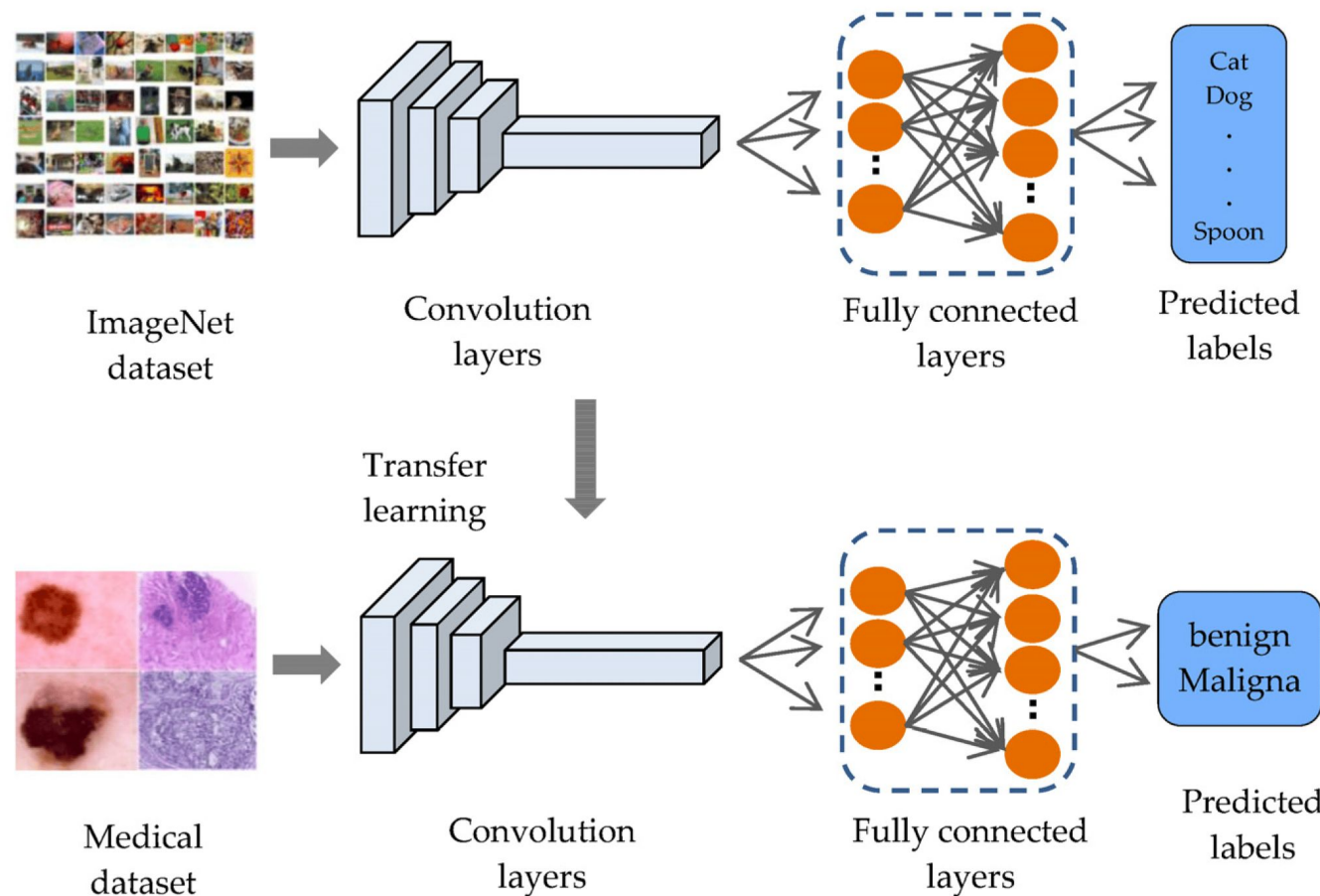
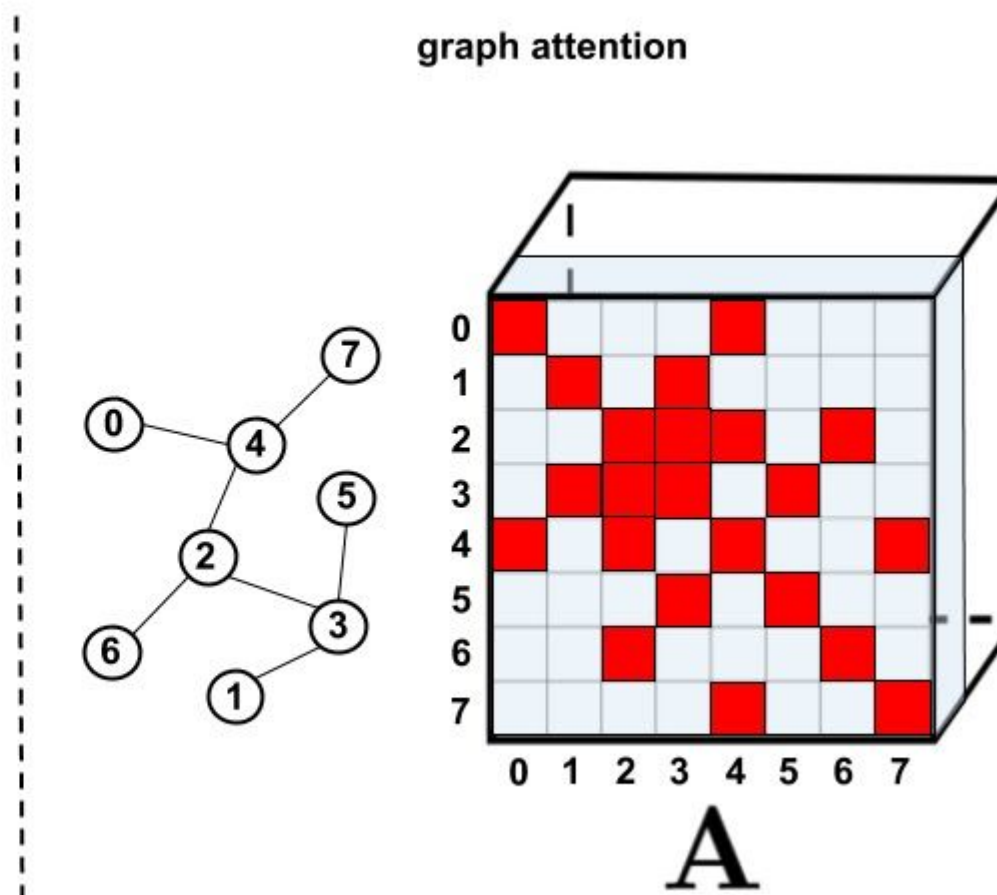
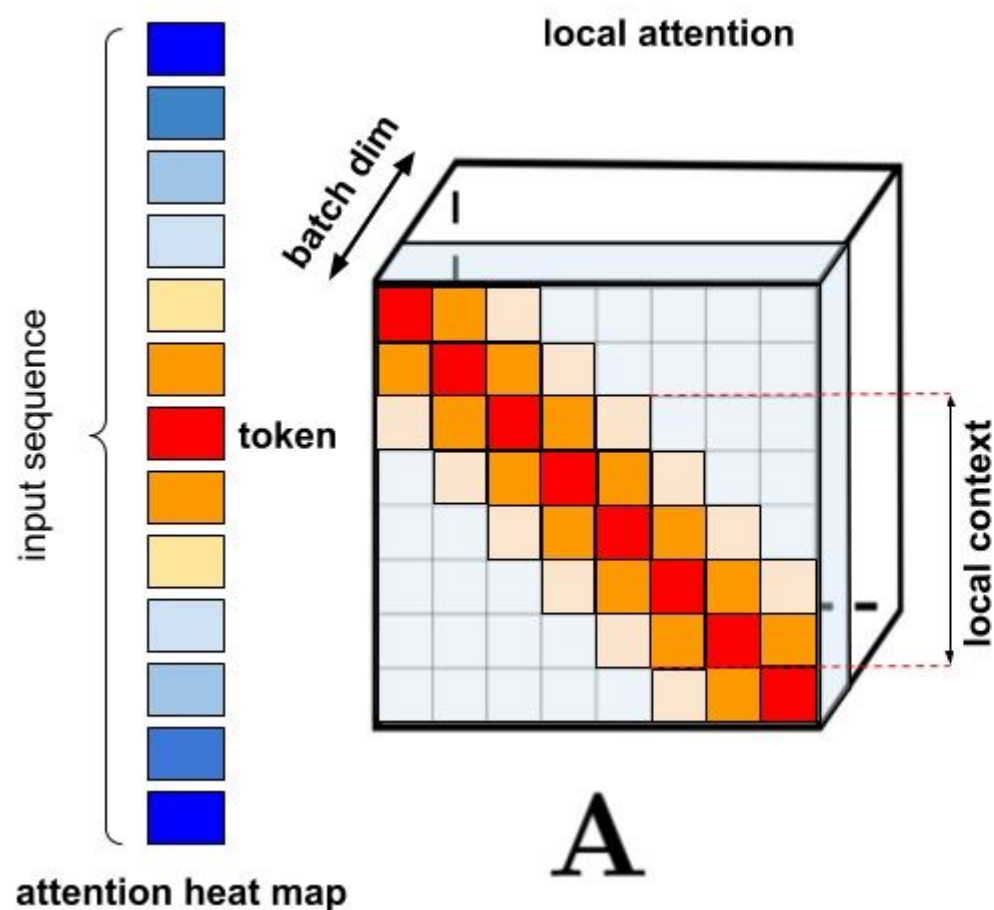


Figure 1: The Transformer - model architecture.



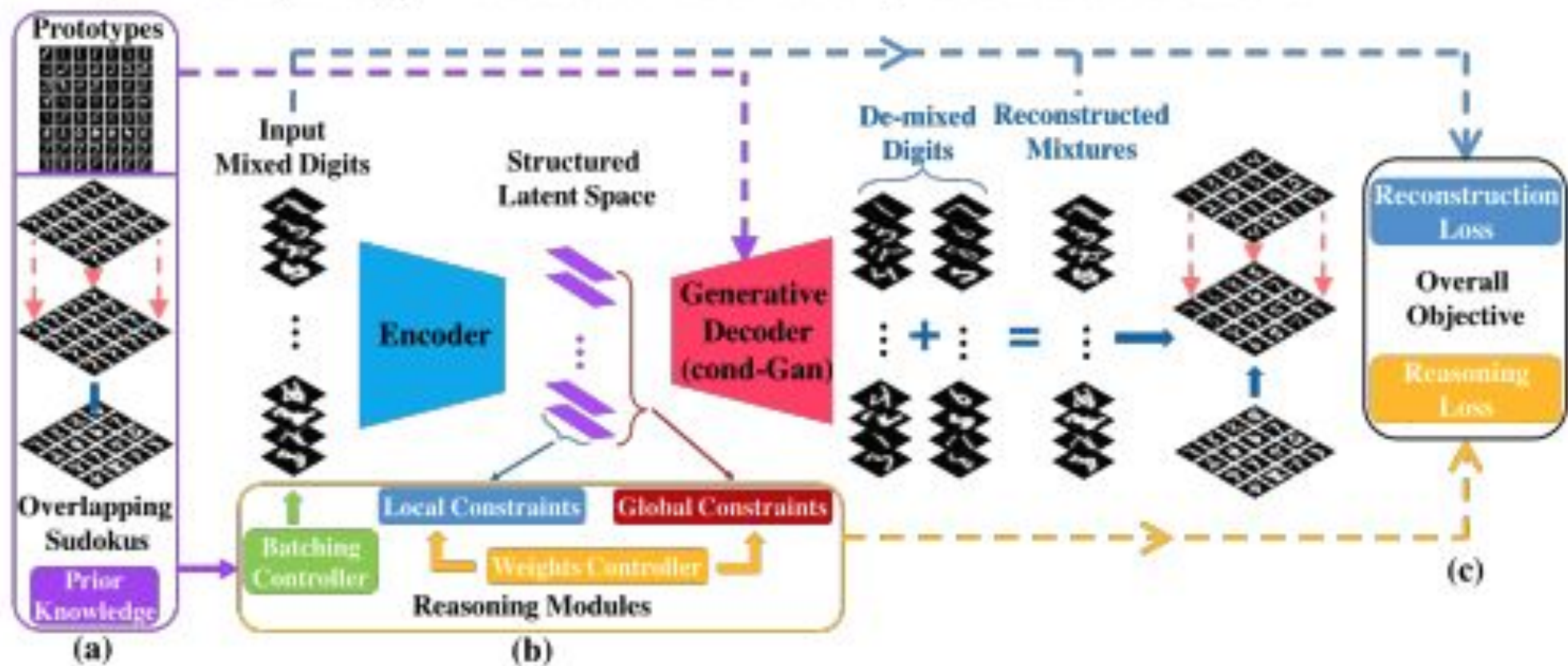
# Atención

Efecto de realzar algunos patrones de los datos de entrada y permite transferir información entre redes neuronales



# Razonamiento

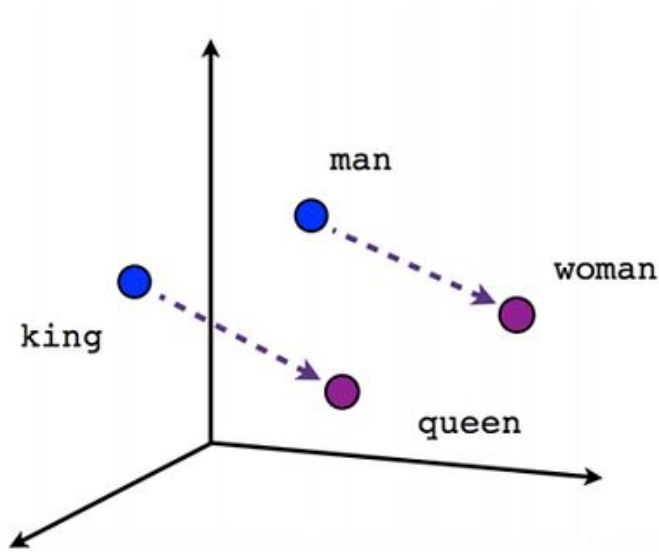
Capacidad de resolver problemas complejos mediante lógica, inducción o deducción.



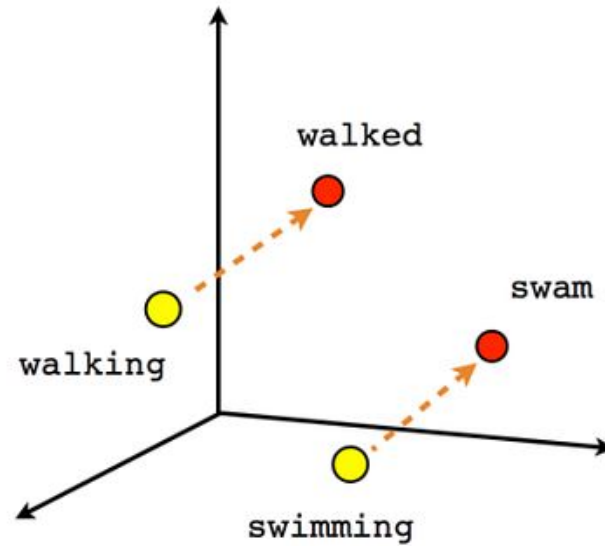


# Razonamiento

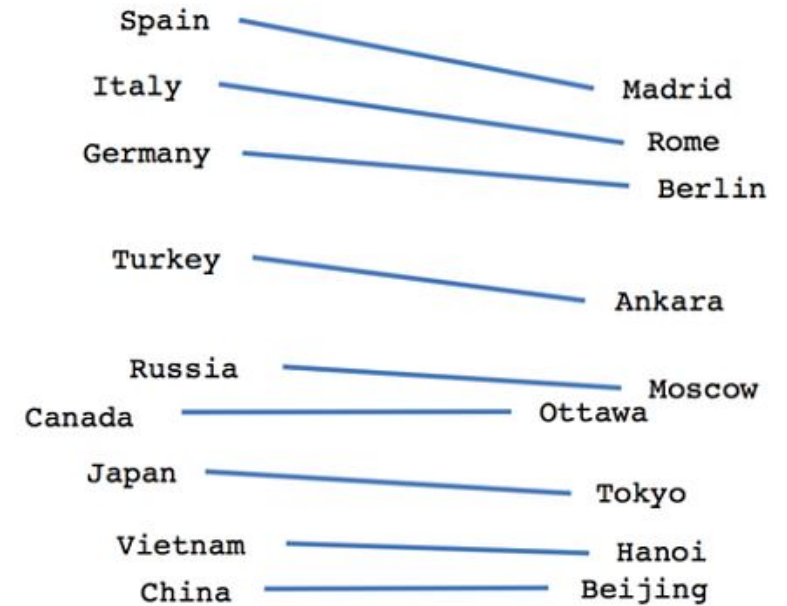
Capacidad de resolver problemas complejos mediante lógica, inducción o deducción.



Male-Female



Verb tense



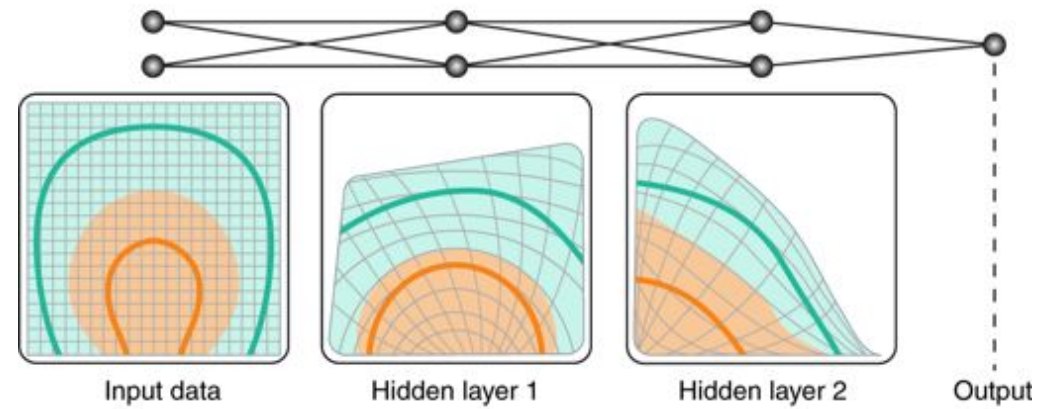
Country-Capital

# Explicatividad vs predictibilidad

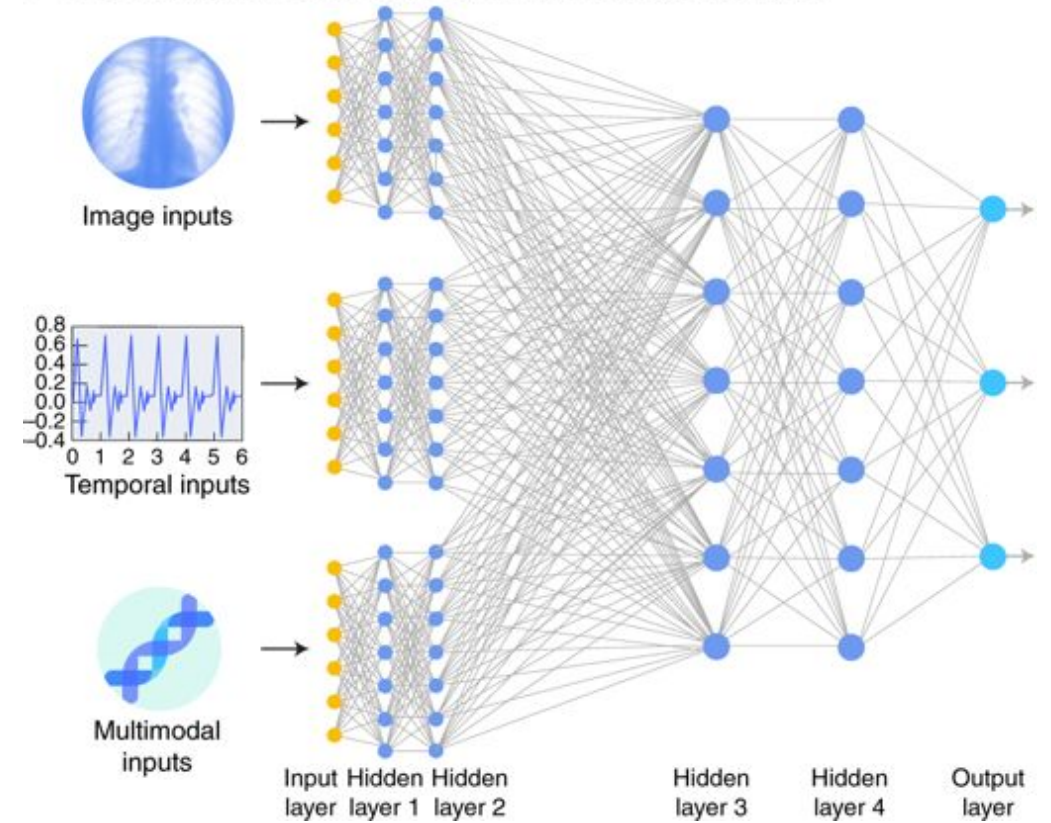


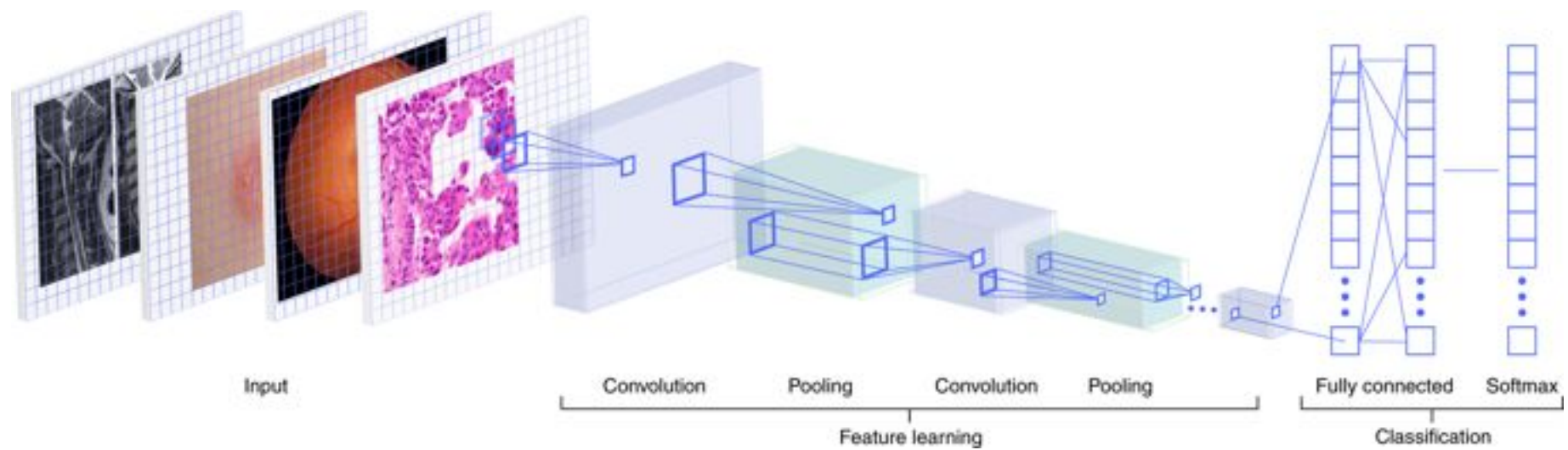
Casos de uso en salud

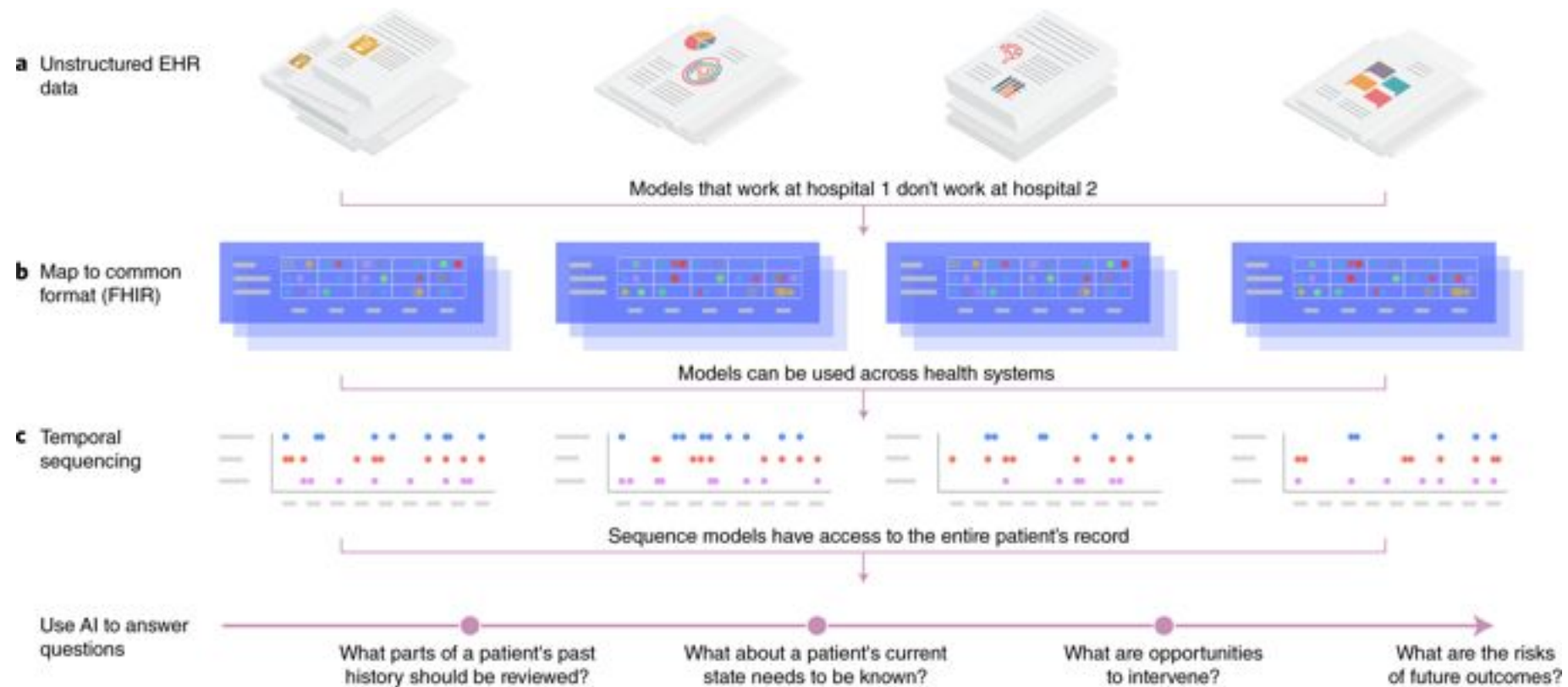
**a** Neural network layers make data linearly separable

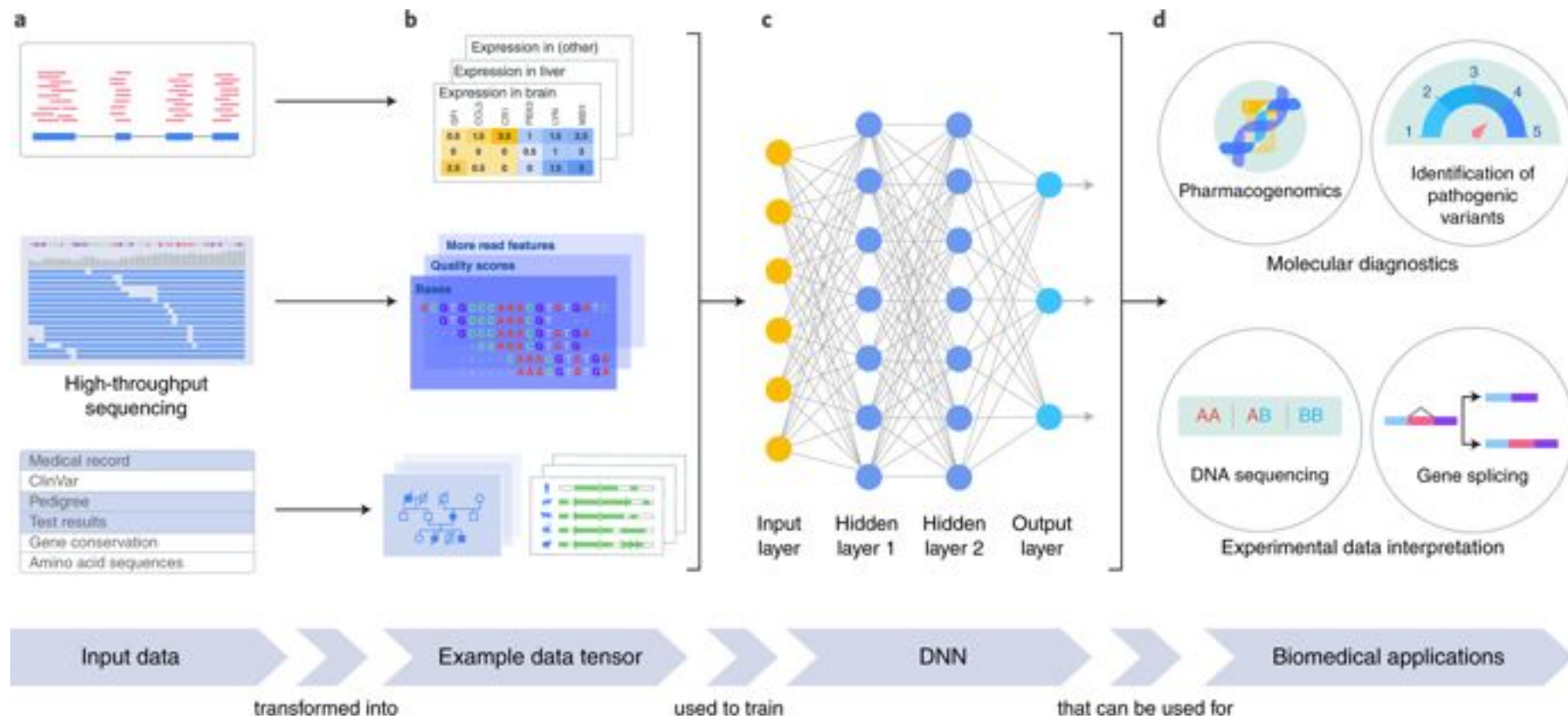


**b** Deep learning can featurize and learn from a variety of data types



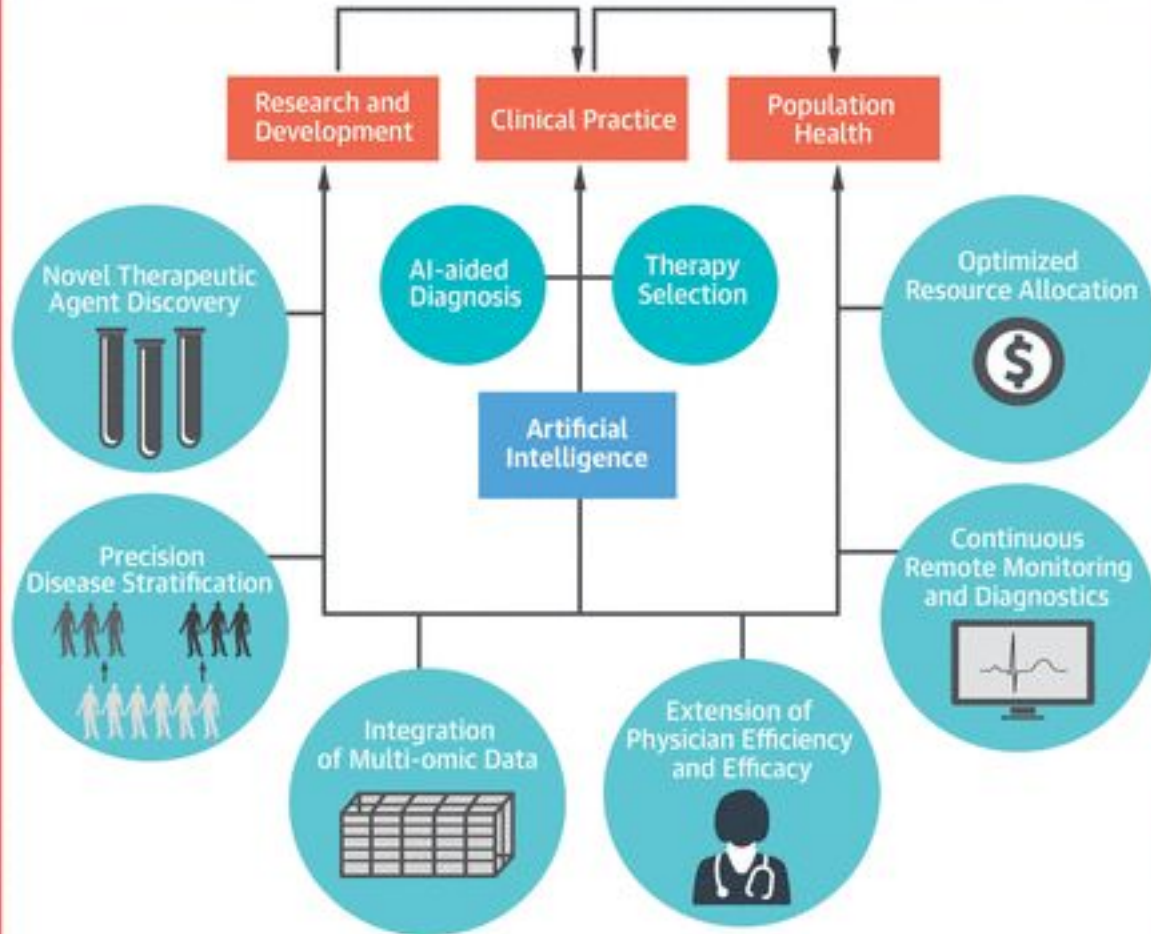






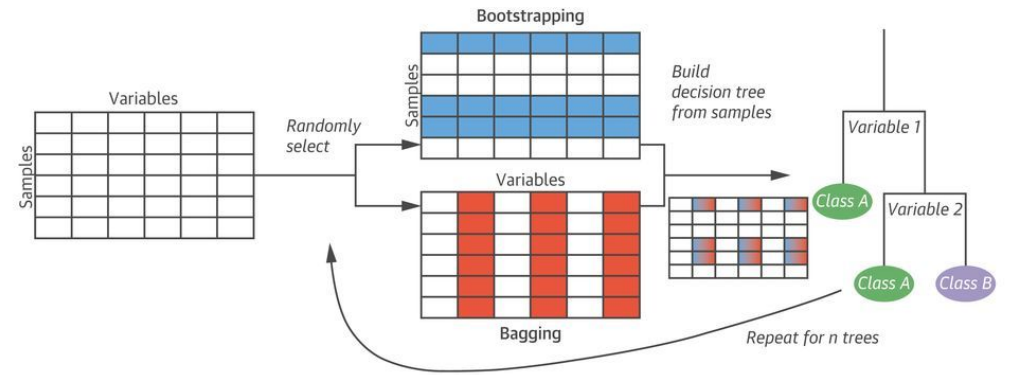


## CENTRAL ILLUSTRATION: Role of Artificial Intelligence in Cardiovascular Medicine

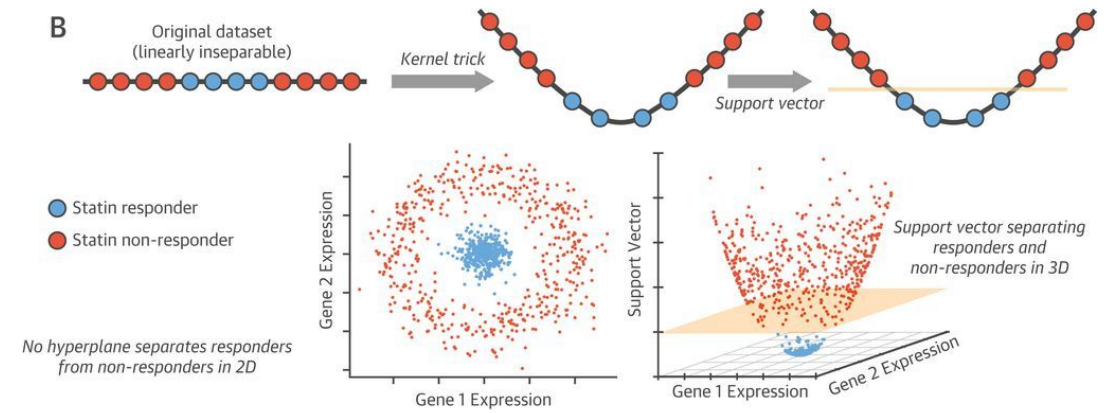


Johnson, K.W. et al. J Am Coll Cardiol. 2018;71(23):2668-79.

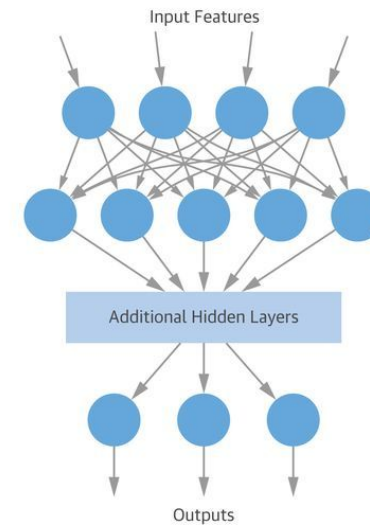
A



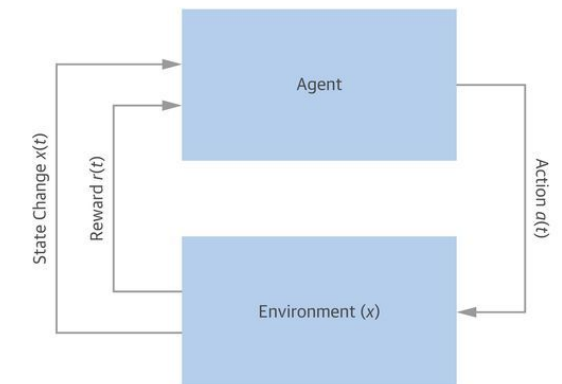
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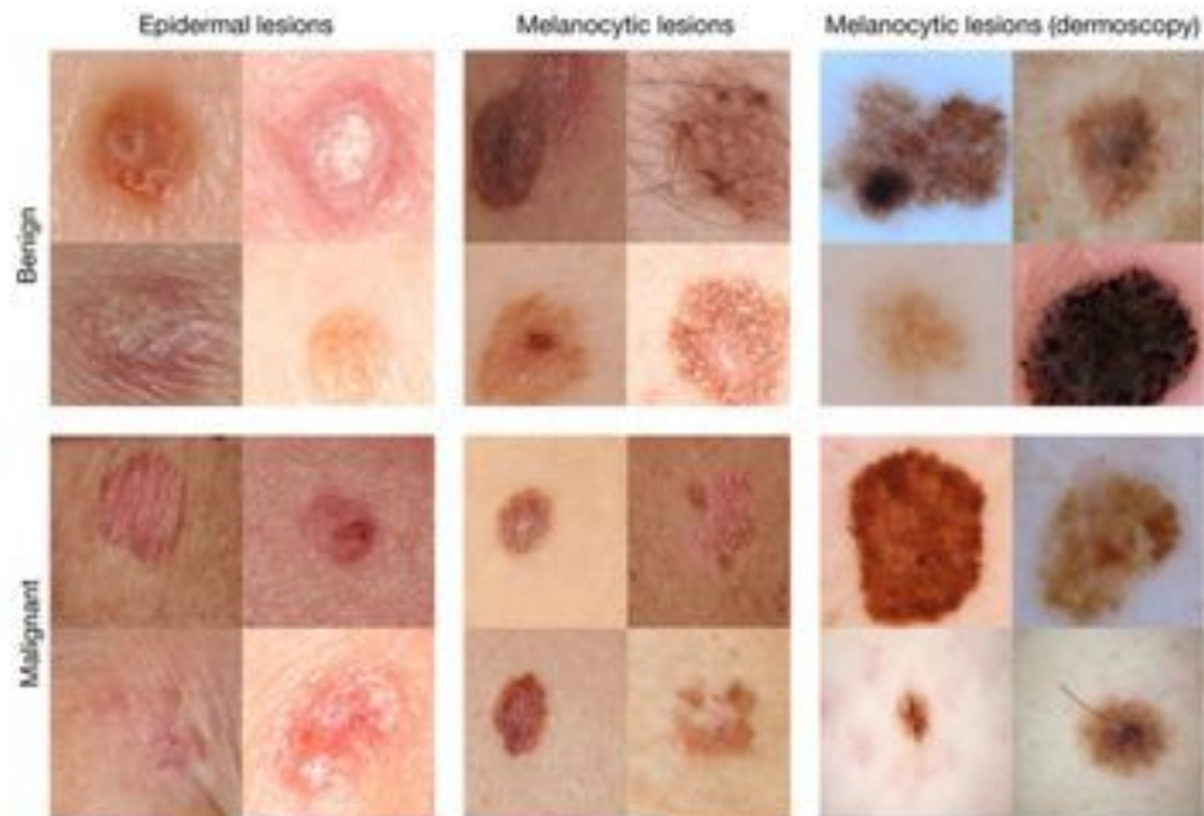
C

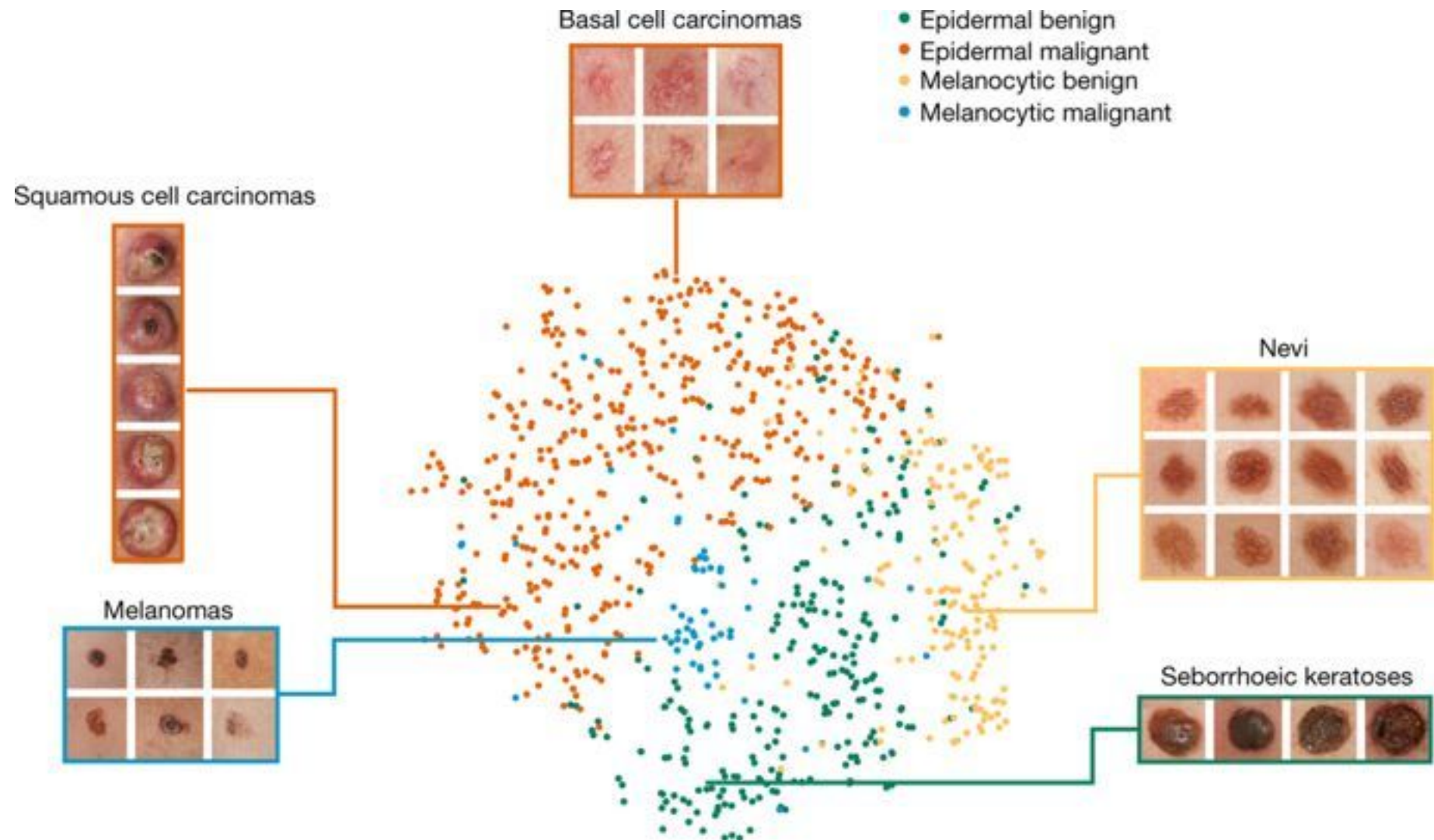


D

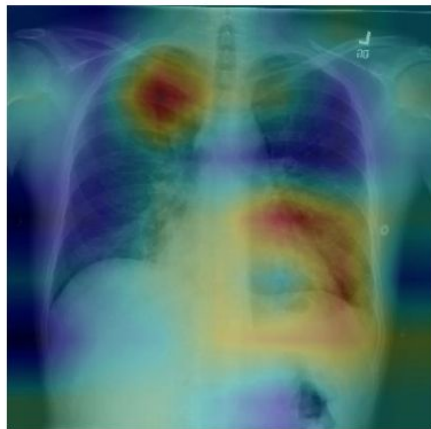




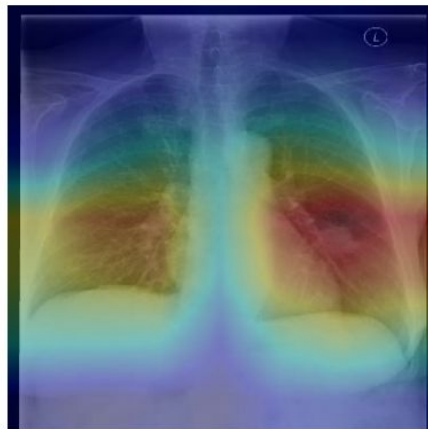
**a****b**



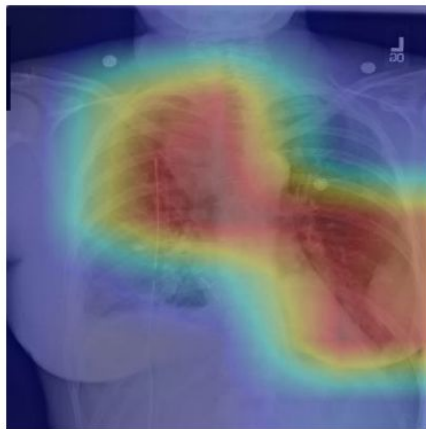




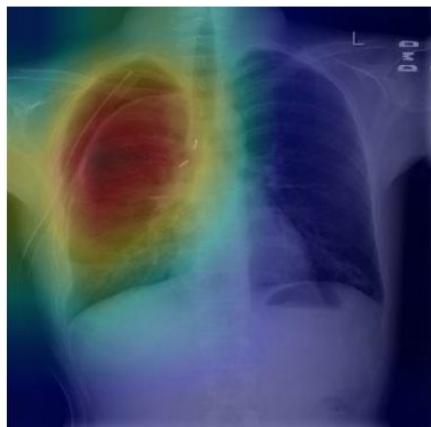
(a) Patient with multifocal community acquired pneumonia. The model correctly detects the airspace disease in the left lower and right upper lobes to arrive at the pneumonia diagnosis.



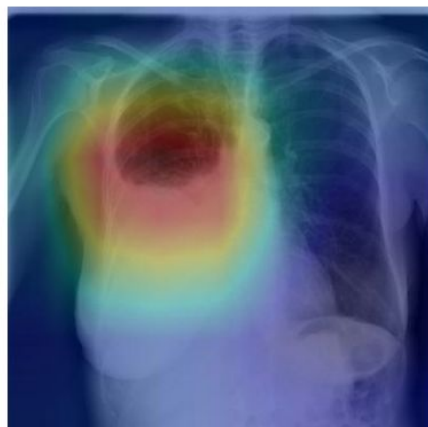
(b) Patient with a left lung nodule. The model identifies the left lower lobe lung nodule and correctly classifies the pathology.



(c) Patient with primary lung malignancy and two large masses, one in the left lower lobe and one in the right upper lobe adjacent to the mediastinum. The model correctly identifies both masses in the X-ray.



(d) Patient with a right-sided pneumothorax and chest tube. The model detects the abnormal lung to correctly predict the presence of pneumothorax (collapsed lung).



(e) Patient with a large right pleural effusion (fluid in the pleural space). The model correctly labels the effusion and focuses on the right lower chest.



(f) Patient with congestive heart failure and cardiomegaly (enlarged heart). The model correctly identifies the enlarged cardiac silhouette.

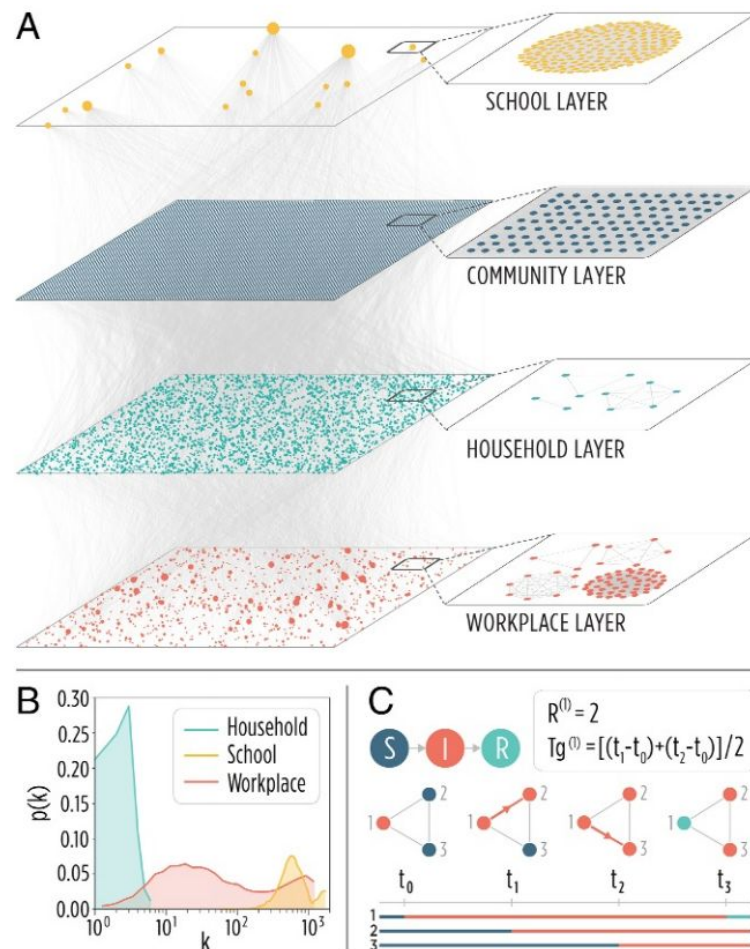


**Input**  
Chest X-Ray Image

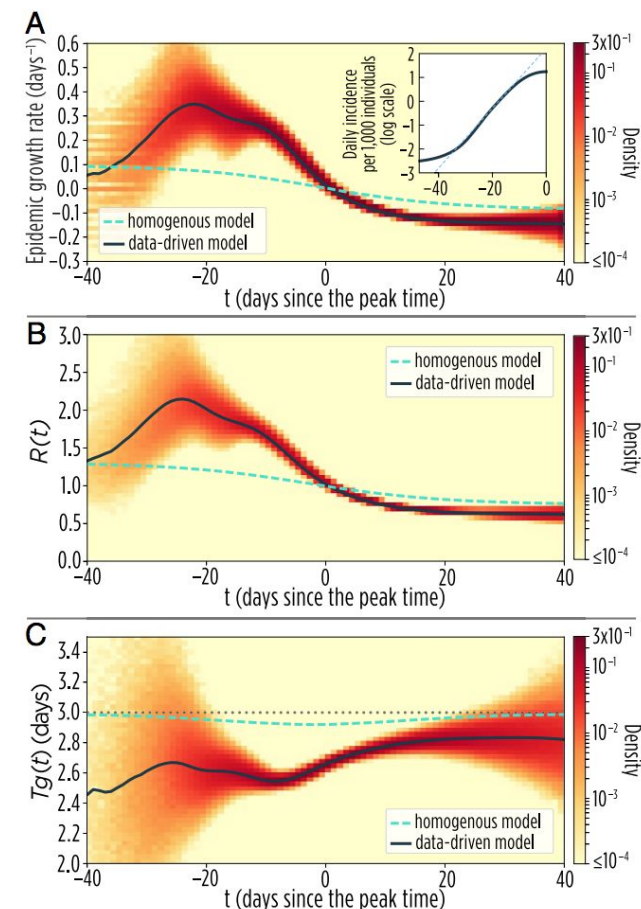
**CheXNet**  
121-layer CNN

**Output**  
Pneumonia Positive (85%)



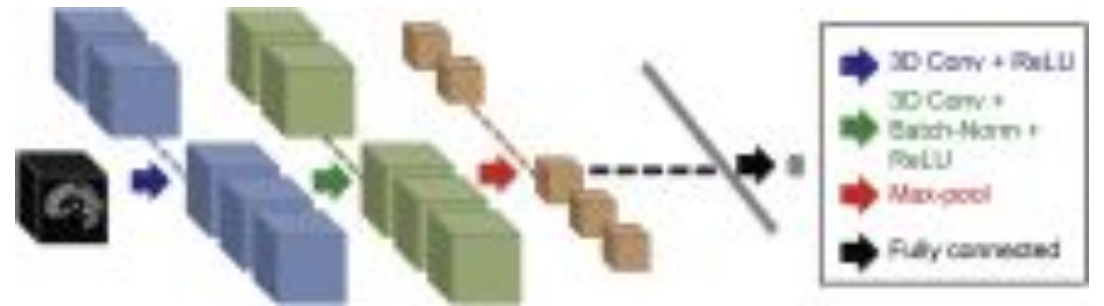
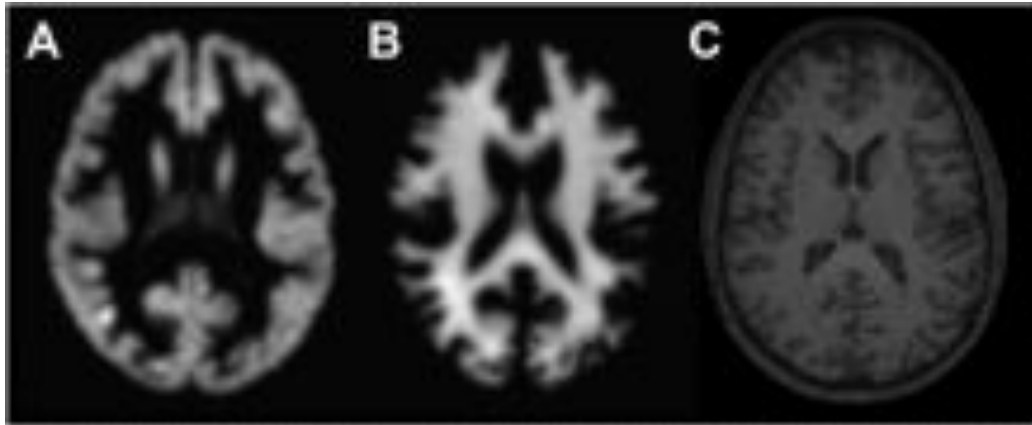


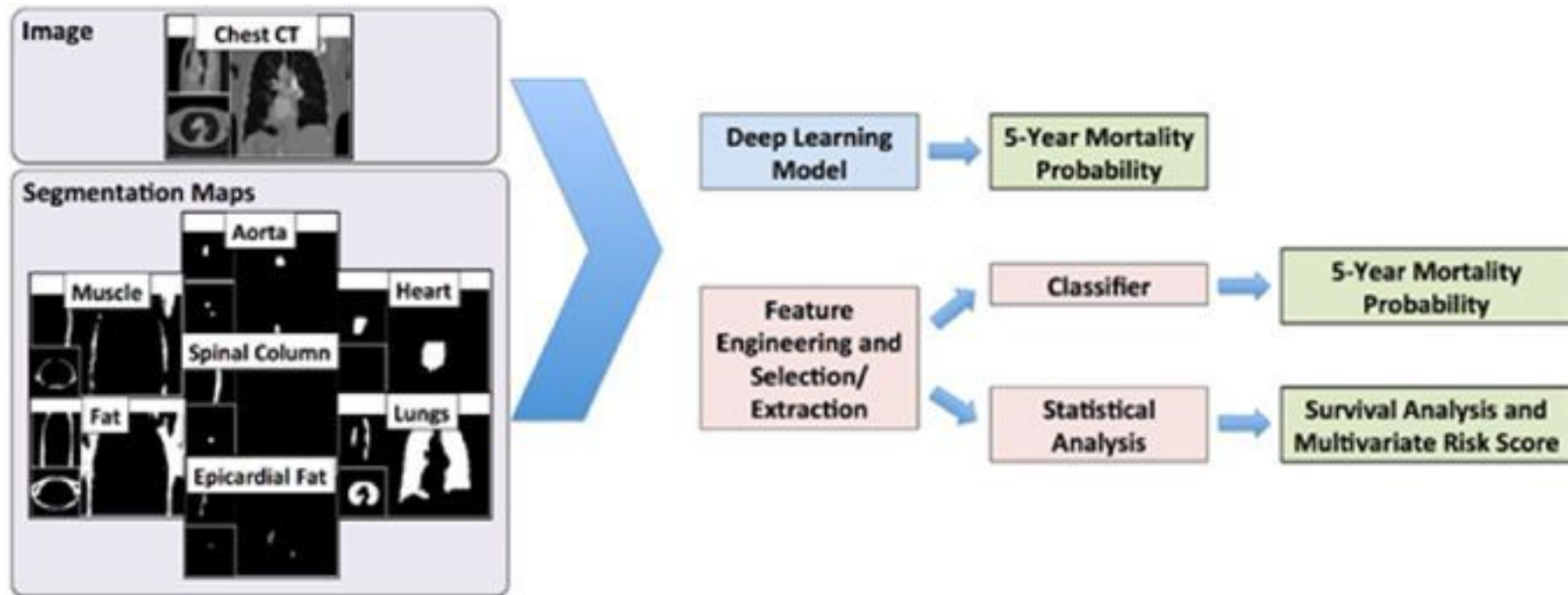
**Fig. 1.** Model structure. (A) Visualization of the multiplex network representing a subsample of 10,000 individuals of the synthetic population. Note that the community layer is a complete graph, although not all edges are visible for the sake of readability of the illustration. (B) Degree distributions in the school, household, and workplace layers. (C) Schematic representation of the infection transmission model along with examples of the computation of individual reproduction number and generation time for the simulated transmission chains. I, infectious; R, removed; S, susceptible.



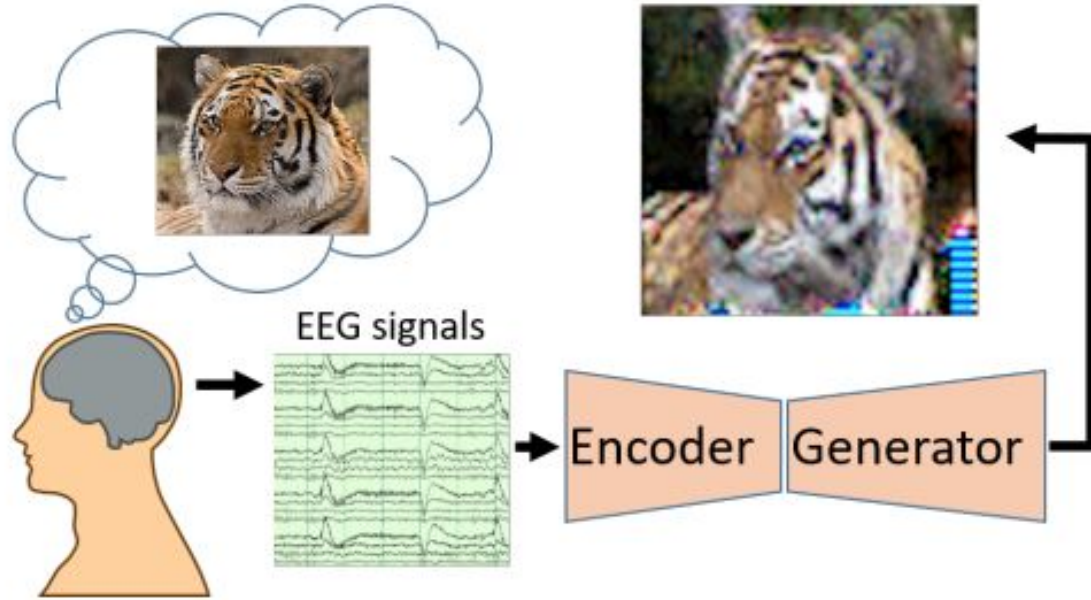
**Fig. 2.** Fundamental epidemiological indicators. (A) Mean daily exponential epidemic growth rate,  $r$ , over time of the data-driven and homogeneous models. The colored area shows the density distribution of  $r(t)$  values obtained in the single realizations of the data-driven model. Results are based on 50,000 realizations of each model. Results are aligned at the epidemic peak, which corresponds to time  $t = 0$ . Inset shows the logarithm of the mean daily incidence of new influenza infections over time, which does not follow a linear trend. (B) Mean  $R(t)$  of data-driven and homogeneous models. The colored area shows the density distribution of  $R(t)$  values obtained in the single realizations of the data-driven model. (C) The three lines represent the mean  $Tg(t)$  of data-driven and homogeneous models. The colored area shows the density distribution of  $Tg(t)$  values obtained in the single realizations of the data-driven model. The horizontal dotted





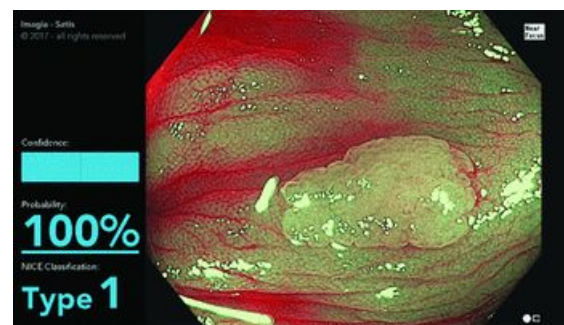
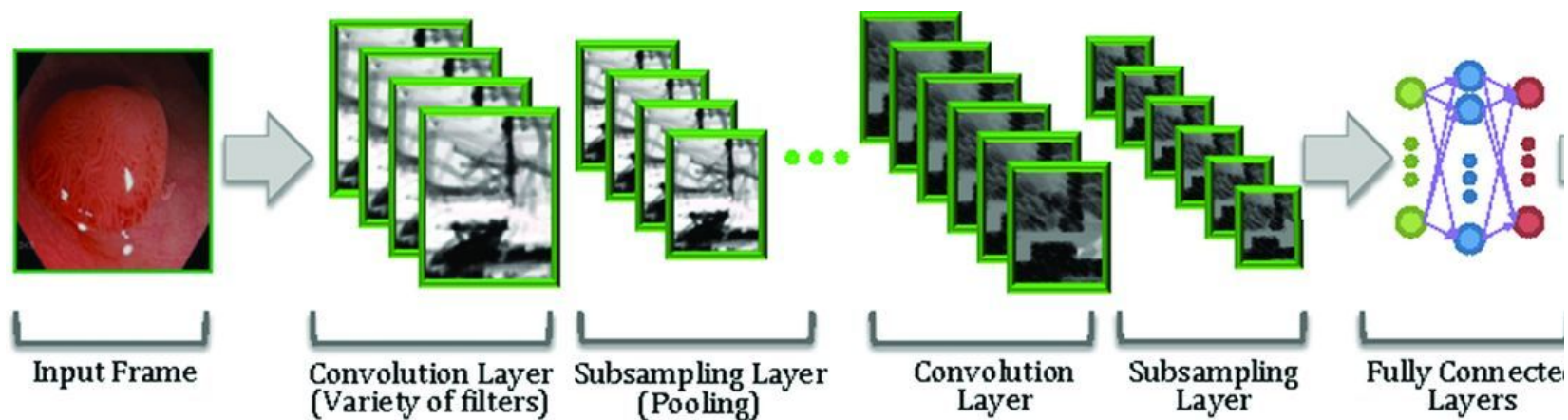




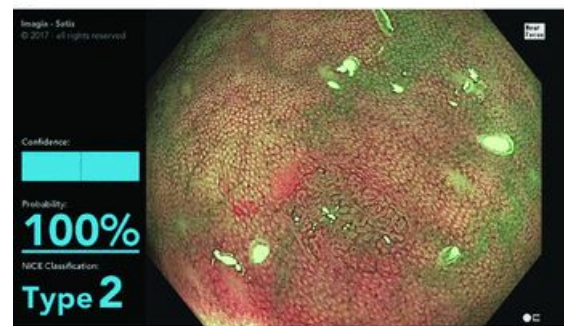


**Figure 1: Overview of our proposed method where an EEG signal from brain is sent to the encoder and the encoded signal is used to generate a visualization corresponding to the captured EEG signal.**





A



B

El CDC tiene una predicción ensamblada de todos los modelos que se han construido. Un gestor nunca puede confiar en el resultado de un solo modelo.

## CDC COVID Data Tracker

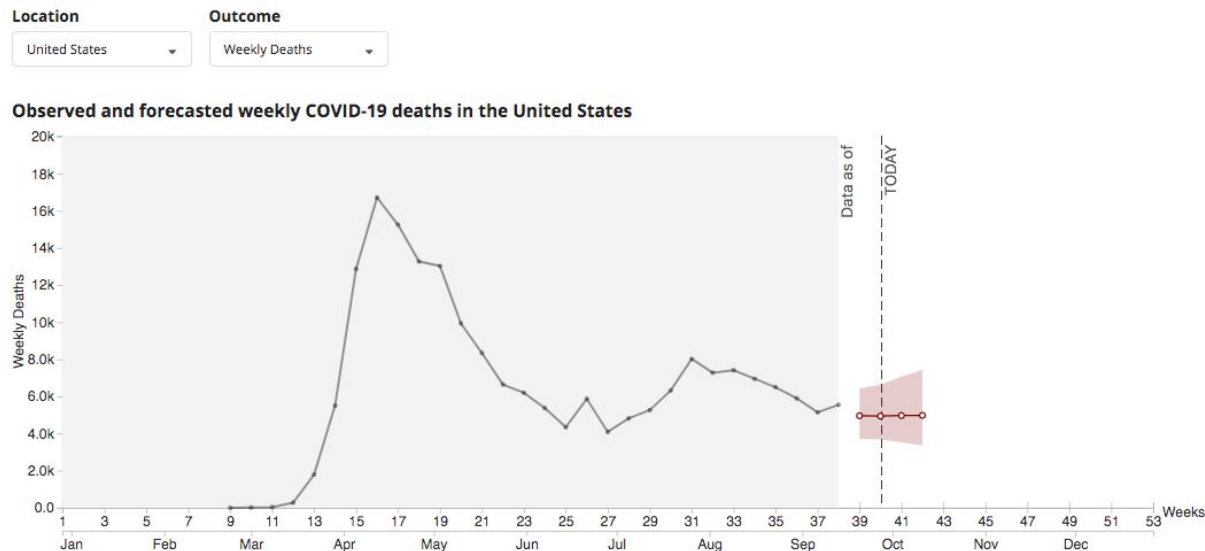
Maps, charts, and data provided by the CDC

Case Trends ▾ Laboratory ▾ Community Impact ▾ Unique Populations ▾ COVID-19 Home

Cases and Deaths by State | US and State Trends | Compare State Trends | Demographics | Trends by Population Factors | Cases and Deaths by County | Forecasting | Trends in ED Visits

### United States Forecasting

Updated September 24, 2020. Observed and forecasted cumulative and weekly reported COVID-19 deaths as of September 21, 2020.  
Use the interactive tool below to explore observed and forecasted cumulative and weekly COVID-19 deaths in the United States. Click on past weeks to see previous forecasts or use the arrows in the legend. For more information, please visit the CDC COVID-19 Forecasting Page.





Por eso el CDC ha recopilado todas las predicciones colaborando con terceros.

## CDC COVID Data Tracker

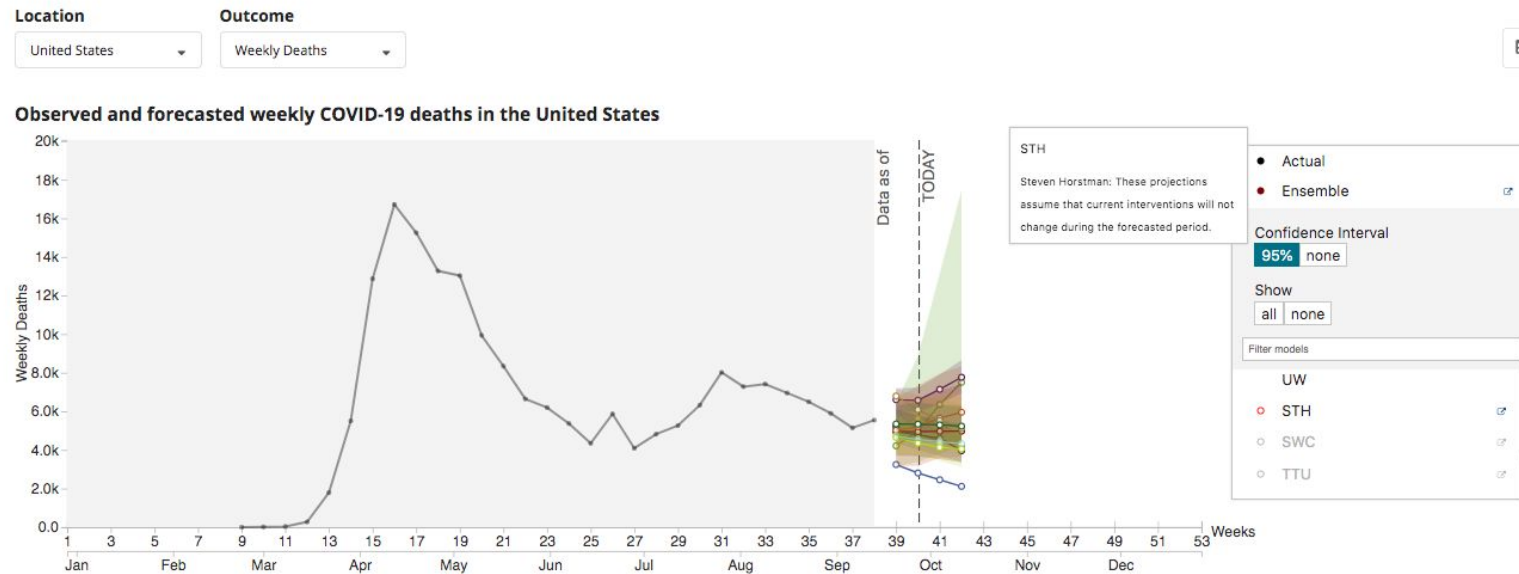
Maps, charts, and data provided by the CDC

Case Trends ▾ Laboratory ▾ Community Impact ▾ Unique Populations ▾ COVID-19 Home

[Cases and Deaths by State](#) | [US and State Trends](#) | [Compare State Trends](#) | [Demographics](#) | [Trends by Population Factors](#) | [Cases and Deaths by County](#) | [Forecasting](#) | [Trends in ED Visits](#)

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Productos basados en datos y  
modelos de negocio



## ¿Productos basados en datos?

Son aquellos en los que el core del negocio son los datos que manejan y permiten.

- ▶ Tomar decisiones
- ▶ Reducción de la incertidumbre.
- ▶ Automatizar o optimizar un proceso

# ¿Productos basados en datos?

Ibai @IbaiLlanos · 19h  
Alguien me explica qué ha sido esto



0:01

1,394 3,194 56.2K 3.4M

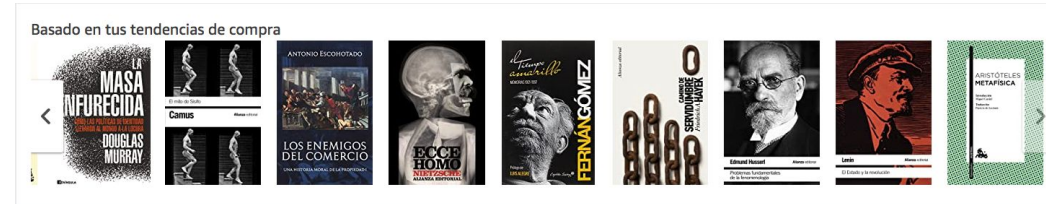
**Push notifications**  
Never miss what's happening on Twitter by enabling push notifications

Turn on notifications

**NEAR Protocol** @NEARProtocol  
200M in total transactions on #NEAR. Probably nothing

Follow us!

**NEAR Protocol** @NEARProtocol  
NEAR is the Blockchain Operating System.  
<https://t.co/RUXPdKERJf>



|         | ↑ 26°<br>↓ 13° | ↑ 30°<br>↓ 16° | ↑ 32°<br>↓ 19° | ↑ 29°<br>↓ 19° | ↑ 27°<br>↓ 16° | ↑ 29°<br>↓ 15° | ↑ 31°<br>↓ 17° |
|---------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 08:00   | 14°            | 17°            | 19°            | 19°            | 16°            | 16°            | 17°            |
| 14:00   | 26°            | 30°            | 32°            | 29°            | 27°            | 29°            | 31°            |
| 20:00   | 26°            | 30°            | 30°            | 27°            | 25°            | 28°            | 30°            |
| Lluvias | 0 mm           | 0 mm           | 0 mm           | 0 mm           | 0 mm           | 0 mm           | 0 mm           |
| Viento  | 15 km/h        | 13 km/h        | 19 km/h        | 17 km/h        | 11 km/h        | 9 km/h         | 9 km/h         |
|         | 06:46          | 06:46          | 06:47          | 06:47          | 06:48          | 06:48          | 06:49          |
|         | 21:49          | 21:49          | 21:49          | 21:49          | 21:49          | 21:49          | 21:49          |

13:06

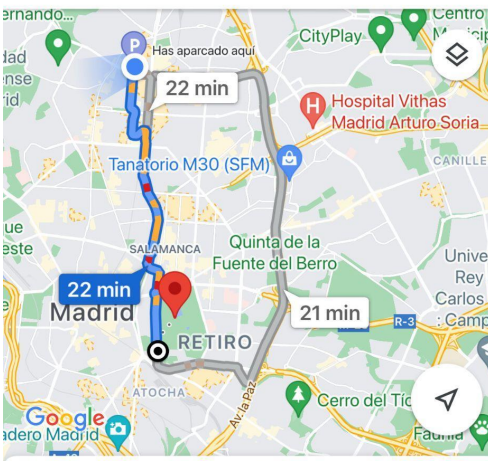
Tu ubicación

Parque de El Retiro

22 min 34 min 1 h 3 22 min

Evitando peajes

Se requiere mascarilla en algunos espacios públicos debido al COVID-19 [Ver más](#)



22 min (6,4 km) Dificil

Actualmente es la mejor ruta debido a las condiciones del tráfico

Pasos Iniciar Fijar

## ¿Productos basados en datos?



# ¿Productos basados en datos?



zebra

PROJECTS DATA SEARCH

6 Docs Hello Tom Goldberg

Reset search SEARCH

http://searchapp.zebra-med.com/?s

186,221 studies

CREATE DATASET

STUDY PARAMETERS

Modality  
CT - Computed Tomography,

Body part  
Chest,

Male (94,251) Female (94,226)

Age 35 - 110

▼ FINDINGS

- ☐ Abdominal Aortic Aneurysm 7,127
- ☐ Aortic Stentgraft 6,422
- ☐ Appendicitis 30
- ☐ Benign appearing Nodules 565

My Tab

Volume Referencing Flip Selection

Layout Studies Series Sort Tile Mode

ROBIN BATES  
F 923472412  
Pos: 195.00 mm  
SI: 1  
Acq#: ZEBRA  
Patient Pos: HFS  
Study Desc: PET-CT SCAN  
Series Desc: Plain Chest 2.5mm Inspiration  
◀ 2 - 1 ▶

Current  
MEDICAL SYSTEMS Discovery STE  
38% Pixel  
DFOV 146.6 x 60.1 cm

Viewer  
10 cm

C 50  
W 500

Link 1

Current  
MEDICAL SYSTEMS Discovery STE  
120kV, 50mAs  
SC 500.00 mm

Current  
GE MEDICAL SYSTEMS Discovery STE  
120kV, 30mAs  
SC 500.00 mm

AI1

Robin Bates Mar 8, 2010

Lung Nodules

No lung nodule has been detected.

Not Detected Detected

Copy To Clipboard



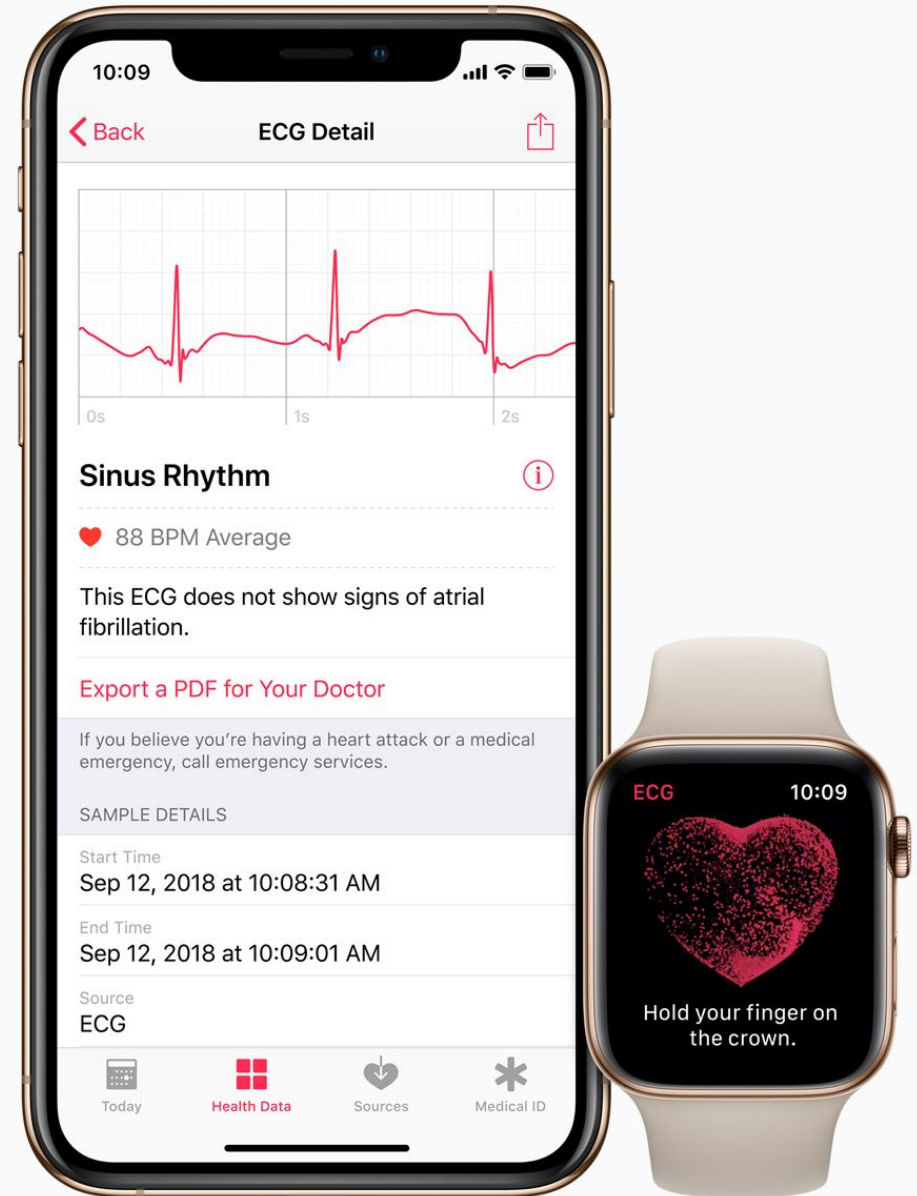
¿Productos basados en datos?

kaia





# ¿Productos basados en datos?

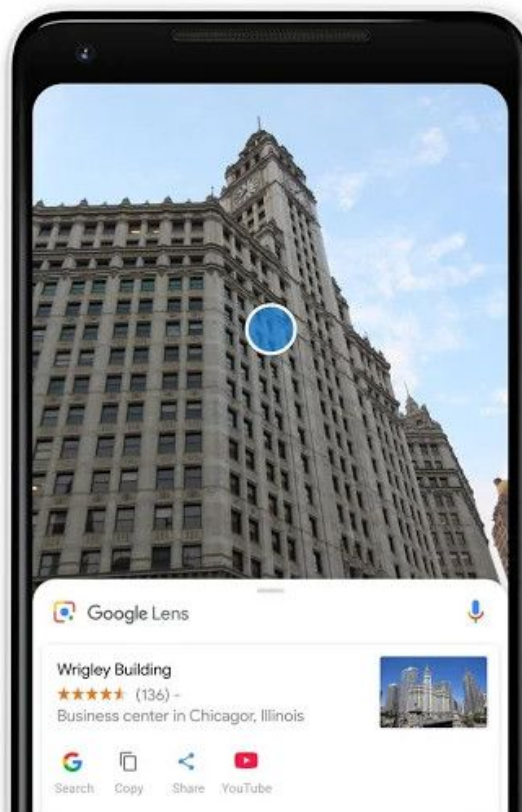


# ¿Productos basados en datos?

Take action on text



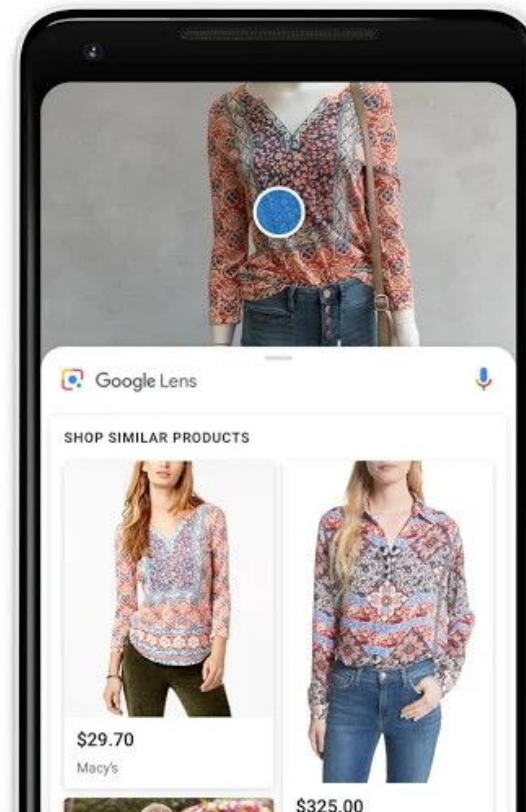
Learn more about the world



Identify plants and animals



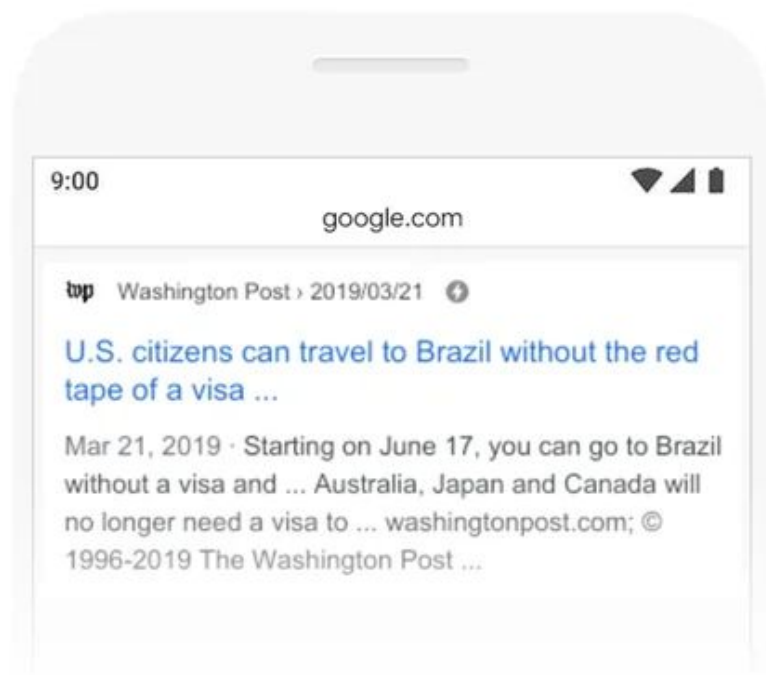
Find a look you like



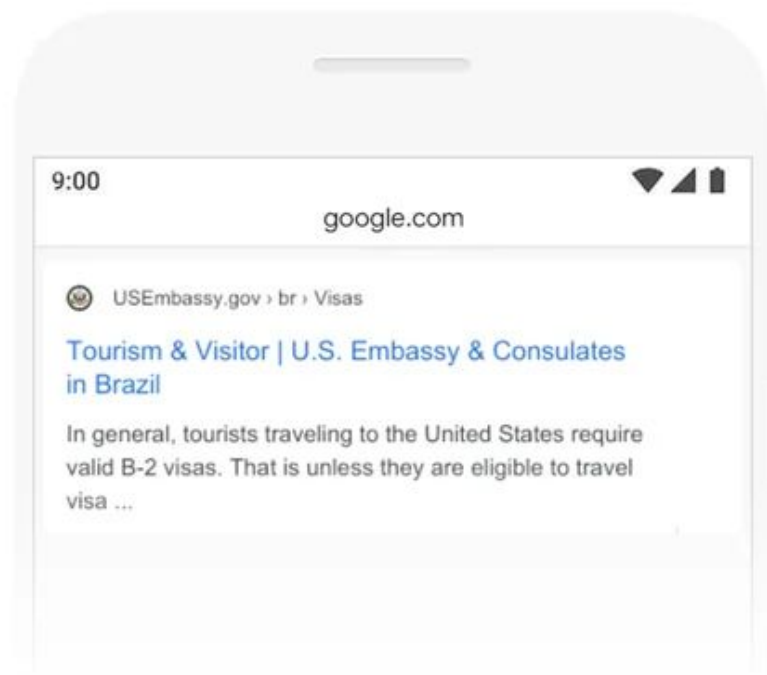
# ¿Productos basados en datos?

2019 brazil traveler to usa need a visa

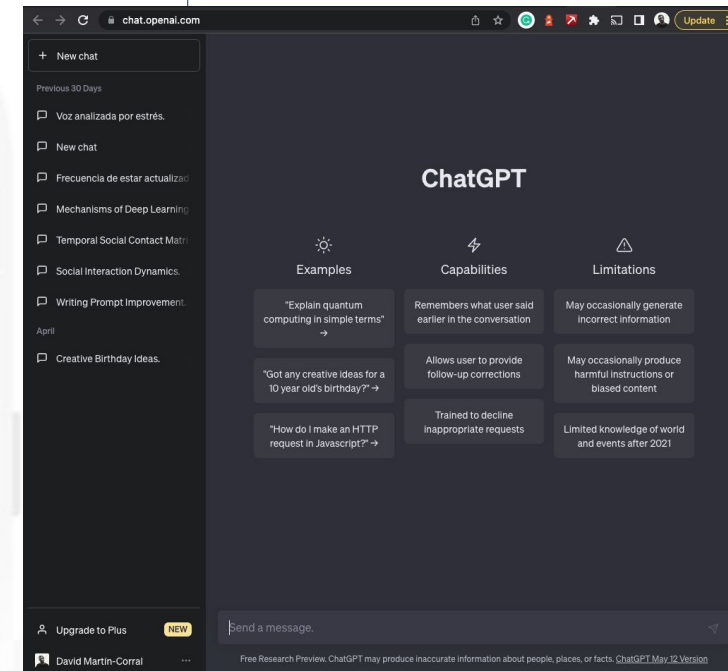
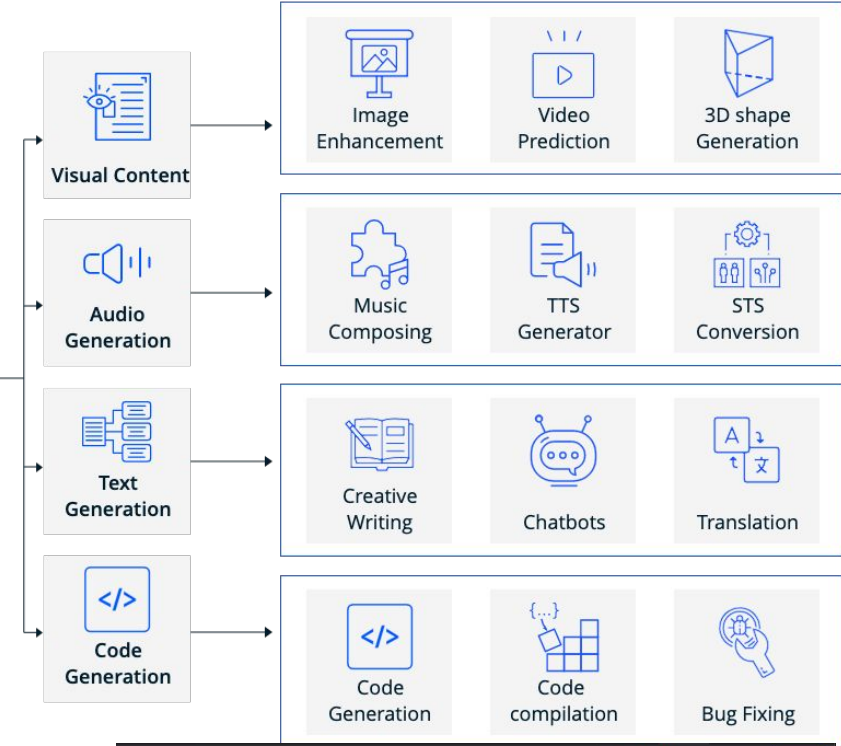
BEFORE



AFTER

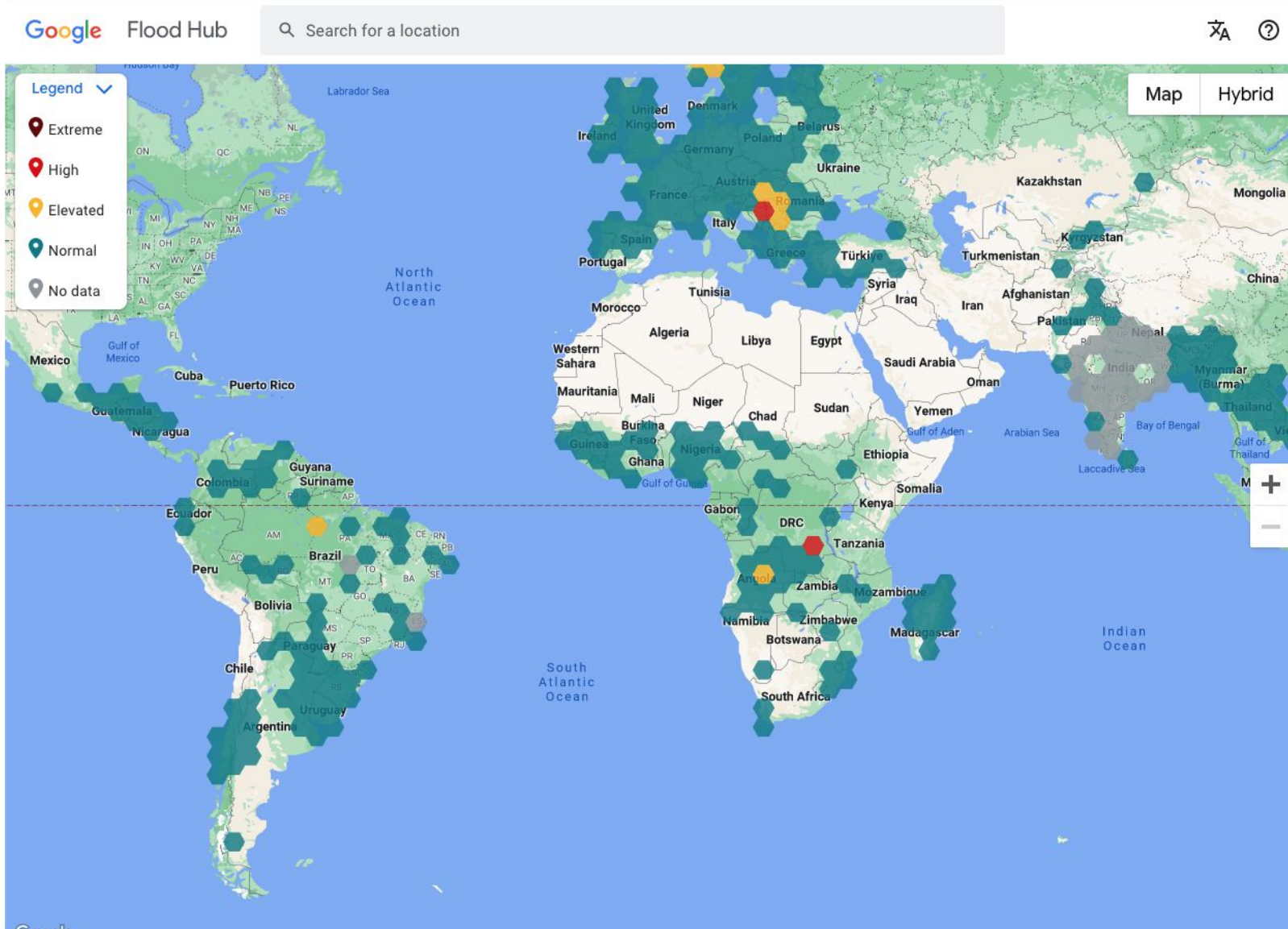


Generative AI use cases





# ¿Productos basados en datos?



## Modelos de negocio

Posibles modelos para monetizar los datos en el entorno actual.

- ▶ **Usuarios de datos:** Empresas que utilizan datos para elaborar estrategias y crear mejores productos.
- ▶ **Proveedores de datos:** Empresas que se dedican principalmente al comercio de datos.
- ▶ **Redes de distribución:** Empresas con un modelo de negocio publicitario.
- ▶ **Facilitadores de infraestructura:** Empresas que participan en el suministro de infraestructura de datos, análisis y consultoría.